

Vision-Language-Action (VLA) Models: Concepts, Progress, Applications and Challenges

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Abstract

Vision-Language-Action (VLA) models mark a transformative advancement in artificial intelligence, aiming to unify perception, natural language understanding, and embodied action within a single computational framework. This foundational review presents a comprehensive synthesis of recent advancements in Vision-Language-Action models, systematically organized across five thematic pillars that structure the landscape of this rapidly evolving field. We begin by establishing the conceptual foundations of VLA systems, tracing their evolution from cross-modal learning architectures to generalist agents that tightly integrate vision-language models (VLMs), action planners, and hierarchical controllers. Our methodology adopts a rigorous literature review framework, covering over 80 VLA models published in the past three years. Key progress areas include architectural innovations, efficient training strategies, and real-time inference accelerations. We explore diverse application domains such as autonomous vehicles, medical and industrial robotics, precision agriculture, humanoid robotics, and augmented reality. The review further addresses major challenges across real-time control, multimodal action representation, system scalability, generalization to unseen tasks, and ethical deployment risks. Drawing from the state-of-the-art, we propose targeted solutions including agentic AI adaptation, cross-embodiment generalization, and unified neuro-symbolic planning. We outline a forward-looking roadmap where VLA models, VLMs, and agentic AI converge to strengthen socially aligned, adaptive, and general-purpose embodied agents. This work, therefore, is expected to serve as a foundational reference for advancing intelligent, real-world robotics and artificial general intelligence. The project repository is available on GitHub (Source Link)

Keywords: Vision-Language-Action, Action Tokenization, Artificial Intelligence, Robotics, Vision-Language Models

1. Introduction

Before Vision-Language-Action (VLA) models were developed, progress in robotics and artificial intelligence happened mostly in separate domains: vision systems that could acquire, interpret and recognize images [55, 87, 176], language systems that could understand and generate text [213, 177], and action systems that could control movement [60]. These isolated systems worked well on their own, but struggled to work together and generalize to novel scenarios or adapt to the complexity and unpredictability of real-world challenges [57, 22].

As illustrated in Figure 1, traditional computer vision models, primarily based on convolutional neural networks (CNNs), were tailored for narrowly specified tasks such as object detection [25, 157, 26, 24, 174] or classification [205, 89, 95, 73], requiring extensive labeled datasets and cumbersome retraining for even slight shifts in environment or objectives [200, 77]. These vision models could “see” (e.g., identifying apples in an orchard, as shown in Figure 1) but lacked any understanding of language or the ability to convert visual insights into desired actions. Language models, particularly large language mod-

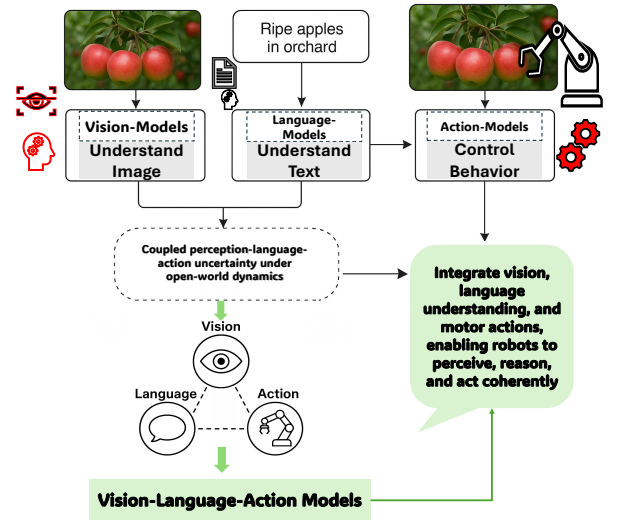


Figure 1: **Evolution from isolated modalities to unified Vision-Language-Action models.** Integrated perception, language, and action enable adaptive, generalizable embodied intelligence.

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els (LLMs), revolutionized text-based understanding and generation [28]; however, they remained restricted to processing

视觉-语言-动作（VLA）模型：概念、进展、应用与挑战

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摘要

视觉-语言-动作（VLA）模型标志着人工智能领域的一项革命性突破，其目标是在单一计算框架内整合感知、自然语言理解与具身化动作。这篇基础性综述全面总结了VLA模型的最新进展，通过五个主题支柱系统性地梳理了这一快速发展的领域现状。我们首先阐述了VLA系统的概念基础，追溯其从跨模态学习架构到集成视觉语言模型（VLM）、动作规划器和分层控制器的通用智能体的演变历程。本研究采用严谨的文献综述方法，涵盖了过去三年发布的80余种VLA模型。主要进展领域包括架构创新、高效训练策略及实时推理加速技术。我们探讨了自动驾驶汽车、医疗与工业机器人、精准农业、人形机器人及增强现实等多样化的应用领域。该综述进一步探讨了实时控制、多模态动作表征、系统可扩展性、对未见任务的泛化能力以及伦理部署风险等主要挑战。基于当前最前沿的研究成果，我们提出了针对性解决方案，包括智能体AI适应性、跨具身化泛化以及统一神经符号规划。我们勾勒出一条前瞻性路线图：VLA模型、VLMs与智能体AI将相互融合，共同打造具备社会适应性、自适应能力及通用功能的具身智能体。因此，本研究有望成为推动智能现实机器人技术与通用人工智能发展的基础性参考文献。项目代码库已发布于GitHub（源代码链接）。

关键词：视觉-语言-动作、动作分词、人工智能、机器人技术、视觉-语言模型

1. 引言

在视觉-语言-动作（VLA）模型问世之前，机器人技术和人工智能领域的进展主要集中在各自独立的领域：能够获取、解析和识别图像的视觉系统[55, 87, 176]、具备文本理解与生成能力的语言系统[213, 177]，以及可控制运动的动作系统[60]。这些孤立的系统虽能独立运行良好，但在协同工作、泛化到新场景或适应现实挑战的复杂性与不可预测性方面仍存在局限[57, 22]。

如图1所示，传统的计算机视觉模型主要基于卷积神经网络（CNN），专为特定任务设计，例如目标检测[25, 157, 26, 24, 174]或分类[205, 89, 95, 73]，需要大量标注数据集，且即使环境或目标发生微小变化也需要繁琐的重新训练[200, 77]。这些视觉模型能够“看见”（例如，在图像中识别苹果）如图1所示，该系统虽具备果园管理功能，却完全缺乏语言理解能力，也无法将视觉洞察转化为预期动作。语言模型（尤其是大型语言模型）

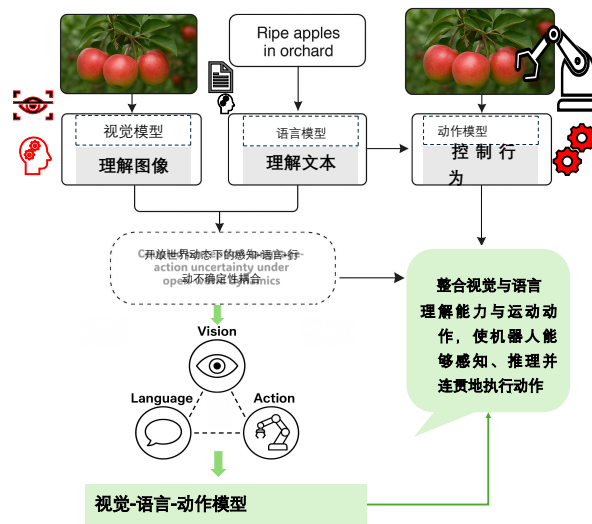


图1：从独立的感知模态到统一的视觉-语言-动作模型的演变。整合感知、语言和动作能力，能够实现具有自适应性和可泛化的具身智能。

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大语言模型（LLMs）彻底革新了基于文本的理解与生成能力[28]；然而，其应用仍局限于特定类型的文本处理任务。

language without the capability to perceive or reason about the physical world [98] (“Ripe apples in orchard” in Figure 1 exemplifies this limitation). Meanwhile, action-based systems in robotics, relying heavily on hand-crafted policies or reinforcement learning [158], enabled specific behaviors like object manipulation but demanded painstaking engineering and failed to generalize beyond specifically designed scenarios [153].

Despite progress with VLMs, which achieved impressive multi-modal understanding by combining vision and language [190, 30, 298, 37, 10, 189], there remained a clear integration gap: the inability to generate or execute coherent actions based on multi-modal input [156, 137]. As further visualized in Figure 1, most AI systems are specialized in one or two modalities such as vision-language, vision-action, or language-action, which struggled to fully integrate all three into a unified, end-to-end frameworks. Consequently, robots could recognize objects visually (“apple”), understand a corresponding textual instruction (“pick the apple”), or perform a predefined motor action (grasping), yet integrating and performing all these abilities into fluid, adaptable behavior has been missing. The result was a pipeline that could not flexibly adapt to new tasks or environments, leading to brittle generalization and labor-intensive engineering efforts. This limitation highlighted a critical bottleneck in embodied AI: without systems that could jointly perceive, understand, and act, intelligent autonomous behavior remained a challenging goal.

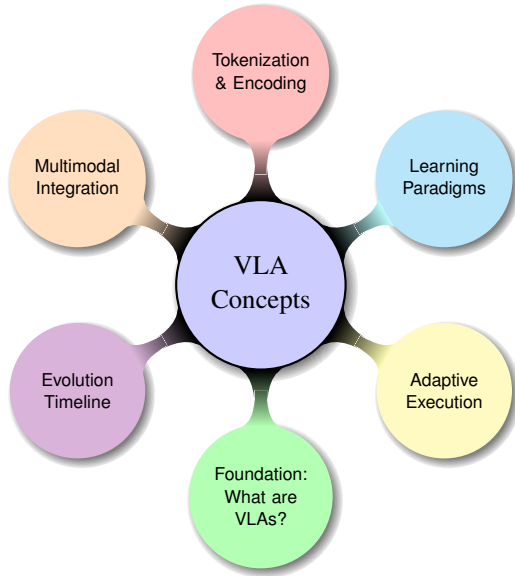


Figure 2: Mind map of core VLA concepts. Each color-coded branch highlights a foundational dimension: definitions (foundation), historical evolution, multi-modal integration, tokenization and encoding, learning paradigms, and adaptive execution in embodied settings.

The pressing need to bridge these gaps catalyzed the emergence of VLA models. VLA models, conceptualized around 2021-2022, and pioneered by efforts such as Google DeepMind’s Robotic Transformer 2 (RT-2) [299], introduced a transformative architecture that unified perception, reasoning, and control within a single framework. As a solution to the limitations outlined in Figure 2, VLAs integrate vision inputs,

language comprehension, and motor control capabilities, enabling embodied agents to perceive their surroundings, understand complex instructions, and execute appropriate actions dynamically. Early VLA approaches achieved this integration by extending vision-language models to include action tokens numerical or symbolic representations of robot motor commands, thereby allowing the model to learn from paired vision, language, and trajectory data [156]. This methodological innovation dramatically improved robots’ ability to generalize to unseen objects, interpret novel language commands, and perform multi-step reasoning in unstructured environments [107].

VLA models represent a transformative step in the development of unified multi-modal intelligence, overcoming the long-standing limitations of treating vision, language, and action as separate domains [156]. By leveraging internet-scale datasets that integrate visual, linguistic, and behavioral information, VLAs empower robots to not only recognize and describe their environments but also to reason contextually and execute appropriate actions in complex, dynamic settings [254]. The progression illustrated in Figures 2 and 3 from isolated vision, language, and action systems to an integrated VLA paradigm-captures a fundamental shift toward the development of truly adaptive and generalizable embodied agents. In light of the transformative potential of this paradigm, a comprehensive and critically informed review of the current literature is both timely and essential. First, such a review is necessary to clarify the foundational concepts and architectural principles that distinguish VLAs from their predecessors. Second, it provides a structured account of the rapid progress and key milestones in the field, enabling researchers and practitioners to appreciate the trajectory of algorithmic and technological advancements. Third, an in-depth review is essential for mapping the diverse range of real-world applications - from household robotics to industrial automation and assistive technologies-where VLAs are already demonstrating transformative potential. Furthermore, by critically examining the current challenges, such as data efficiency, safety, generalization, and ethical considerations, the review identifies barriers that must be addressed for widespread deployment. Finally, synthesizing these insights helps to inform the broader AI and robotics communities about emerging research directions and practical considerations, fostering collaboration and innovation.

In this review, we systematically analyze the foundational principles of VLA models. Additionally, we discuss their developmental progress and technical challenges. Our objective is to consolidate the current understanding and applications of VLAs while identifying limitations and proposing future directions for their evolution. The review begins with a detailed examination of key conceptual foundations (Figure 2), including the definition of VLA models, their historical evolution, mechanisms for multimodal integration, and unified tokenization and representation strategies spanning vision, language, and action. These conceptual descriptions set the stage for understanding how VLAs are structured and function across modalities.

Building upon this description, we present a unified view of recent progress and training efficiency strategies (Figure 3). This includes architectural innovations adopted and extended

这种语言缺乏感知或推理物理世界的能力[98]（图1中的“果园里的熟苹果”正是这一局限性的典型例证）。与此同时，机器人领域的基于动作的系统虽然高度依赖人工设计的策略或强化学习[158]，能够实现诸如物体操控等特定行为，但需要繁琐的工程设计，且无法在特定设计场景之外进行泛化[153]。

尽管视觉语言模型（VLMs）通过结合视觉与语言能力实现了令人印象深刻的多模态理解[190, 30, 298, 37, 10, 189]，但仍然存在明显的整合短板：无法基于多模态输入生成或执行连贯动作[156, 137]。如图1所示，大多数人工智能系统仅专注于单一或两种模态（如视觉-语言、视觉-动作或语言-动作），难以将三种模态完全整合到统一的端到端框架中。因此，机器人虽能通过视觉识别物体（“苹果”）、理解对应文本指令（“摘苹果”）或执行预设运动动作（抓取），却始终未能将这些能力整合为流畅灵活的行为模式。这种局限导致系统无法灵活适应新任务或环境，造成泛化能力薄弱且需要大量人工工程投入。该瓶颈凸显了具身人工智能的核心挑战：若缺乏能协同感知、理解与执行的系统，智能自主行为仍将是难以实现的目标。

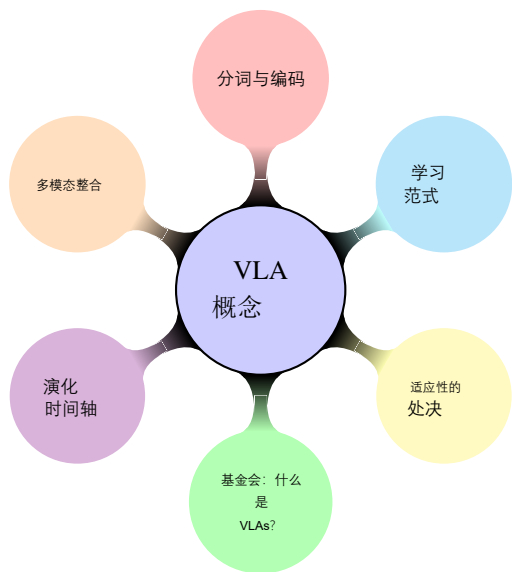


图2：核心VLA概念思维导图。各颜色编码分支分别突出基础维度：定义（基础）、历史演变、多模态整合、分词与编码、学习范式，以及具身化场景中的自适应执行。

弥合这些技术鸿沟的迫切需求催生了视觉语言架构（VLA）模型的诞生。这类模型概念化于2021-2022年间，由谷歌DeepMind的机器人Transformer 2（RT-2）[299]等研究团队率先提出，其创新架构将感知、推理与控制功能整合于单一框架之中。为解决图2所示的技术瓶颈，VLA模型通过整合视觉输入、语言理解与运动控制能

力，使具身智能体能够感知环境、理解复杂指令并动态执行相应动作。早期VLA方案通过将视觉-语言模型扩展至包含机器人运动指令的数值或符号表示形式（即动作标记），使模型能够从视觉数据、语言指令及运动轨迹的三重数据对中学习[156]。这一方法论创新显著提升了机器人在非结构化环境中对未知物体的泛化能力、对新式语言指令的解析能力以及多步骤推理能力[107]。

VLA模型标志着统一多模态智能发展中的重大突破，成功克服了长期以来将视觉、语言和动作视为独立领域的局限性[156]。通过整合视觉、语言和行为信息的互联网规模数据集，VLA使机器人不仅能识别和描述环境，还能在复杂动态场景中进行情境推理并执行恰当动作[254]。图2和3展示的从孤立视觉系统、语言系统和动作系统向集成VLA范式的演进过程，体现了向真正自适应、可泛化具身智能体发展的根本转变。鉴于该范式具有变革潜力，对现有文献进行全面且批判性综述既具有时效性又至关重要。首先，此类综述有助于厘清区分VLA与其前身的基础概念和架构原理。其次，本书系统梳理了该领域快速发展的历程及关键里程碑，帮助研究人员和从业者清晰把握算法与技术进步的轨迹。第三，深入的综述对于梳理虚拟现实增强技术在各类实际应用场景中的表现至关重要——从家用机器人到工业自动化及辅助技术，这些领域均已展现出其变革潜力。此外，通过批判性审视当前面临的挑战（如数据效率、安全性、泛化能力及伦理考量），本综述明确了实现广泛应用所需克服的障碍。最后，这些研究成果的整合有助于向更广泛的人工智能与机器人学界阐明新兴研究方向与实践考量，从而促进合作与创新。

在本综述中，我们系统地分析了视觉语言模型（VLA）的基本原理。此外，我们还探讨了其发展进程及技术挑战。我们的目标是整合当前对VLA的理解与应用成果，同时识别其局限性并提出未来发展的方向。本文首先详细阐述了VLA模型的核心概念基础（图2），包括模型定义、历史演变历程、多模态整合机制，以及涵盖视觉、语言和动作领域的统一分词与表征策略。这些概念性阐述为理解VLA在不同模态下的结构与功能机制奠定了基础。

基于上述描述，我们提出了近期进展与训练效率策略的统一观点（图3）。该观点涵盖所采用及扩展的架构创新方案。

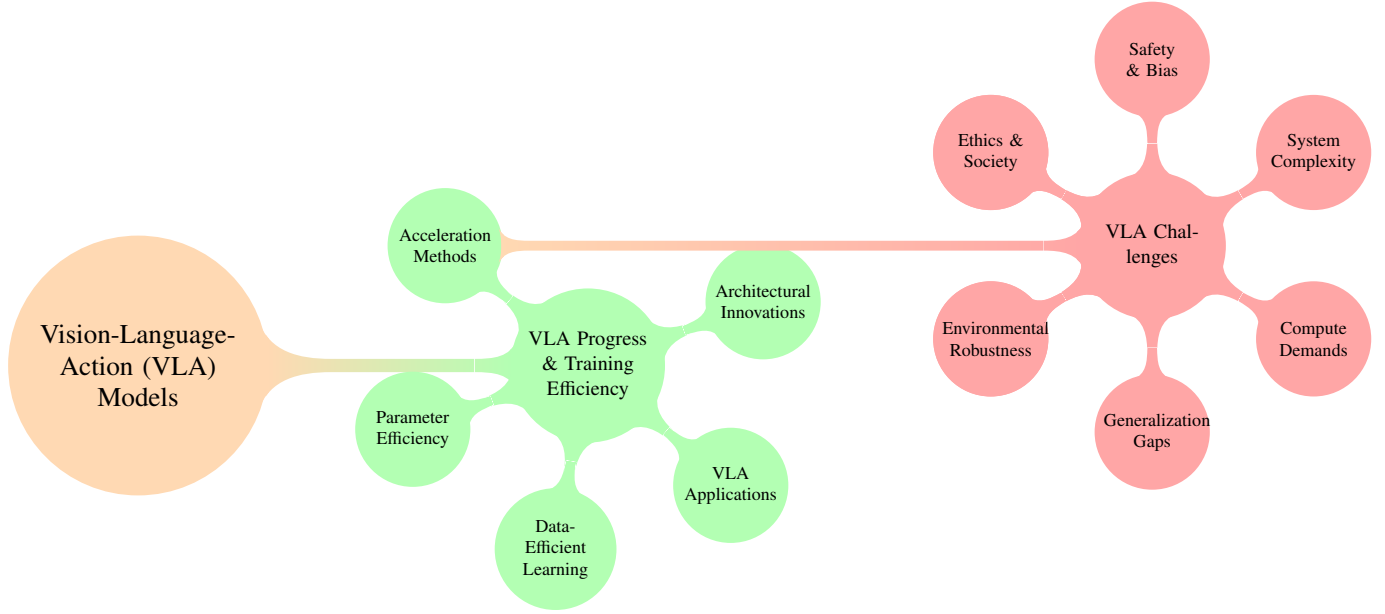


Figure 3: Mind map illustrating VLA model ecosystem: progress in training efficiency (architectural innovations, data/parameter efficiency, acceleration) alongside key challenges (inference constraints, multimodal action, safety, generalization, ethics) that must be overcome for scalable real-world deployment.

within VLA models, along with data-efficient learning frameworks, parameter-efficient modeling techniques, and model acceleration strategies originally developed in broader machine learning and robotics contexts. Together, these advances are critical for scaling VLA systems to real-world applications.

Following this, we present a comprehensive discussion of the current limitations encountered by VLA systems (Figure 3), many of which reflect broader challenges in embodied AI and robotics but arise in distinct and compounded forms due to the tight integration of vision, language, and action. The limitations to be discussed include inference bottlenecks, safety concerns, high computational demands, limited generalization, and ethical implications. We not only highlight these pressing challenges but also provide an analytical discussion on potential solutions to address them.

Together, these three figures offer a visual depiction that lays out the framework and supports the textual analysis presented in this review. By outlining the conceptual landscape, recent innovations, and open challenges, this work aims to guide future research and encourage the development of more robust, efficient, and ethically grounded VLA systems.

Figure 4 summarizes the overall structure and logical flow of this review and illustrates how the manuscript is organized to provide a comprehensive and systematic analysis of VLA research. As depicted in the figure, the paper progresses from foundational concepts to recent advances, applications, challenges, and future research directions, ensuring a coherent narrative across sections. To construct this architecture, an extensive and rigorous literature search was conducted using the primary keywords “Vision–Language–Action” and “Vision–Language Models,” together with the commonly used abbreviation “VLA.” These keywords were employed to retrieve candidate studies from major academic and technical reposi-

tories, including Hugging Face, arXiv, ScienceDirect, Nature, IEEE Xplore, Wiley, and Springer Nature. The resulting corpus was further refined through manual screening to ensure relevance, technical depth, and alignment with the scope of this review. Only articles that directly contributed to the understanding of VLA concepts, methodological progress, application domains, and open challenges were retained. This multi-stage filtering process enabled a balanced coverage of both foundational and state-of-the-art works while avoiding peripheral or loosely related studies. As a result, Figure 4 not only reflects the thematic organization of the paper but also embodies the underlying review methodology.

This paper follows a structured, hierarchical organization that systematically develops the foundations, evolution, and implications of VLA models, as summarized in Figure 4. The presentation begins with an introduction that motivates embodied intelligence and the emergence of VLAs, followed by a concepts section that establishes core principles including multimodal integration, tokenization, learning paradigms, and real-time control. Building on these foundations, the progress section examines architectural innovations, training and efficiency advancements, and parameter-efficient acceleration strategies. The applications section then grounds these developments in real-world domains, including humanoid robotics, autonomous vehicles, healthcare, agriculture, industrial systems, and interactive AR navigation. This is followed by a focused analysis of key challenges, including real-time inference, safety, generalization, system integration, and ethical considerations. The paper concludes with a future roadmap that distills cross-cutting research directions in continual learning, scalability, interpretability, and embodied intelligence.

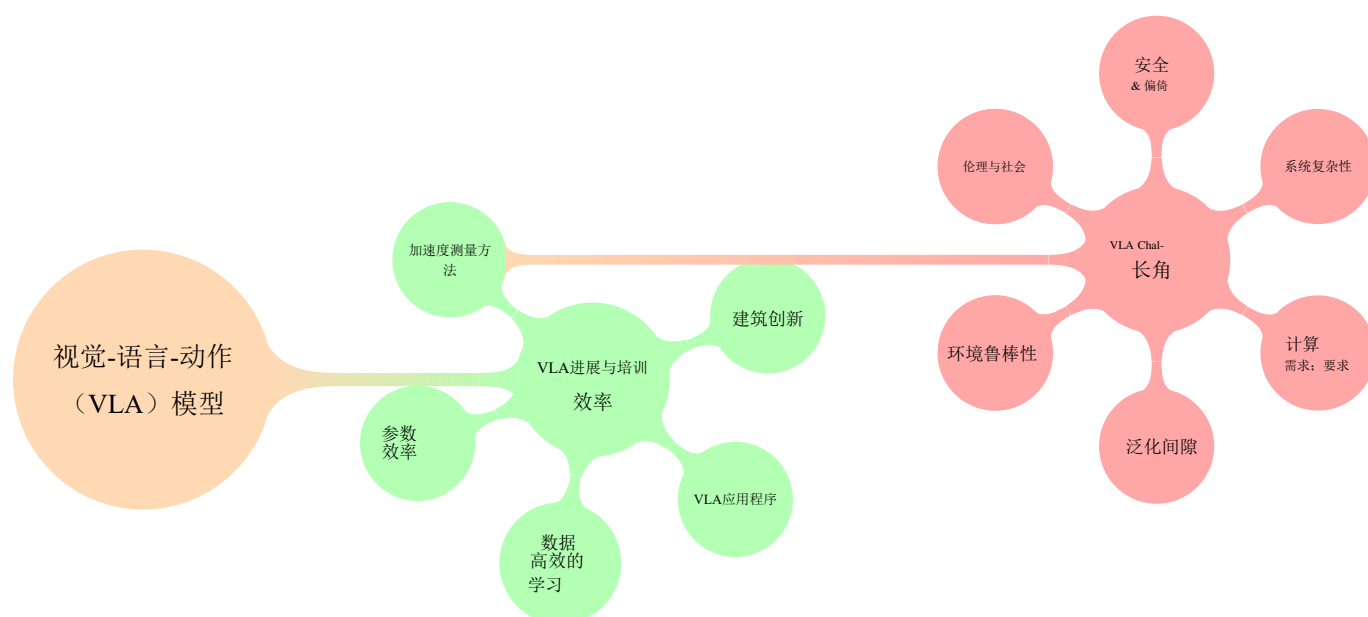


图3: VLA模型生态系统思维导图: 展示训练效率提升进展（架构创新、数据/参数效率优化、加速技术应用）与关键挑战（推理限制、多模态动作处理、安全性、泛化能力、伦理考量），这些要素是实现可扩展实际部署必须攻克的核心问题。

在VLA模型中，结合了最初在更广泛的机器学习和机器人技术领域开发的数据高效学习框架、参数高效建模技术以及模型加速策略。这些进展共同对于将VLA系统扩展应用于实际场景至关重要。

基于此，我们对甚大天线系统（VLA）当前面临的局限性进行了全面探讨（图3）。这些局限性不仅反映了具身人工智能与机器人技术领域的普遍挑战，更因视觉、语言与动作系统的深度整合而呈现出独特且相互交织的特征。需要重点讨论的局限性包括推理瓶颈、安全风险、高计算需求、泛化能力不足以及伦理影响等。本文不仅着重剖析这些紧迫挑战，还针对潜在解决方案展开了系统性分析讨论。

这三幅图表共同构成了可视化框架，为本综述中提出的文本分析提供了理论支撑。通过梳理概念框架、最新创新成果及现存挑战，本研究旨在为未来研究指明方向，推动更稳健、高效且符合伦理标准的VLA系统发展。

图4总结了本综述的整体结构与逻辑脉络，清晰展示了论文如何通过系统化组织架构实现对视觉语言分析（VLA）研究的全面解析。如图所示，本文从基础概念切入，逐步展开最新研究成果、实际应用、现存挑战及未来研究方向，确保各章节内容逻辑连贯。为构建这一研究框架，我们采用“视觉-语言-动作”和“视觉-语言模型”作为核心检索关键词，并结合常用缩写“VLA”，对Hugging Face、arXiv、ScienceDirect、Nature、IEEE Xplore、Wiley及Springer Nature等主流学术数据库进行了大规模文

献检索。通过人工筛选对检索结果进行精简处理，确保文献与综述主题高度契合，涵盖技术深度与研究广度。最终仅保留那些对VLA概念阐释、方法论突破、应用场景拓展及开放性挑战研究具有直接贡献的学术成果。这种多阶段筛选流程既实现了基础性研究与前沿成果的均衡覆盖，又有效规避了边缘性或关联性较弱的研究。因此，图4不仅清晰呈现了论文的主题结构，更完整体现了本文采用的综述方法论框架。

本文采用结构化、分层化的组织架构，系统阐述了VLA模型的基础理论、发展历程及其应用意义（详见图4）。文章开篇通过引言部分阐述了具身智能的理论背景及VLA模型的诞生动机，随后在概念章节中确立了多模态整合、标记化处理、学习范式及实时控制等核心原则。基于这些基础，进展部分深入探讨了架构创新、训练效率提升及参数高效加速策略。应用部分将这些研究成果应用于人形机器人、自动驾驶汽车、医疗健康、农业、工业系统及交互式增强现实导航等实际领域。接着对关键挑战进行了重点分析，包括实时推理、安全性、泛化能力、系统集成及伦理考量等问题。文末提出的未来路线图提炼了持续学习、可扩展性、可解释性及具身智能领域的跨学科研究方向。

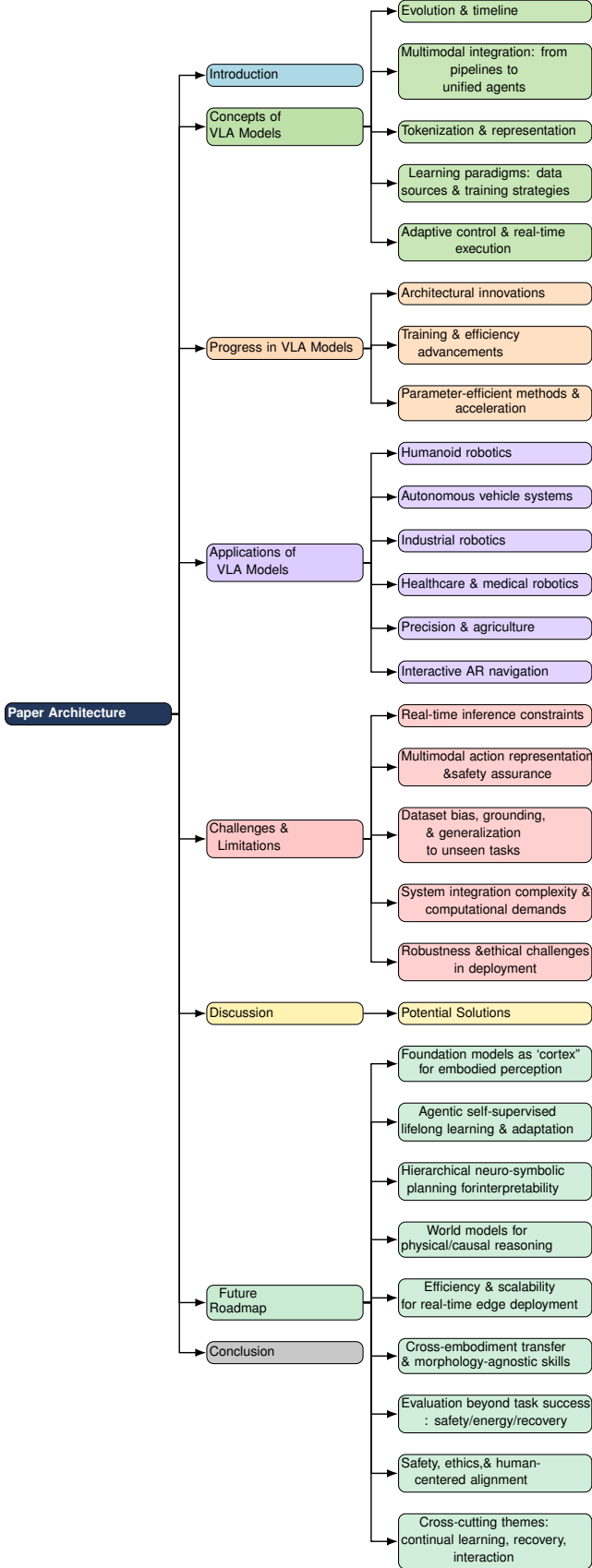


Figure 4: Flow and structure of this paper, from conclusion and future roadmap back through applications, progress, concepts, to introduction.

2. Concepts of Vision-Language-Action Models

VLA models represent a class of intelligent systems that jointly process visual inputs, interpret natural language instructions, and generate executable action representations that can be instantiated on physical robotic hardware operating in dynamic environments. Technically, VLAs combine vision encoders (e.g., CNNs, ViTs), language models (e.g., LLMs, transformers), and policy modules or planners to achieve task-conditioned control. These models typically build upon multimodal fusion techniques established in vision-language models, such as cross-attention, concatenated embeddings, or token unification, and extend them to align sensory observations, linguistic instructions, and action representations.

Unlike traditional visuomotor pipelines, VLAs support semantic grounding [154], enabling context-aware reasoning [228], affordance detection [79], and temporal planning [141]. A typical VLA model observes the environment through camera or sensor data, interprets goals expressed in language (e.g., “pick up the red apple”) (Figure 5), and outputs low-level or high-level action sequences that could be implemented by automated systems to perform the action. Recent advancements integrate imitation learning, reinforcement learning, or retrieval-augmented modules to improve sample efficiency and generalization. This review examines how VLA models have evolved from foundational fusion architectures to general-purpose agents capable of real-world deployment across robotics, navigation, and human-AI collaboration.

VLA models are multi-modal artificial intelligence systems that unify visual perception, language comprehension, and physical action generation into a single framework. These models enable robots or AI agents to interpret sensory inputs (e.g., images, text), understand contextual meaning, and autonomously execute tasks in real-world environments - all through end-to-end learning and action rather than isolated sub-systems. As shown conceptually in Figure 5, VLA models bridge the historical disconnect between visual recognition, language comprehension, and/or motor execution that limited the capabilities of earlier robotic and AI systems.

2.1. Evolution and Timeline

The rapid development of VLA models from 2022-2025 demonstrates three distinct evolutionary phases:

1. **Foundational Integration (2022–2023).** Early VLAs established basic visuomotor coordination through multimodal fusion architectures. [202] first combined CLIP embeddings with motion primitives, while [181] demonstrated generalist capabilities across 604 tasks. [19] achieved 97% success rates in manipulation through scaled imitation learning, and [112] introduced temporal reasoning via transformer-based planners. By 2023, [299] enabled visual chain-of-thought reasoning, and [40] advanced stochastic action prediction through diffusion processes. These foundations addressed low-level control but lacked compositional reasoning, the ability to decompose complex tasks into reusable, semantically grounded

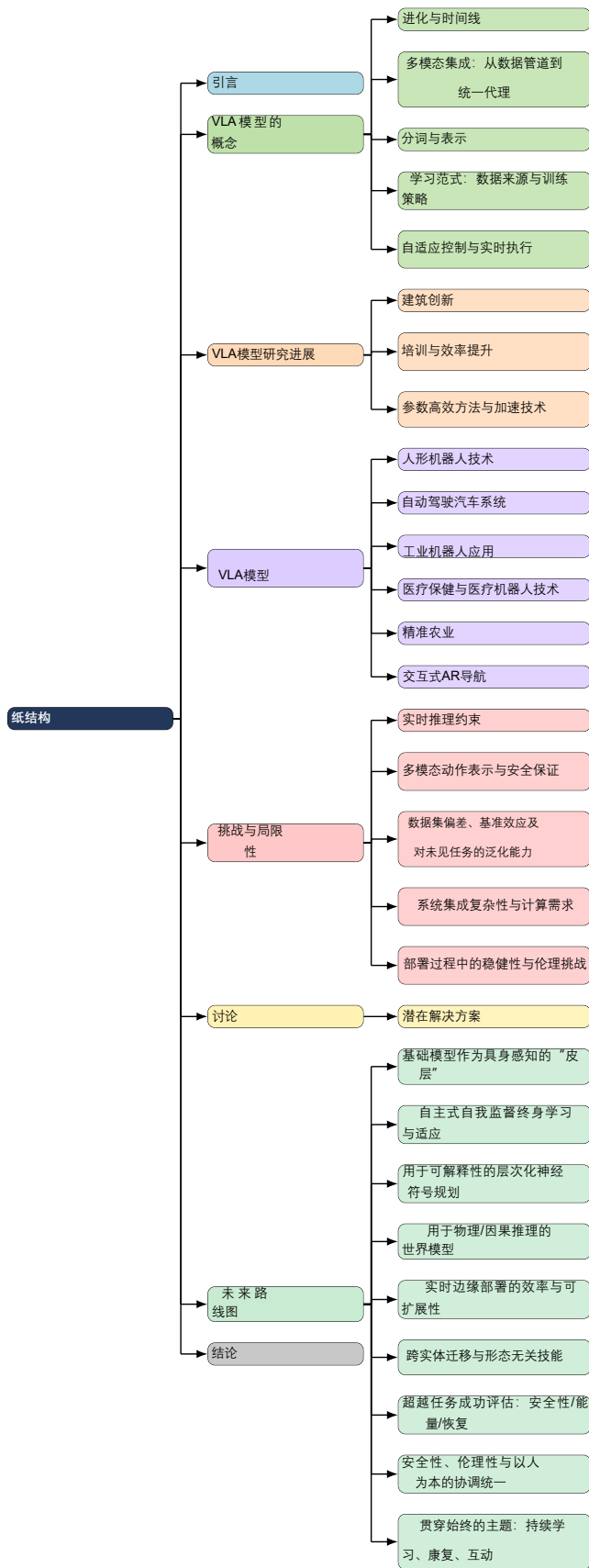


图4: 本文的结构与论述流程, 从结论与未来路线图出发, 依次涵盖应用案例、研究进展、核心概念直至引言部分。

2. 视觉-语言-动作模型的概念

视觉语言模型（VLA）是一类智能系统，能够协同处理视觉输入、解析自然语言指令，并生成可在动态环境中运行的物理机器人硬件上实现的可执行动作表征。从技术层面来看，VLA通过整合视觉编码器（如卷积神经网络、视觉Transformer）、语言模型（如大语言模型、Transformer模型）以及策略模块或规划器，实现任务驱动型控制。这类模型通常基于视觉语言模型中成熟的多模态融合技术（如交叉注意力机制、拼接嵌入向量或标记统一技术），并将其扩展应用于感官观测数据、语言指令与动作表征的协同处理。

与传统视觉运动处理流程不同，视觉语言代理（VLAs）支持语义接地[154]，能够实现上下文感知推理[228]、功能项检测[79]以及时间规划[141]。典型的VLA模型通过摄像头或传感器数据观察环境，解析语言表达的目标（例如“捡起红苹果”）（图5），并输出可由自动化系统执行的低级或高级动作序列。最新研究通过整合模仿学习、强化学习或检索增强模块来提升样本效率与泛化能力。本文综述了VLA模型如何从基础融合架构演进为通用智能体，实现机器人技术、导航系统及人机协作等领域的实际应用。

VLA模型是多模态人工智能系统，将视觉感知、语言理解与物理动作生成整合为统一框架。这类模型使机器人或AI智能体能够解析感官输入（如图像、文本）、理解上下文语义，并在真实环境中自主执行任务——所有功能均通过端到端学习与动作执行实现，而非依赖孤立子系统。如图5所示，VLA模型有效弥合了视觉识别、语言理解与运动执行之间的历史鸿沟，这种技术突破显著提升了早期机器人系统与人工智能系统的整体性能。

2.1. 进化与时间线

2022年至2025年间VLA模型的快速发展呈现出三个明显的进化阶段：

- 1. 基础整合阶段（2022–2023年）。**早期视觉语言助理（VLA）通过多模态融合架构建立了基础视觉运动协调能力。[202] 首次将CLIP嵌入与运动基元相结合，同时[181]在604项任务中展现出通用能力。[19]通过规模化模仿学习实现了97%的操作成功率，并[112]通过基于Transformer的规划器引入了时序推理机制。截至2023年，[299]该系统实现了可视化思维链推理功能，并通过扩散过程实现了[40]先进的随机动作预测。这些基础架构虽解决了低层级控制问题，但尚缺乏组合推理能力——即无法将复杂任务分解为可复用且具有语义基础的模块。

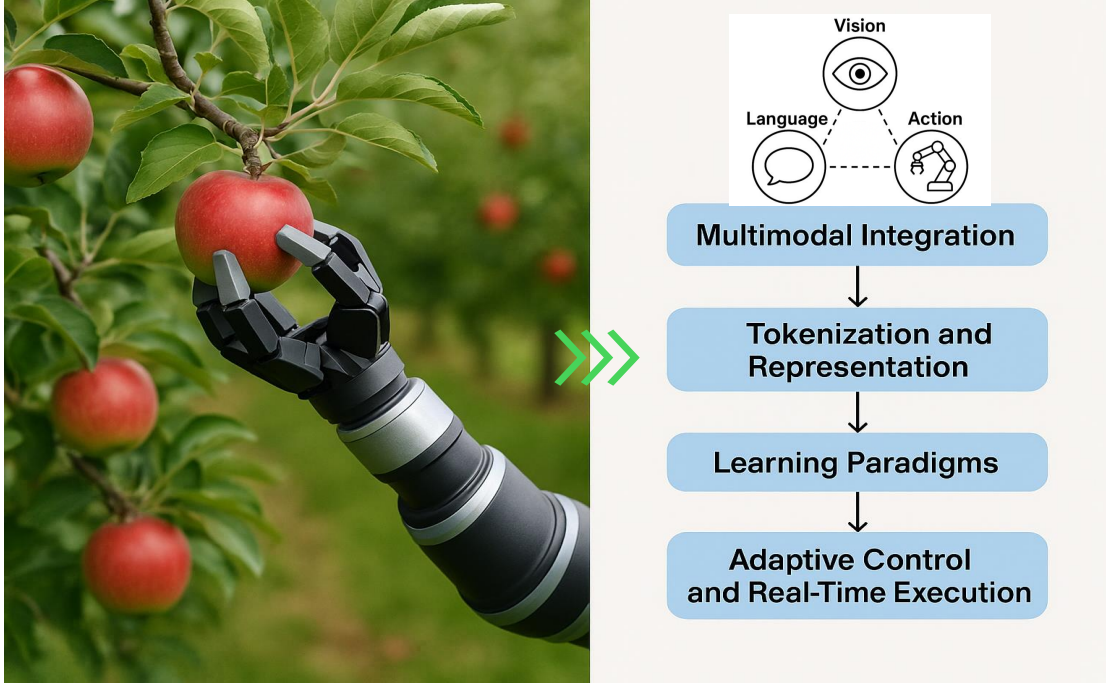


Figure 5: **Foundational Concept of VLA Models (in an Apple-Picking Scenario)** This illustration depicts a robotic arm autonomously picking a ripe apple in an orchard, guided by a VLA model. On the right, a flowchart outlines the four key stages of VLA models: Multimodal Integration, Tokenization and Representation, Learning Paradigms, and Adaptive Control and Real-Time Execution.

sub-actions and recombine them across novel contexts—prompting subsequent innovations in affordance grounding [287, 100].

2. **Specialization and Embodied Reasoning (2024).** Second-generation VLAs incorporated domain-specific inductive biases. [269] enhanced few-shot adaptation through retrieval-augmented training, while [280] optimized navigation via 3D scene-graph integration. [48] introduced reversible architectures for memory efficiency, and [239] addressed partial observability with physics-informed attention. Simultaneously, [5] improved compositional understanding through object-centric disentanglement, and [293] extended applications to autonomous driving via multi-modal sensor fusion. However, these advances required new benchmarking methodologies [254].
3. **Generalization and Safety-Critical Deployment (2025).** Latest systems prioritize robustness and human alignment. [274] integrated formal verification for risk-aware decisions, while [53] demonstrated whole-body control through hierarchical VLAs. [20] optimized compute efficiency for embedded deployment, and [131] combined neural-symbolic reasoning for causal inference. Emerging paradigms like [129]’s affordance chaining and [14]’s sim-to-real transfer learning address cross-embodiment challenges, while [139] bridges VLAs with human-in-the-loop interfaces through natural language grounding.

Figure 6 presents a comprehensive timeline highlighting the evolution of 45 VLA models developed between 2022 and 2025. The earliest VLA systems, including CLIPort [202],

Gato [181], RT-1 [19], and VIMA [112], laid the foundation by combining pretrained vision-language representations with task-conditioned policies for manipulation and control. These early VLA systems were followed by ACT [287], RT-2 [299], and VoxPoser [100], which integrated visual chain-of-thought reasoning and affordance grounding. Models like Diffusion Policy [40] and Octo [218] introduced stochastic modeling and scalable data pipelines. In 2024, systems such as Deer-VLA [269], ReVLA [48], and Uni-NaVid [280] added domain specialization and memory-efficient designs, while Occllama [239] and ShowUI [139] tackled partial observability and user interaction. The trajectory continued with robotics-focused VLAs like Quar-VLA [54] and Robo-Mamba [143]. Recent innovations emphasize generalization and deployment: SafeVLA [274], Humanoid-VLA [53], and MoManipVLA [246] incorporate verification, full-body control, and memory systems. Models such as Gr00t N1 [14] and SpatialVLA [175] further bridge sim-to-real transfer and spatial grounding. This timeline illustrates how VLAs have advanced from modular learning to general-purpose, safe, and embodied intelligence.

2.2. Multimodal Integration: From Isolated Pipelines to Unified Agents

A central advancement in the emergence of VLA models lies in their ability to perform multi-modal integration, the joint processing of vision, language, and action within a unified architecture. Traditional robotic systems treated perception, natural language understanding, and control as discrete modules, often linked through manually defined interfaces or data trans-

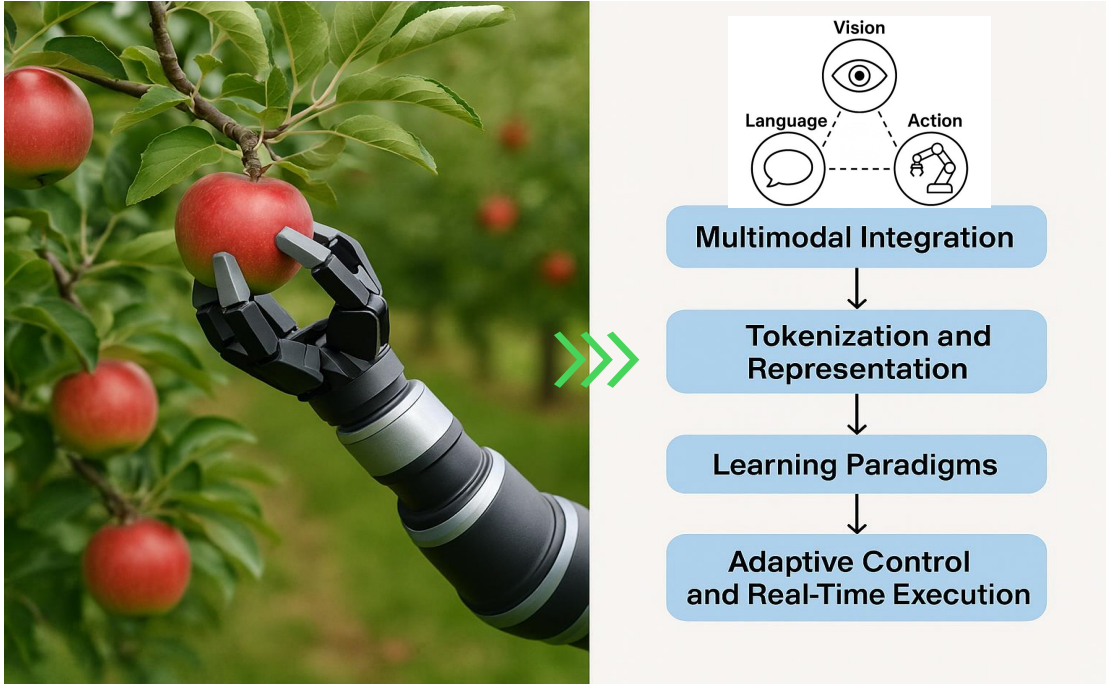


图5: VLA模型的基本概念（在苹果采摘场景中）该示意图展示了机械臂在果园中自主采摘成熟苹果的过程，整个过程由VLA模型引导。右侧流程图概述了VLA模型的四个关键阶段：多模态整合、分词与表征、学习范式以及自适应控制与实时执行。

- 子动作并将其重组到新的语境中，从而推动 affordance grounding 领域后续的创新[287, 100]。
2. **专业化与具身推理（2024年）**。第二代虚拟实验室自动化系统（VLAs）整合了领域特定的归纳偏见。[269] 通过检索增强训练实现了增强的少样本适应能力；同时，[280]通过三维场景图整合优化了导航性能。[48]提出了可逆架构以提升内存效率；[239]则采用基于物理机制的注意力机制解决了部分可观测性问题。与此同时，[5]通过以物体为中心的解耦技术提升了成分理解能力，并通过多模态传感器融合将[293]扩展应用于自动驾驶领域。然而，这些进展需要新的基准测试方法[254]。
 3. **泛化能力与安全关键部署（2025年）**。最新系统优先考虑系统的鲁棒性及与人类需求的一致性。[274] 针对风险感知决策的集成形式验证，同时[53]通过分层虚拟逻辑架构（VLAs）实现了全身控制。[20] 针对嵌入式部署优化计算效率，并采用[131]神经符号推理技术进行因果推断。新兴范式如[129]的 affordance chaining 和[14]的 simto-real transfer learning 可有效解决跨表征迁移挑战，同时[139]通过自然语言接地技术将虚拟现实增强系统（VLAs）与人机交互界面相连接。

图6展示了一个全面的时间线，突显了2022年至2025年间开发的45个VLA模型的演进历程。最早的VLA系统包括 CLIPort[202]、Gato[181]、RT-1[19]和 VIMA[112]，它们

通过将预训练视觉语言表征与任务条件化操作控制策略相结合奠定了基础。随后出现的ACT[287]、RT-2[299]和 VoxPoser[100]集成了视觉思维链推理与功能位点定位。Difusion Policy[40]和Octo[218]等模型引入了随机建模与可扩展数据管道。2024年，Deer-VLA[269]、ReVLA[48]和 Uni-NaVid[280]等系统增加了领域专业化与内存高效设计，而Occllama[239]和ShowUI[139]则解决了部分可观测性与用户交互问题。该发展轨迹延续至以机器人技术为核心的VLA系统，如Quar-VLA[54]和RoboMamba[143]。近期的创新重点在于泛化能力和部署能力：SafeVLA[274]、Humanoid-VLA[53]和MoManipVLA[246]均集成了验证机制、全身控制系统及记忆系统。而Gr00t N1[14]和Spatia-IVLA[175]等模型则进一步实现了仿真到现实的迁移能力与空间定位能力。这一发展脉络清晰展现了VLAs如何从模块化学习逐步演进为具备通用性、安全性及具身智能的系统。

2.2. 多模态整合：从孤立管道到统一代理

VLA模型在发展过程中取得的核心突破在于其具备多模态整合能力，即能够在统一架构中实现视觉、语言与动作的协同处理。传统机器人系统将感知、自然语言理解及控制功能视为独立模块，通常通过人工定义的接口或数据传输方式实现模块间连接。

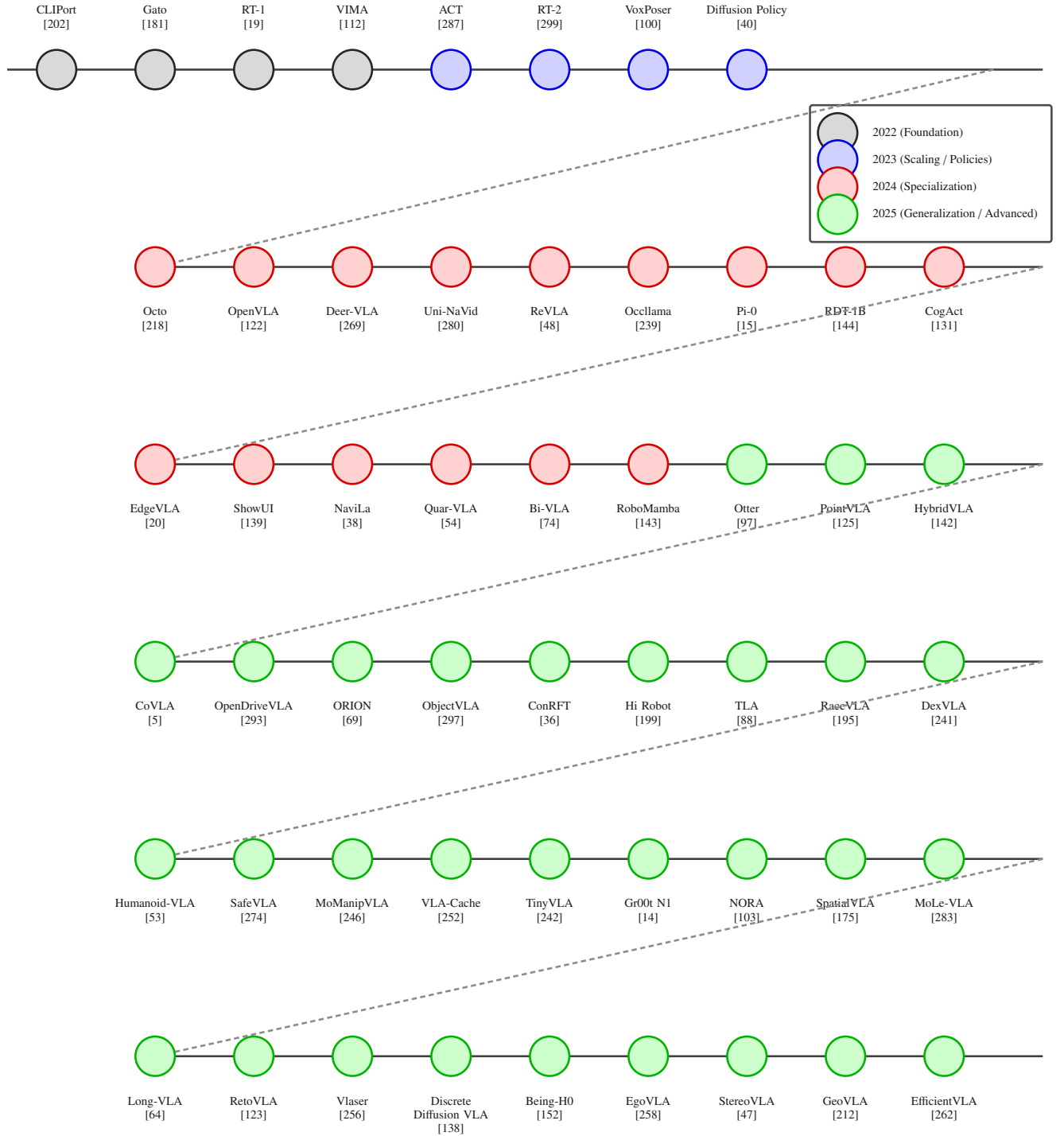


Figure 6: Comprehensive timeline of Vision–Language–Action (VLA) models (2022–2025), organized by year with high-contrast color coding and thematic grouping.

formations [140, 21, 219]. For instance, classic pipeline-based frameworks required a perception model to output symbolic labels, which were then mapped by a planner to specific actions frequently with domain-specific hand engineering [178, 117]. These approaches lacked adaptability, failed in ambiguous or unseen environments, and could not generalize instructions beyond pre-defined templates.

In contrast, modern VLAs fuse modalities end-to-end using

large-scale pretrained encoders and transformer-based architectures [244]. This shift enables the model to interpret visual observations and linguistic instructions within the same computational space, allowing flexible, context-aware reasoning [128]. For example, in the task “Pick the ripe apples” (Figure 5), the vision encoder—typically a Vision Transformer (ViT) or Con-vNeXt—parses the scene to localize and categorize relevant objects (e.g., fruit, foliage, background) and infer ripeness-related

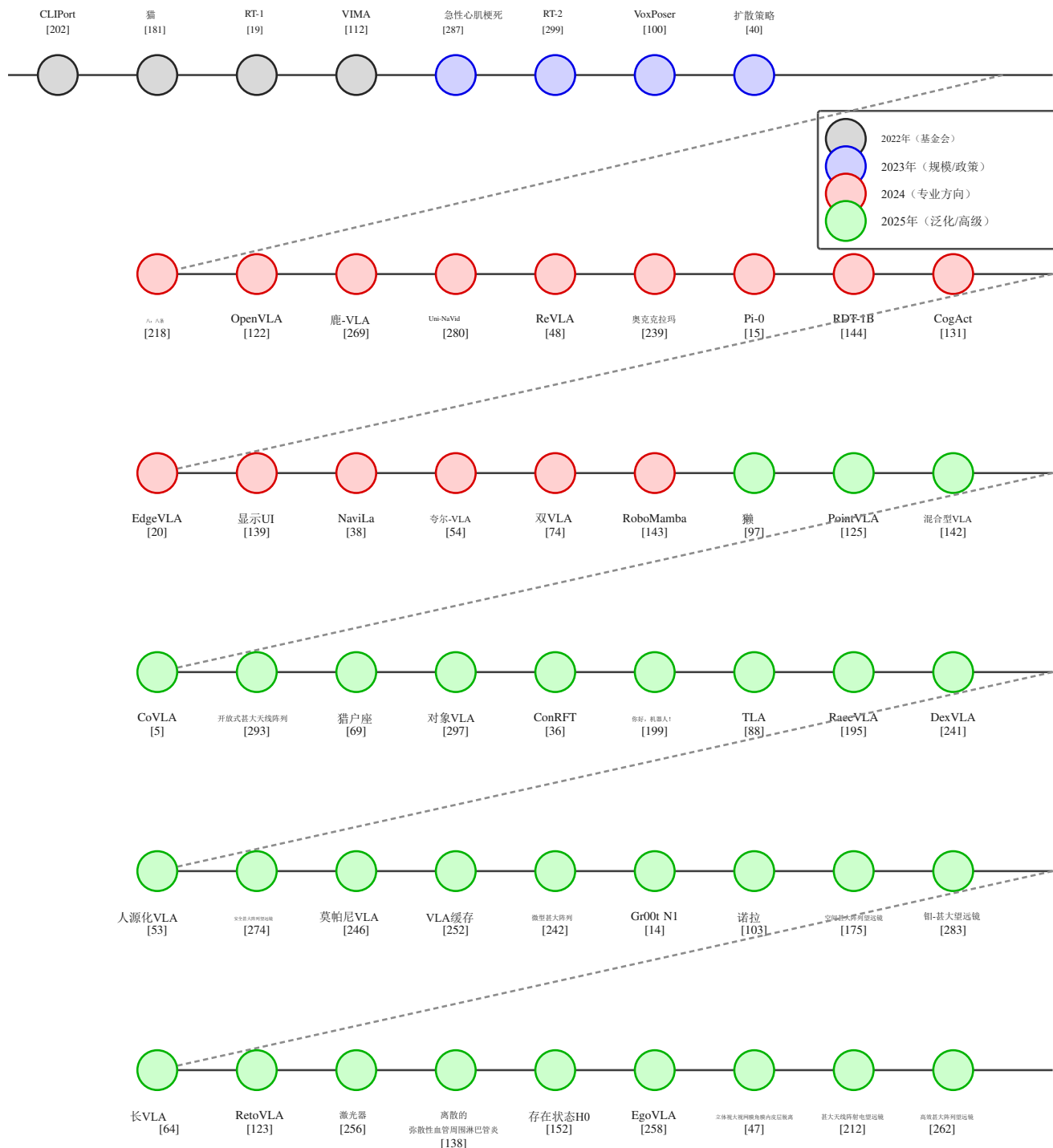


图6：视觉-语言-动作（VLA）模型综合时间线（2022-2025年），按年份组织并采用高对比度颜色编码及主题分组。

研究文献[140, 21, 219]表明，传统基于管道架构的框架需要感知模型输出符号化标签，随后由规划器将这些标签映射为具体动作——这一过程通常需要领域特定的人工工程干预[178, 117]。这类方法存在三大局限：适应性不足、在模糊或未知环境中表现欠佳，且无法将指令泛化至预定义模板之外。

相比之下，现代视觉语言模型通过大规模预训练编码器和基于Transformer架构的端到端融合技术[244]，实现了模态

间的无缝衔接。这种架构转变使模型能够在同一计算空间内解析视觉观测数据与语言指令，从而实现灵活且具有上下文感知能力的推理[128]。以“挑选熟苹果”任务（图5）为例，视觉编码器（通常采用视觉Transformer ViT或ConvNeXt）通过解析场景来定位并分类相关物体（如水果、树叶、背景元素），同时推断果实成熟度相关特征。

visual cues based on learned texture, shape, and contextual features rather than fixed color assumptions [243]. Meanwhile, the language model, often a variant of T5, GPT, or BERT, encodes the instruction into a high-dimensional embedding. These representations are then fused via cross-attention or joint tokenization schemes, producing a unified latent space that informs the action policy [86].

This multimodal synergy was first effectively demonstrated in CLIPort [202], which takes an RGB image of a tabletop scene and a natural language instruction (e.g., “place the blue block on the red square”) as inputs, encodes them using CLIP for semantic grounding, and outputs pixel-level pick-and-place action distributions via a convolutional transport decoder. By directly conditioning visuomotor policies on language embeddings, CLIPort eliminates explicit language parsing and enables end-to-end language-conditioned manipulation. Similarly, VIMA [112] advanced this approach by employing a transformer encoder to jointly process object-centric visual tokens and instruction tokens, enabling few-shot generalization across spatial reasoning tasks.

Recent developments push this fusion further by incorporating temporal and spatial grounding. VoxPoser [100] employs voxel-level reasoning to resolve ambiguities in 3D object selection by composing pretrained vision-language models and classical motion planners, notably achieving zero-shot manipulation without task-specific training data. In contrast, RT-2 [299] fuses visual-language tokens and action representations within a unified transformer, co-trained on large-scale internet vision-language corpora and over 100,000 real-robot demonstrations from the RT-1 dataset, enabling zero-shot generalization to unseen instructions. Another noteworthy contribution is Octo [218], which introduces a memory-augmented transformer trained on more than four million robot trajectories collected across diverse robots and environments via the Open X-Embodiment dataset, supporting long-horizon decision-making and demonstrating the scalability of joint perception–language–action learning.

Crucially, VLAs offer robust solutions to challenges faced in real-world grounding. For example, Occllama [239] handles occluded object references through attention-based mechanisms, while ShowUI [139] demonstrates natural language interfaces that allow non-expert users to command agents through voice or typed input. These capabilities are only possible because the integration is not limited to surface-level fusion; rather, it captures semantic, spatial, and temporal alignment across modalities.

2.3. Tokenization and Representation: How VLAs Encode the World

A core innovation that sets VLA models apart from conventional vision-language architectures lies in their token-based representation framework, which enables holistic reasoning over perceptual [161, 286], linguistic, and physical action spaces [136]. Inspired by autoregressive generative models like transformers, modern VLAs encode the world using discrete tokens that unify all modalities vision, language, state, and action into a shared embedding space [142]. This allows the model to

not only understand “what needs to be done” (semantic reasoning), but also “how to do it” (control policy execution) in a fully learnable and compositional way [248, 150, 221].

- **Prefix Tokens: Encoding Context and Instruction:** Prefix tokens serve as the contextual backbone of VLA models [252, 107]. These tokens encode the environmental scene (via images or video) and the accompanying natural language instruction into compact embeddings that prime the model’s internal representations [17].

For instance, as depicted in Figure 7 in a task such as “stack the green blocks on the red tray,” the image of a cluttered tabletop is processed through a vision encoder like ViT or ConvNeXt, while the instruction is embedded by an LLM (e.g., T5 or LLaMA). These are then transformed into a sequence of prefix tokens that establish the model’s initial understanding of the goal and environmental layout. This shared representation enables cross-modal grounding, allowing the system to resolve spatial references (e.g., “on the left,” “next to the blue cup”) and object semantics (“green blocks”) across both modalities.

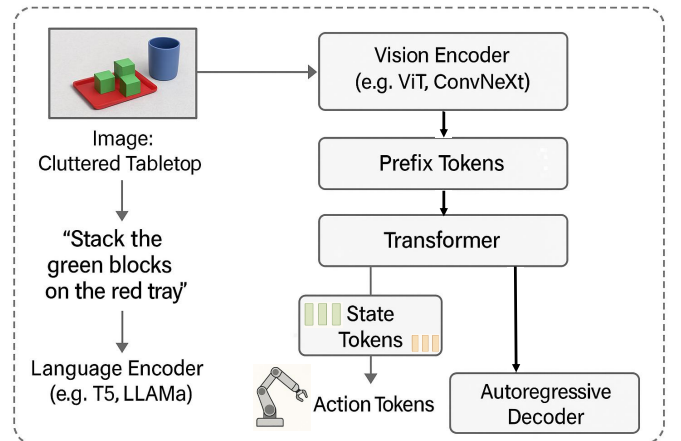


Figure 7: A diagram illustrating the end-to-end tokenization and representation process in VLA models. Visual input (e.g., cluttered tabletop) is encoded by a vision encoder (e.g., ViT), while natural language instructions (e.g., “stack the green blocks”) are processed by a language encoder (e.g., T5). The system fuses prefix, state, and action tokens through a transformer and autoregressively predicts motor actions.

- **State Tokens: Embedding the Robot’s Configuration:** In addition to perceiving external stimuli, VLAs must be aware of their internal physical state [242, 143]. This is achieved through the use of state tokens, which encode real-time information about the agent’s configuration joint positions, force-torque readings, gripper status, end-effector pose, and even the locations of nearby objects [126]. These tokens are crucial for ensuring situational awareness and safety, especially during manipulation or locomotion [211, 105].

Figure 8 illustrates how VLA models utilize state tokens to enable dynamic, context-aware decision-making in both manipulation and navigation settings. In Figure 8a, a robot

视觉线索基于学习到的纹理、形状和上下文特征，而非固定的颜色假设[243]。与此同时，语言模型（通常采用T5、GPT或BERT等变体）会将指令编码为高维嵌入向量。这些表征随后通过交叉注意力机制或联合分词方案进行融合，形成统一的潜在空间，从而为动作策略提供决策依据[86]。

这种多模态协同效应最初在CLIPort[202]中得到了有效验证：该模型以桌面场景的RGB图像和自然语言指令（例如“将蓝色方块放置在红色方块上”）作为输入，利用CLIP进行语义编码，并通过卷积传输解码器输出像素级别的拾取-放置动作分布。通过直接基于语言嵌入来约束视觉运动策略，CLIPort消除了显式的语言解析步骤，实现了端到端的语言条件化操作。类似地，VIMA[112]进一步改进了该方法，采用Transformer编码器同时处理以物体为中心的视觉标记和指令标记，从而实现了跨空间推理任务的少样本泛化能力。

最新研究通过整合时空信息进一步推动了融合技术的发展。VoxPoser[100]采用体素级推理方法，通过整合预训练视觉语言模型与经典运动规划器来解决三维物体选择中的歧义问题，尤其实现了无需任务特定训练数据的零样本操作能力。相比之下，RT-2[299]在统一Transformer架构中融合视觉语言标记与动作表征，该模型基于大规模互联网视觉语言语料库及RT-1数据集中的10万余个真实机器人演示数据进行联合训练，实现了对未见指令的零样本泛化能力。另一重要突破是Octo[218]提出的记忆增强Transformer模型，该模型基于Open XEmbodiment数据集收集的超过400万条机器人轨迹数据（涵盖多种机器人及环境场景），支持长时域决策能力，充分展现了感知-语言-动作联合学习的可扩展性。

关键在于，视觉语言模型（VLAs）为现实世界中的定位挑战提供了稳健的解决方案。例如，Occllama[239]通过基于注意力机制处理遮挡物体参照，而ShowUI[139]则实现了自然语言交互界面，使非专业用户能够通过语音或文本输入操控智能体。这些功能之所以得以实现，是因为其整合机制不仅限于表面级融合，而是实现了跨模态的语义、空间与时间对齐。

2.3. 分词与表征：虚拟语言资产如何编码现实世界

VLA模型与传统视觉语言架构的核心创新在于其基于标记的表征框架，该框架能够对感知[161, 286]、语言及物理动作空间进行整体推理[136]。受Transformer等自回归生成模型的启发，现代视觉语言模型通过离散标记对世界进行编码，将视觉、语言、状态和动作等多模态信息统一到共享的嵌入空间[142]。这种设计使模型不仅能理解“需要

做什么”（语义推理），还能以完全可学习且组合式的方式实现“如何执行”（控制策略执行）[248, 150, 221]。

- **前缀标记：编码上下文与指令：**前缀标记是VLA模型的上下文基础框架[252, 107]。这些标记将环境场景（通过图像或视频）及相应的自然语言指令编码为紧凑的嵌入表示，从而为模型的内部表征提供初始输入[17]。

例如，在图7所示的“将绿色积木堆放在红色托盘上”任务中，杂乱桌面的图像通过ViT或ConvNeXt等视觉编码器进行处理，而指令则由大语言模型（如T5或LLaMA）进行嵌入。这些信息随后被转化为一系列前缀标记序列，帮助模型初步理解任务目标和环境布局。这种共享表征实现了跨模态定位机制，使系统能够跨模态解析空间参照（如“左侧”、“蓝色杯子旁边”）和物体语义（“绿色积木”）等信息。

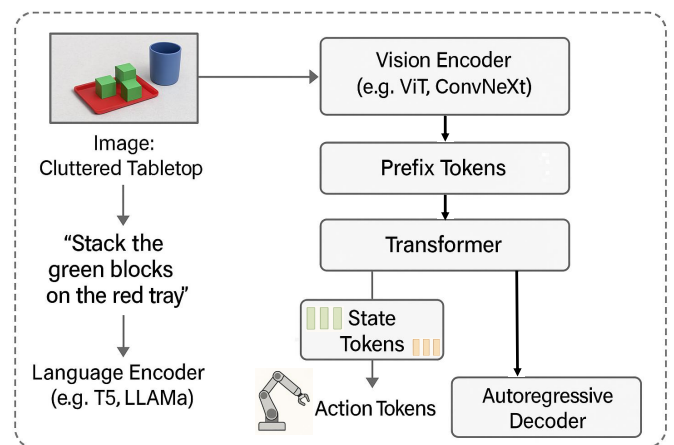


图7：示意图展示了VLA模型中的端到端分词与表示过程。视觉输入（例如杂乱的桌面）由视觉编码器（如ViT）进行编码，而自然语言指令（例如“堆叠绿色积木”）则由语言编码器（如T5）处理。系统通过Transformer融合前缀、状态和动作令牌，并采用自回归机制预测运动动作。

- **状态标记：嵌入机器人配置信息：**除了感知外部刺激外，虚拟现实代理（VLAs）还需了解自身的内部物理状态[242, 143]。这一功能通过状态标记实现，这些标记实时编码了代理的配置信息，包括关节位置、力矩读数、夹爪状态、末端执行器姿态，甚至邻近物体的位置信息。[126]这些标记对于确保情境感知与安全至关重要，尤其是在物体被操控或移动过程中[211, 105]。

图8展示了VLA模型如何利用状态标记，在操作和导航场景中实现动态、上下文感知的决策能力。在图8a中，机器人

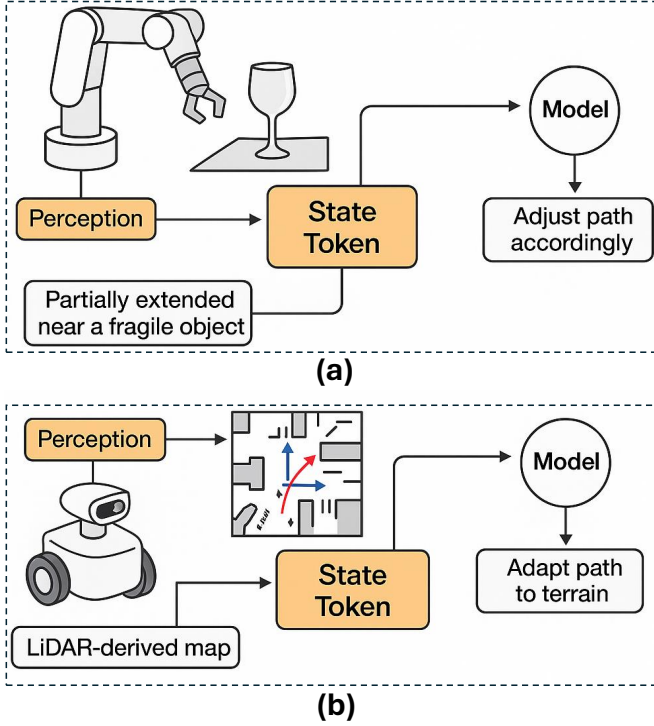


Figure 8: Illustrating how VLA models utilize prefix, state, and action tokens in real-world scenarios. In robotic manipulation, state tokens detect arm extension near fragile objects, enabling path adjustment. In navigation, they represent LiDAR and odometry data. The apple-picking task shows how prefix tokens guide goal understanding, while action tokens generate motion sequences for targeted grasping and execution.

arm is shown partially extended near a fragile object. In such scenarios, state tokens play a critical role by encoding real-time proprioceptive information, such as joint angles, gripper pose, and end-effector proximity. These tokens are continuously fused with visual and language-based prefix tokens, allowing the transformer to reason about physical constraints. The model can thus infer that a collision is imminent and adjust the motor commands accordingly e.g., rerouting the arm trajectory or modulating force output. In mobile robotic platforms, as depicted in Figure 8b, state tokens encapsulate spatial features such as odometry, LiDAR scans, and inertial sensor data. These are essential for terrain-aware locomotion and obstacle avoidance. The transformer model integrates this state representation with environmental and instructional context to generate navigation actions that dynamically adapt to changing surroundings. Whether grasping objects in cluttered environments or autonomously navigating uneven terrain, state tokens provide a structured mechanism for situational awareness, enabling the autoregressive decoder to produce precise, context-informed action sequences that reflect both internal robot configuration and external sensory data.

- **Action Tokens: Autoregressive Control Generation:** The final layer of the VLA token pipeline involves action tokens [121, 122], which are autoregressively generated

by the model to represent the next step in motor control [242]. Each token corresponds to a low-level control signal, such as joint angle updates, torque values, wheel velocities, or high-level movement primitives [81]. During inference, the model decodes these tokens one step at a time, conditioned on prefix and state tokens, effectively turning VLA models into language-driven policy generators [67, 209]. This formulation allows seamless integration with real-world actuation systems, supports variable-length action sequences [11, 99], and enables model fine-tuning via reinforcement or imitation learning frameworks [285]. Notably, models like RT-2 [299] and PaLM-E [58] exemplify this design, where perception, instruction, and embodiment are merged into a unified token stream.

For instance, in the apple-picking task as depicted in Figure 9, the model may receive prefix tokens that include the image of the orchard and the text instruction. The state tokens describe the robot’s current arm posture and whether the gripper is open or closed. Action tokens are then predicted step by step to guide the robotic arm toward the apple, adjust the gripper orientation, and execute a grasp with appropriate force. The beauty of this approach is that it allows transformers, which are traditionally used for text generation, to now generate sequences of physical actions in a manner similar to generating a sentence only here, the sentence is the motion.

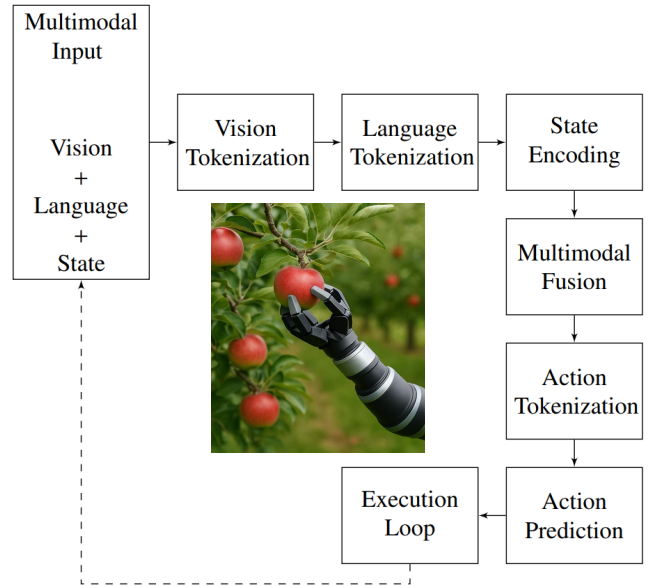


Figure 9: **Illustrating the process of how VLAs Encode the World.** VLAs encode the world by converting vision, language, and sensor inputs into tokens, fusing them through cross-attention, predicting action sequences via transformers, and executing tasks with real-time feedback - enabling robots to interpret scenes, follow instructions, and adapt actions dynamically.

To operationalize the VLA paradigm in robotics, we present in Figure 9 a structured pipeline that demonstrates how multimodal information specifically vision, language, and proprioceptive state is encoded, fused, and converted into executable

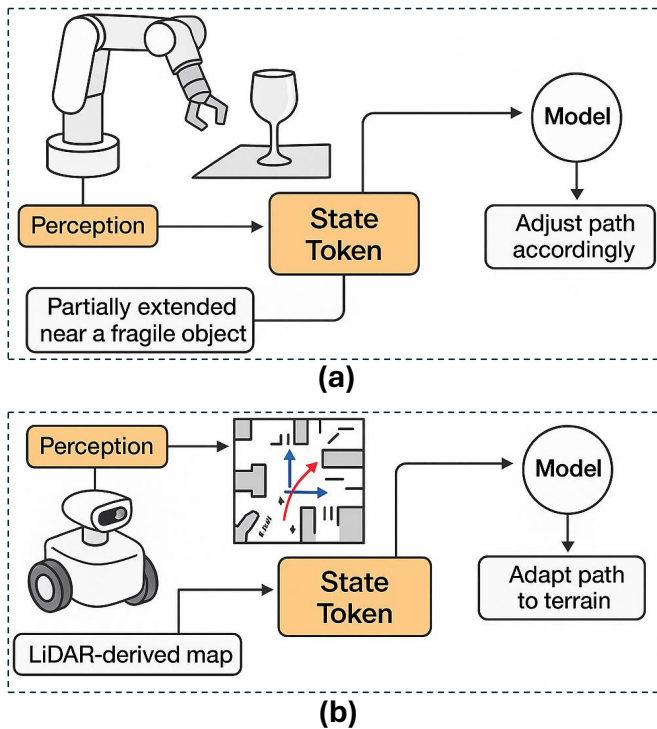


图8：展示VLA模型在实际应用场景中如何运用前缀标记、状态标记与动作标记。在机器人操作场景中，状态标记可检测机械臂接近易碎物体时的伸展状态，从而实现路径调整；在导航系统中，这些标记对应激光雷达（LiDAR）与里程计数据。苹果采摘任务中，前缀标记引导目标理解机制，而动作标记则生成精准抓取与执行的动作序列。

图中展示了机械臂部分伸展并靠近一个易碎物体。在此类场景中，状态标记通过编码实时本体感觉信息（如关节角度、夹爪姿态及末端执行器位置）发挥着关键作用。这些标记会持续与基于视觉和语言的前缀标记进行融合，使Transformer模型能够推理物理约束条件。因此，模型可推断碰撞即将发生，并相应调整运动指令——例如重新规划机械臂轨迹或调节力输出。如图8b所示，在移动机器人平台上，状态标记封装了里程计数据、激光雷达扫描数据及惯性传感器数据等空间特征，这些数据对地形感知式移动和障碍物规避至关重要。Transformer模型将这种状态表征与环境及指令上下文相结合，生成能动态适应环境变化的导航动作。无论是应对杂乱环境中的物体抓取任务，还是自主穿越崎岖地形，状态标记都为情境感知提供了结构化机制。这使得自回归解码器能够生成精准且基于上下文的动作序列，既体现机器人内部配置特征，又融合外部传感器数据信息。

- **动作标记：自回归控制生成：**VLA标记流程的最终层涉及动作标记[121, 122]，这些标记由模型通过自回归生成来表征运动控制的下一步骤[242]。每个标记对

应一个低级控制信号，例如关节角度更新、扭矩值、车轮速度或高级运动原语[81]。在推理过程中，模型根据前缀标记和状态标记逐步解码这些标记，从而将VLA模型转化为语言驱动的策略生成器[67, 209]。该架构可与真实世界驱动系统无缝集成，支持变长动作序列[11, 99]，并通过强化学习或模仿学习框架实现模型微调[285]。值得注意的是，RT-2[299]和PaLM-E[58]等模型正是这种设计的典范，其将感知、指令和具身化融合为统一的标记流。

例如，在图9所示的苹果采摘任务中，模型可能接收包含果园图像和文本指令的前缀标记。状态标记描述了机器人当前机械臂的姿态以及夹爪是张开还是闭合的状态。随后，系统会逐步预测动作标记，以引导机械臂朝向苹果移动、调整夹爪方向，并以适当的力度执行抓取动作。这种方法的精妙之处在于：它使得传统上用于文本生成的Transformer模型，能够以类似于生成句子的方式生成物理动作序列——只不过这里的“句子”指的是具体的运动动作本身。

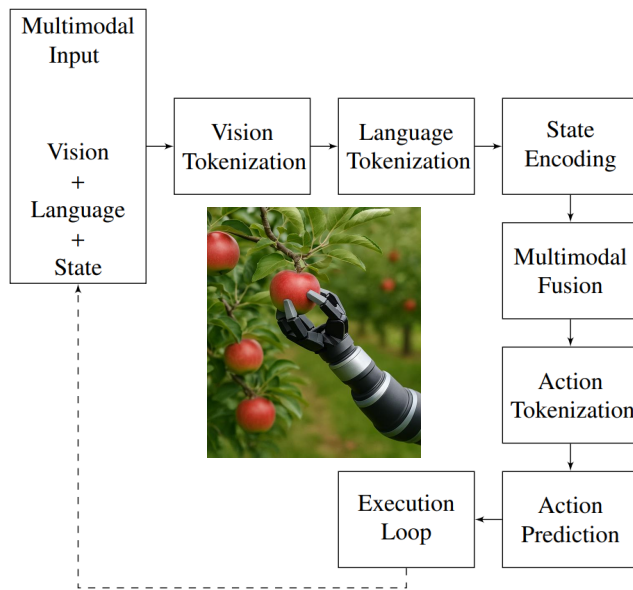


图9：展示VLAs编码世界的过程。VLA通过将视觉、语言和传感器输入转换为令牌，利用交叉注意力机制进行融合，通过Transformer预测动作序列，并实现实时反馈的任务执行——从而让机器人能够解读场景、遵循指令并动态调整动作。

为在机器人领域实现VLA范式，我们在图9中提出了一套结构化流程，该流程具体展示了如何对视觉、语言及本体感觉状态等多模态信息进行编码、融合，并最终转化为可执行指令。

action sequences. This end-to-end loop allows a robot to interpret complex tasks like “pick the ripe apple near the green leaf” and execute precise, context-sensitive manipulations. The system begins with **multimodal input acquisition**, where three distinct data streams are collected: visual observations (e.g., RGB-D frames), natural language commands, and real-time robot state information (e.g., joint angles or velocity). These are independently tokenized into discrete embeddings using pre-trained modules [51, 282]. As depicted in the diagram, the image is processed through a Vision Transformer (ViT) backbone to generate *vision tokens*, the instruction is parsed by a language model such as BERT or T5 to produce *language tokens*, and state inputs are transformed via a lightweight MLP encoder into compact *state tokens*.

These tokens are then fused using a cross-modal attention mechanism, where the model jointly reasons over object semantics, spatial layout, and physical constraints [76]. This fused representation forms the contextual basis for decision-making [96, 149]. In Figure 9, this is denoted as the **multi-modal fusion** step. The fused embedding is passed into an autoregressive decoder typically a transformer that generates a series of action tokens. These tokens may correspond to joint displacements, gripper force modulation, or high-level motor primitives (e.g., “move to grasp pose”, “rotate wrist”). The predicted action tokens are subsequently translated into low-level control commands and executed by an external, hardware-dependent execution loop, which interfaces with the robot controller to close the perception-action cycle by feeding back updated state observations for the next VLA inference step. This closed-loop mechanism enables the model to dynamically adapt to perturbations, object shifts, or occlusions in real time [276, 155, 251].

To provide clear and specific implementation details, Algorithm 1 formalizes the VLA tokenization process. Given an RGB-D frame I , natural language instruction T , and joint angle vector θ , the algorithm produces a set of action tokens that can be executed in sequence. The image I is processed via a ViT to produce V , a set of 400 visual tokens. In parallel, the instruction T is encoded by a BERT model to yield L , a sequence of 12 semantic language tokens. Simultaneously, the robot state θ —including joint angles, end-effector pose, and proprioceptive signals—is encoded by a multilayer perceptron into a compact 64-dimensional state embedding S , providing the model with real-time awareness of the robot’s configuration and physical constraints during action generation. These tokens are then fused via a cross-attention module to produce a shared 512-dimensional representation F , capturing the semantics, intent, and situational awareness needed for grounded action. Finally, a policy decoder such as FAST [171] maps the fused features into 50 discrete action tokens, which can then be decoded into motor commands $\tau_{1:N}$.

The decoding process is implemented using a transformer-based architecture, as shown in the code snippet titled *Action Prediction Code*. A Transformer decoder is instantiated with 12 layers, a model dimension of 512, and 8 attention heads. The fused multimodal tokens are provided as context, and the decoder autoregressively predicts action tokens one step at a time, where each predicted token represents the next control decision

conditioned on the full multimodal context and all previously generated actions. The resulting action-token sequence is then detokenized into a continuous motor command trajectory for execution. This implementation mirrors how text generation works in LLMs, but here the “sentence” is a motion trajectory a novel repurposing of natural language generation techniques for physical action synthesis.

Together, Figure 9, Algorithm 1, and the pseudocode illustrate how VLAs unify perception, instruction, and embodiment within a coherent and interpretable token space. This modularity allows the framework to generalize across tasks and robot morphologies, facilitating rapid deployment in real-world applications like apple picking, household tasks, and mobile navigation. Importantly, the clarity and separability of the tokenization steps make the architecture extensible, enabling further research on token learning, hierarchical planning, or symbolic grounding in VLA systems.

Algorithm 1 VLA Tokenization Pipeline

- 1: **Input:** RGB-D frame I , text command T , joint angles θ
 - 2: $V \leftarrow \text{ViT}(I)$ ▷ 400 vision tokens
 - 3: $L \leftarrow \text{BERT}(T)$ ▷ 12 language tokens
 - 4: $S \leftarrow \text{MLP}(\theta)$ ▷ 64-dim state encoding
 - 5: $F \leftarrow \text{CrossAttention}(V, L, S)$ ▷ 512-dim fused token
 - 6: $A \leftarrow \text{FAST}(F)$ ▷ 50 action tokens
 - 7: **Output:** Motor commands $\tau_{1:N}$
-

Action Prediction Code

```
# Python-like pseudocode
def predict_actions(fused_tokens):
    transformer = Transformer(
        num_layers=12,
        d_model=512,
        nhead=8
    )
    action_tokens = transformer.decode(
        fused_tokens,
        memory=fused_tokens
    )
    return detokenize(action_tokens)
```

2.4. Learning Paradigms: Data Sources and Training Strategies

Training VLA models requires a hybrid learning paradigm that integrates both semantic knowledge from the web and task-grounded information from robotics datasets [35]. As shown in prior sections, the multi-modal architecture of VLAs must be exposed to diverse forms of data that support language understanding, visual recognition, and motor control. This is typically achieved through two primary data sources.

First, as depicted in figure 10, large-scale internet-derived corpora form the backbone of the model’s semantic prior. These datasets include image-caption pairs (e.g., COCO, LAION-400M), instruction-following datasets (e.g., HowTo100M, WebVid), and visual question-answering corpora (e.g., VQA,

动作序列。这种端到端循环使机器人能够理解诸如“从绿叶旁采摘熟苹果”之类的复杂任务，并执行精确且具有上下文敏感性的操作。系统首先进行**多模态输入采集**，收集三种不同的数据流：视觉观测数据（例如RGB-D帧）、自然语言指令以及实时机器人状态信息（例如关节角度或速度）。这些数据通过预训练模块[51, 282]被独立分词为离散的嵌入表示。如图所示，图像通过视觉Transformer（ViT）主干网络生成**视觉令牌**；指令由BERT或T5等语言模型解析生成**语言令牌**；状态输入则通过轻量级MLP编码器转换为紧凑的**状态令牌**。

这些标记随后通过跨模态注意力机制进行融合，模型在此过程中对物体语义、空间布局及物理约束条件进行联合推理[76]。这种融合表征构成了决策制定的上下文基础[96, 149]。如图9所示，该过程被标记为**多模态融合**步骤。融合后的嵌入向量将输入自回归解码器（通常为Transformer模型），生成一系列动作标记。这些标记可能对应联合位移、夹爪力调节或高级运动原语（例如“移动至抓取姿态”、“腕部旋转”）。预测的动作标记随后被转化为低级控制指令，由外部硬件依赖型执行循环执行，并通过与机器人控制器交互反馈更新状态观测值，从而完成感知-动作闭环以进行下一阶段VLA推理。这种闭环机制使模型能够实时动态适应扰动、物体位移或遮挡等环境变化[276, 155, 251]。

为提供清晰具体的实现细节，算法1对VLA分词过程进行形式化描述。给定RGB-D帧 I 、自然语言指令 T 及关节角度向量 θ ，该算法生成可按顺序执行的动作标记集。图像 I 通过ViT处理得到 V （400个视觉标记集）。同时，指令 T 通过BERT模型编码生成 L （12个语义语言标记序列）。与此同时，机器人状态 θ （包括关节角度、末端执行器姿态及本体感觉信号）通过多层感知机编码为紧凑的64维状态嵌入 S ，使模型在动作生成过程中实时感知机器人构型与物理约束。这些标记通过交叉注意力模块融合生成共享的512维表征 F ，捕捉执行具体动作所需的语义信息、意图及情境感知能力。最后，像FAST[171]这样的策略解码器将融合特征映射到50个离散动作标记中，这些标记随后可被解码为运动指令 $\tau_{1:N}$ 。

解码过程采用基于Transformer架构的实现方案，具体代码片段标题为**动作预测代码**。该Transformer解码器配置了12层网络结构、512维模型维度及8个注意力头。融合后的多模态标记作为上下文输入，解码器通过自回归机制逐步预测动作标记——每个预测标记均基于完整多模态上下文及所有先前生成的动作，预示后续控制决策。最终生成的

动作标记序列经解码处理后，转化为连续的运动指令轨迹以供执行。该实现方式与大语言模型中的文本生成机制相似，但此处的“句子”实为运动轨迹，这标志着自然语言生成技术在物理动作合成领域实现了创新性应用。

图9、算法1及伪代码共同展示了VLA如何在一个连贯且可解释的标记空间内整合感知、指令与实体化功能。这种模块化设计使该框架能够跨任务和机器人形态进行通用化应用，便于在苹果采摘、家务任务及移动导航等实际场景中快速部署。尤为重要的是，标记化步骤的清晰性与可分离性使得该架构具有可扩展性，为VLA系统在标记学习、层次化规划或符号基础构建方面的进一步研究提供了可能。

算法1 VLA分词流水线

```
1: Input: RGB-D frame  $I$ , text command  $T$ , joint angles  $\theta$ 
2:  $V \leftarrow \text{ViT}(I)$  ▷ 400 vision tokens
3:  $L \leftarrow \text{BERT}(T)$  ▷ 12 language tokens
4:  $S \leftarrow \text{MLP}(\theta)$  ▷ 64-dim state encoding
5:  $F \leftarrow \text{CrossAttention}(V, L, S)$  ▷ 512-dim fused token
6:  $A \leftarrow \text{FAST}(F)$  ▷ 50 action tokens
7: Output: Motor commands  $\tau_{1:N}$ 
```

动作预测代码

类Python伪代码

预测动作（融合词元）：Transformer模

```
型（层数=12）
    d_model=512,
    nhead=
    8)
    action_tokens = transformer.decode(fused
    sed_tokens,
    memory=
    fusedtokens )
    返回解令牌化（action_tokens）
```

2.4. 学习范式：数据来源与训练策略

训练视觉语言模型（VLA）需要采用混合学习范式，该范式需整合来自网络的语义知识与机器人数据集中的任务相关信息[35]。如前文所述，VLA的多模态架构必须接触多种类型的数据，这些数据需支持语言理解、视觉识别及运动控制功能。通常通过两大主要数据源实现这一目标。

首先，如图10所示，大规模互联网来源的语料库构成了模型语义先验的核心基础。这些数据集包括图像-描述对（例如COCO、LAION -400M）、指令遵循数据集（例如HowTo100M、WebVid）以及视觉问答语料库（例如VQA）。

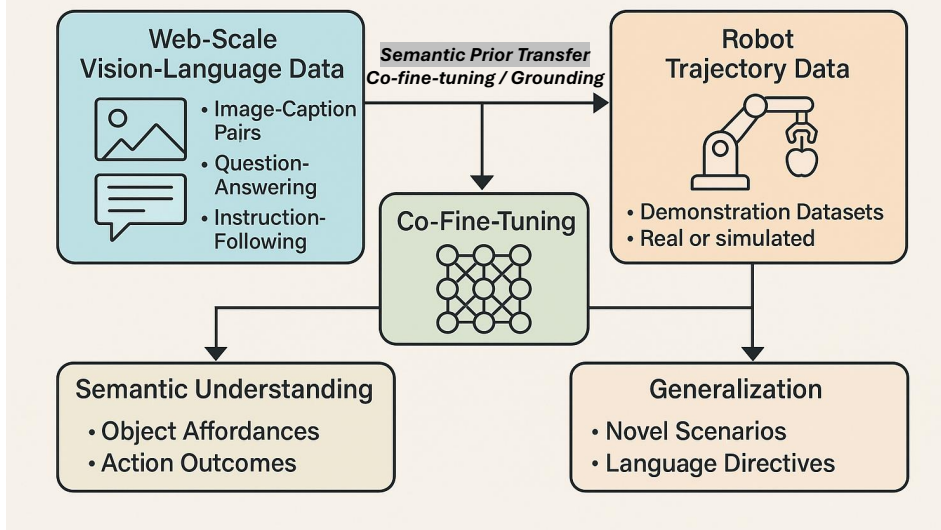


Figure 10: Learning Paradigms: Data Sources and Training Strategies for VLAs.

GQA). Such datasets enable pretraining of the visual and language encoders, helping the model acquire general representations of objects, actions, and concepts [2]. This phase often uses contrastive or masked modeling objectives, such as CLIP-style contrastive learning or language modeling losses, to align vision and language modalities within a shared embedding space [186, 260]. Importantly, this stage gives VLAs a foundational “understanding of the world” that facilitates compositional generalization, object grounding, and zero-shot transfer [33, 16].

However, semantic understanding alone is insufficient for physical task execution [44, 231, 137]. Thus, the second phase focuses on grounding the model in embodied experience [231]. Robot trajectory datasets collected either from real-world robots or high-fidelity simulators are used to teach the model how language and perception translate into action [67]. These include datasets like RoboNet [45], BridgeData [61], and RT-X [227], which provide video-action pairs, joint trajectories, and environment interactions under natural language instructions [159]. Demonstration data may come from kinesthetic teaching, teleoperation, or scripted policies [115, 13]. This phase typically employs supervised learning (e.g., behavior cloning) [68], reinforcement learning (RL), or imitation learning to train the autoregressive policy decoder to predict action tokens based on fused visual-language-state embeddings [83].

Recent works increasingly adopt *multistage or multitask training strategies*. For example, models are often pretrained on vision-language datasets using masked language modeling, then fine-tuned on robot demonstration data using token-level autoregressive loss [122, 295, 252]. Others use curriculum learning, where simpler tasks (e.g., object pushing) precede more complex ones (e.g., multistep manipulation) [288]. Some approaches further leverage domain adaptation such as in OpenVLA [122] or sim-to-real transfer to bridge the gap between synthetic and real-world distributions [125]. By unifying semantic priors with task execution data, these learning

paradigms allow VLA models to generalize across tasks, domains, and embodiments forming the backbone of scalable, instruction-following agents capable of robust real-world operation.

Through co-fine-tuning, these datasets are brought into alignment [232, 63]. The model learns to map from visual and linguistic inputs to appropriate action sequences [175]. This training paradigm not only helps the model understand object affordances (e.g., apples can be grasped) and action outcomes (e.g., lifting requires force and trajectory), but also promotes generalization to novel scenarios [129]. A model trained on kitchen manipulation tasks may be able to infer how to pick an apple in an outdoor orchard if it has learned general principles of object localization, grasping, and following language directives.

Recent architectures, such as Google DeepMind’s RT-2 (Robotic Transformer 2) [299], have demonstrated this principle in action. RT-2 treats action generation as a form of text generation, where each action token corresponds to a discrete command in a robot’s control space. Because the model is trained on both web-scale multi-modal data and thousands of robot demonstrations, it can flexibly interpret novel instructions and perform zero-shot generalization to new objects and tasks something that was largely impossible with traditional control systems or even with early multi-modal models.

2.5. Adaptive Control and Real-Time Execution

Another strength of VLAs lies in their ability to perform adaptive control, using real-time feedback from sensors to adjust behavior on the fly [195]. This is particularly important in dynamic, unstructured environments like orchards, homes, or hospitals, where unexpected changes (e.g., wind moving an apple, lighting changes, human presence) can alter the task parameters. During execution, state tokens are updated in real-time, reflecting sensor inputs and joint feedback [252]. The model can then revise its planned actions accordingly. For instance, in the apple-picking scenario, if the target apple shifts slightly or

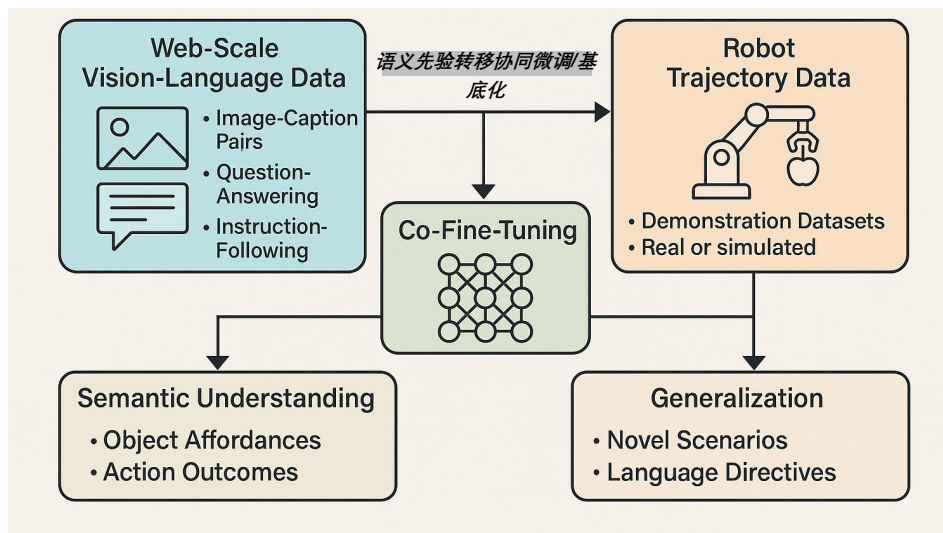


图10: 学习范式: VLAs的数据源与训练策略。

GQA) 这类数据集能够对视觉编码器和语言编码器进行预训练, 帮助模型获取物体、动作和概念的通用表征[2]。该阶段通常采用对比学习或掩码建模目标, 例如CLIP风格的对比学习或语言建模损失函数, 以使视觉模态与语言模态在共享嵌入空间中实现对齐[186, 260]。重要的是, 这一阶段为视觉语言模型(VLA)奠定了基础性的“世界认知”, 从而促进组合泛化能力、物体定位能力以及零样本迁移能力的提升[33, 16]。

然而, 仅凭语义理解不足以完成物理任务执行[44, 231, 137]。因此, 第二阶段专注于将模型建立在具身经验之上[231]。从真实机器人或高保真模拟器收集的机器人轨迹数据集被用于教会模型如何将语言和感知转化为动作[67]。这些数据集包括RoboNet[45]、BridgeData[61]和RT-X[227]等数据集, 它们提供了视频-动作对、关节轨迹以及在自然语言指令下的环境交互[159]。演示数据可能来自动觉教学、远程操作或脚本化策略[115, 13]。该阶段通常采用监督学习(例如行为克隆)[68]、强化学习(RL)或模仿学习, 以训练自回归策略解码器根据融合后的视觉-语言-状态嵌入来预测动作标记。

[83]。

近年来的研究越来越多地采用多阶段或多任务训练策略。例如, 模型通常先在视觉-语言数据集上使用掩码语言模型进行预训练, 随后在机器人演示数据上采用词元级自回归损失进行微调[122, 295, 252]。其他研究则采用课程学习方法, 即让简单任务(如物体推送)先于复杂任务(如多步骤操作)进行训练[288]。部分方法进一步利用领域适应技术(如OpenVLA[122])或仿真到真实迁移技术, 以弥合合成数据与真实世界分布之间的差距[125]。通过将语义先验与任务执行数据相结合, 这些学习范式使VLA模型能够跨任务、跨领域、跨形态实现泛化, 从而构

建出具备鲁棒现实操作能力的可扩展指令遵循智能体核心架构。

通过协同微调技术, 这些数据集实现了精准对齐[232, 63]。模型通过学习将视觉输入与语言指令映射为对应的动作序列[175]。这种训练范式不仅能帮助模型理解物体可用性特征(例如苹果可被抓取)和动作结果(例如提举动作需要施加力道并遵循轨迹), 还能提升对新场景的泛化能力[129]。若模型已掌握物体定位、抓取动作及遵循语言指令等通用原理, 那么在厨房操作任务训练后, 它就能推断出如何在户外果园中采摘苹果。

近年来的多种架构, 例如谷歌DeepMind的RT-2(机器人Transformer 2)[299], 已将这一原理付诸实践。RT-2将动作生成视为一种文本生成任务, 其中每个动作标记对应机器人控制空间中的一个离散指令。由于该模型同时基于海量网络规模的多模态数据和数千个机器人演示数据进行训练, 因此能够灵活解读新颖指令, 并对新对象和任务实现零样本泛化——这一能力在传统控制系统甚至早期多模态模型中几乎无法实现。

2.5. 自适应控制与实时执行

虚拟智能体(VLA)的另一优势在于其具备自适应控制能力, 能够通过传感器实时反馈动态调整行为[195]。这一特性在果园、住宅或医院等动态非结构化环境中尤为重要, 因为突发变化(如风力导致苹果移动、光照条件改变、人员进入)可能改变任务参数。执行过程中, 状态标记会实时更新以反映传感器输入和关节反馈信息[252], 模型可据此动态修正预设动作。例如在苹果采摘场景中, 若目标苹果发生轻微位移或

another apple enters the field of view, the model dynamically reinterprets the scene and adjusts the grasp trajectory. This capability mimics human-like adaptability and is a core advantage of VLA systems over pipeline-based robotics.

3. Progress in Vision-Language-Action Models

The inception of VLA models was catalyzed by the remarkable success of transformer-based LLMs, notably ChatGPT, released in November 2022, which demonstrated unprecedented semantic reasoning capabilities (ChatGPT) [179]. This breakthrough inspired researchers to extend language models to multimodal domains, integrating perception and action for robotics. By 2023, GPT-4 introduced multimodal capabilities by processing both text and images, which spurred subsequent efforts to extend language-centric multimodal foundation models toward incorporating physical action representations and control interfaces [1]. Concurrently, VLMs like CLIP (2022) [202] and Flamingo (2022) [3] had established robust visual-text alignment through contrastive learning, enabling zero-shot object recognition and laying the groundwork for VLM models (such as CLIP). These models leveraged large-scale labeled datasets to align images with textual descriptions, a critical precursor to integrating actions.

A pivotal development was the creation of large-scale robotic datasets, such as RT-1’s 130,000 demonstrations, which provided action-grounding data essential for co-training vision, language, and action components [19]. These datasets captured diverse tasks and environments, enabling models to learn generalizable behaviors. Architectural breakthroughs followed with Google’s RT-2 in 2023 [18], a landmark VLA model that unified vision, language, and action tokens, treating robotic control as an autoregressive sequence prediction task (RT-2 Blog(Source Link)). RT-2 discretized actions using Discrete Cosine Transform (DCT) compression and Byte-Pair Encoding (BPE), achieving a 63% improvement in performance on novel objects. Multimodal fusion techniques, such as cross-attention transformers, integrated Vision Transformer (ViT)-processed images (e.g., 400 patch tokens) with language embeddings, enabling robots to execute complex commands like “Pick the red cup left of the bowl.” Additionally, UC Berkeley’s Octo model (2023) introduced an open-source approach with 93M parameters and diffusion decoders, trained on 800,000 robot demonstrations from the OpenX-Embodiment dataset, further broadening the research landscape [218].

3.1. Architectural Innovations in VLA Models

From 2023 to 2024, VLA models underwent significant architectural advancements and refined training methodologies. Dual-system architectures emerged as a key innovation, exemplified by NVIDIA’s GR00T N1 (2025) [14], which combined System 1 (fast diffusion policies with 10ms latency for low-level control) and System 2 (LLM-based planners for high-level task decomposition). This separation enabled efficient coordination between strategic planning and real-time execution, enhancing adaptability in dynamic environments. Other

models, like Stanford’s OpenVLA (2024) [122], introduced a 7B-parameter open-source VLA trained on 970k real-world robot demonstrations, using dual vision encoders (DINOv2 [164] and SigLIP [271]) and a Llama 2 language model [223], outperforming larger models like RT-2-X (55B) [122]. Training paradigms evolved to leverage co-fine-tuning on web-scale vision-language data (e.g., LAION-5B) [194] and robotic trajectory data (e.g., RT-X) [227], aligning semantic knowledge with physical constraints [194]. Synthetic data generation tools like UniSim addressed data scarcity by creating photorealistic scenarios, such as occluded objects, crucial for robust training (UniSim [264]). Parameter efficiency was enhanced through Low-Rank Adaptation (LoRA) adapters for fine-tuning [93], which allowed domain adaptation without full retraining, reducing GPU hours by 70%. The introduction of diffusion-based policies, as seen in Physical Intelligence’s pi 0 model (2024) [15], offered improved action diversity but required significant computational resources. These advancements democratized VLA technology, fostering collaboration and accelerating innovation.

Recent VLA models have converged toward three major architectural paradigms that balance efficiency, modularity, and robustness: early fusion models, dual-system architectures, and self-correcting frameworks. Each of these innovations addresses specific challenges in grounding, generalization, and action reliability in real-world robotic systems.

1. Early Fusion Models: One class of VLA approaches focuses on fusing vision and language representations at the input stage before passing them to the policy module. Huang et al.’s EF-VLA model [96], presented at International Conference on Learning Representations (ICLR 2025), exemplifies this trend by retaining the representational alignment established by CLIP [202]. EF-VLA accepts image-text pairs, encodes them with CLIP’s frozen encoders, and fuses the resulting embeddings early in the transformer backbone prior to action prediction. This design ensures that the semantic consistency learned during CLIP pretraining is preserved, reducing overfitting and enhancing generalization. Notably, EF-VLA demonstrated a 20% performance improvement on compositional manipulation tasks and reached 85% success on previously unseen goal descriptions. By keeping the vision-language backbone frozen, this approach preserves computational efficiency and avoids catastrophic forgetting, while domain-specific training is confined to lightweight policy or action modules, enabling task adaptation without sacrificing the model’s general-purpose visual-semantic representations.

2. Dual-System Architectures: Inspired by dual-process theories of human cognition, models like NVIDIA’s GR00T N1 (2025) [14] implement two complementary subsystems: a fast-reactive module (System 1) and a slow-reasoning planner (System 2). System 1 comprises a diffusion-based control policy that operates at 10 ms latency, ideal for fine-grained, low-level control such as end-effector stabilization or adaptive grasping. In contrast, System 2 uses a LLM for task planning, skill composition, and high-level sequencing. The planner parses long-horizon goals (e.g., “clean the table”) into atomic subtasks, while the low-level controller ensures real-time execution. This

当另一个苹果进入视野范围时，模型会动态重新解析场景并调整抓取轨迹。该能力模拟了类人适应性，是VLA系统相较于基于流水线的机器人技术的核心优势。

3. 视觉-语言-动作模型的研究进展

VLA模型的诞生得益于基于Transformer架构的大语言模型（LLMs）取得的显著成功，尤其是2022年11月发布的ChatGPT，它展现了前所未有的语义推理能力（ChatGPT）[179]。这一突破激励研究人员将语言模型扩展到多模态领域，将感知与动作整合应用于机器人技术。到2023年，GPT-4通过同时处理文本和图像实现了多模态能力，推动了后续将语言中心型多模态基础模型向融合物理动作表征和控制接口方向发展的努力[1]。与此同时，CLIP（2022）[202]和Flamingo（2022）[3]等视觉语言模型通过对比学习建立了强大的视觉-文本对齐能力，实现了零样本物体识别，并为VLM模型（如CLIP）奠定了基础。这些模型利用大规模标注数据集实现图像与文本描述的对齐，这是整合动作的关键前提。

一个关键进展是大规模机器人数据集的创建，例如RT-1的13万次演示数据，这些数据为视觉、语言和动作组件的协同训练提供了关键的动作定位数据[19]。这些数据集涵盖了多样化任务和环境，使模型能够学习可泛化行为。随后在2023年，谷歌推出了RT-2架构突破[18]，这一里程碑式的视觉语言动作（VLA）模型统一了视觉、语言和动作标记，将机器人控制视为自回归序列预测任务（RT-2博客（来源链接））。RT-2采用离散余弦变换（DCT）压缩和字节对编码（BPE）对动作进行离散化处理，在新物体任务上实现了63%的性能提升。多模态融合技术（如交叉注意力变换器）将视觉变换器（ViT）处理的图像（例如400个补丁标记）与语言嵌入相结合，使机器人能够执行诸如“拿起碗左侧的红色杯子”等复杂指令。此外，加州大学伯克利分校的Octo模型（2023年）采用开源方法，包含9300万参数和扩散解码器，基于OpenX-Embodiment数据集中的80万机器人演示数据进行训练，进一步拓展了研究领域[218]。

3.1. VLA模型中的建筑创新

2023至2024年间，VLA模型在架构设计上取得重大突破并优化了训练方法。双系统架构成为关键创新点，NVIDIA的GR00T N1（2025）[14]便是典型代表——该模型将系统1（具备10毫秒延迟的快速扩散策略用于底层控制）与系统2（基于大语言模型的高层任务分解规划器）相结合。这种分离架构实现了战略规划与实时执行的高效协同，显著提升了动态环境下的适应能力。其他模型如斯坦福大学的OpenVLA（2024）[122]，基于97万条真实机器

人演示数据训练的70亿参数开源VLA模型，采用双视觉编码器（DINOv2[164]与SigLIP[271]）及Llama 2语言模型[223]，其性能超越了RT-2-X（550亿参数）等大型模型[122]。训练范式持续演进，通过利用网络规模视觉语言数据（如LAION-5B）[194]与机器人轨迹数据（如RT-X）[227]进行协同微调，将语义知识与物理约束条件精准对齐[194]。UniSim等合成数据生成工具通过创建逼真场景（如遮挡物体）来解决数据短缺问题，这对实现稳健训练至关重要（UniSim[264]）。通过低秩自适应（LoRA）适配器进行微调[93]，参数效率得到显著提升，无需完全重新训练即可实现领域适应，GPU运行时间减少70%。基于扩散策略的创新方法在物理智能公司pi 0模型（2024年）中得到充分体现。[15]该方案提升了动作多样性，但需要大量计算资源。这些进展使甚大天线阵列（VLA）技术得以普及，促进了协作并加速了创新进程。

最新版视觉语言模型（VLA）已形成三大架构范式，这些范式在效率、模块化与鲁棒性之间取得平衡：早期融合模型、双系统架构及自校正框架。这些创新技术分别针对现实机器人系统中定位精度、泛化能力及动作可靠性等具体挑战提出解决方案。

1. 早期融合模型：一类视觉语言建模（VLA）方法专注于在输入阶段融合视觉与语言表征，再将其传递至策略模块。黄等人在国际学习表征会议（ICLR 2025）上提出的EF-VLA模型[96]正是这一趋势的典型代表，该模型保留了CLIP[202]建立的表征对齐机制。EF-VLA接收图像-文本对，通过CLIP的冻结编码器进行编码，并在Transformer主干网络早期阶段融合生成的嵌入向量以进行动作预测。这种设计确保了CLIP预训练过程中习得的语义一致性得以保留，从而降低过拟合风险并提升泛化能力。值得注意的是，EF-VLA在组合操作任务中实现了20%的性能提升，并在未见过的目标描述任务中达到了85%的成功率。通过保持视觉-语言主干结构不变，该方法既保持了计算效率，又避免了灾难性遗忘；同时将领域特定训练限制在轻量级策略或动作模块内，从而能够在不牺牲模型通用视觉-语义表征的前提下实现任务适应。

2. 双系统架构：受人类认知双过程理论启发，NVIDIA的GR00T N1模型（2025年）[14]实现了两个互补子系统：快速反应模块（系统1）与慢速推理规划器（系统2）。系统1采用基于扩散机制的控制策略，具有10毫秒的延迟响应时间，特别适用于末端执行器稳定控制或自适应抓取等精细级低层次控制任务。相比之下，系统2通过大型语言模型（LLM）实现任务规划、技能组合及高层级动作序列编排。规划器将长期目标（如“清理桌面”）分解为原子级子任务，而低层控制器则确保实时执行。该架构通过分层设计实现了多层级控制系统的协同运作，既保持了系统响应速度，又提升了任务执行效率。

decomposition enables multi-timescale reasoning and improved safety, especially in environments where rapid reaction and deliberation must co-exist. In benchmark tests on multi-stage household manipulation, *GR00T N1* outperformed monolithic models such as *RT-1*, *RT-2*, and *OpenVLA* by 17% in success rate and reduced collision failures by 28%.

3. Self-Correcting Frameworks: A third architectural evolution is the emergence of self-correcting VLA models, which augment conventional inference pipelines with explicit failure detection and recovery mechanisms. *SC-VLA* (2024) retains a standard fast inference path similar to earlier end-to-end or hierarchical VLA designs, but introduces an additional, slower correction path that is selectively activated to re-evaluate decisions and generate recovery actions when execution failures or inconsistencies are detected. In this framework, the default behavior is to predict poses or actions directly from the fused embedding using a lightweight transformer. When failures are detected e.g., unsuccessful grasps or obstacle collisions, the model invokes a secondary process that performs chain-of-thought reasoning [281, 270]. This path queries an internal LLM (or external expert system) to diagnose failure modes and generate correction strategies [59]. For example, if the robot repeatedly misidentifies an occluded object, the LLM may suggest an active viewpoint change or gripper reorientation. In closed-loop experiments, *SC-VLA* reduced task failure rates by 35% and significantly improved recoverability in cluttered and adversarial environments.

4. Architectural Design Space of VLA Models VLA models exhibit a rich diversity of architectural designs and functional emphases, which can be systematically organized along the dimensions of end-to-end versus modular pipelines, hierarchical versus flat policy structures, and the balance between low-level control and high-level planning (Table 1). End-to-end VLAs, such as *CLIPort* [202], *RT-1* [19], and *OpenVLA* [122], process raw sensory inputs directly into motor commands via a single unified network. By contrast, component-focused models like *VLA-Test* [237] and *Chain-of-Affordance* [129] decouple perception, language grounding, and action modules, enabling targeted improvements in individual submodules.

Hierarchical architectures have emerged to tackle complex, long-horizon tasks by separating strategic decision making from reactive control. For instance, *CogACT* [131] and *NaVILA* [38] employ a two-tier hierarchy where an LLM-based planner issues subgoals to a low-level controller, thereby combining the strengths of System 2 reasoning and System 1 execution. Similarly, *ORION* [69] integrates a *QT-Former* for long-term context aggregation with a generative trajectory planner in a cohesive framework.

Low-level policy emphasis is typified by diffusion-based controllers (e.g. *Pi-0* [15], *DexGraspVLA* [291]), which excel at producing smooth, diverse motion distributions but often incur higher computational cost. In contrast, high-level planners (e.g. *FAST Pi-0 Fast* [171], *CoVLA* [5]) focus on rapid subgoal generation or coarse trajectory prediction, delegating fine-grained control to specialized modules or traditional motion planners. End-to-end dual-system models like *HybridVLA* [142] and *Helix* [217] blur these distinctions by

jointly training both components while preserving modular interpretability.

Table 1 further highlights how recent VLAs balance these trade-offs. Models such as *OpenDriveVLA* [293] and *CombatVLA* [34] prioritize hierarchical planning in dynamic, safety-critical domains, whereas lightweight, edge-targeted systems like *Edge VLA* [20] and *TinyVLA* [242] emphasize real-time low-level policies at the expense of high-level reasoning. This classification framework not only clarifies the design space of VLAs but also guides future development by pinpointing under-explored combinations such as fully end-to-end, hierarchical models optimized for embedded deployment that promise to advance both the capabilities and the applicability of VLA systems across robotics, autonomous driving, and beyond.

The classification in Table 1 is significant because it provides a clear framework for comparing diverse VLA architectures, highlighting how design choices such as end-to-end integration versus hierarchical decomposition impact task performance, scalability, and adaptability. By categorizing models along dimensions such as low-level policy execution and high-level planning, researchers can more clearly identify the strengths and limitations of existing approaches and uncover opportunities for architectural innovation. For example, agricultural robotics tasks such as high-speed fruit harvesting or precision spraying benefit from architectures emphasizing fast, reactive low-level controllers, whereas applications like orchard navigation, multi-row coverage planning, or long-horizon crop monitoring require stronger high-level planning and reasoning capabilities. This taxonomy therefore aids in selecting appropriate VLA architectures for specific use cases and guides future development toward hybrid systems that balance responsiveness with cognitive planning, ultimately accelerating progress in embodied AI.

Additionally, to synthesize recent advancements in VLA models, Table 2 presents a summary of notable systems developed from 2022 through 2025. Building upon architectural innovations such as early fusion, dual-system processing, and self-correcting feedback loops, these models incorporate diverse design philosophy and training strategies. Each table entry explicitly enumerates the model’s architectural components—namely the vision encoder, language encoder, and action decoder together with the primary training datasets used to ground and evaluate the model’s capabilities. Models like *CLIPort* [202] and *RT-2* [299] laid early foundations by aligning semantic embeddings with action policies, while more recent frameworks like *Pi-Zero*, *CogACT* [131], and *GR00T N1* [14] introduce scalable architectures with diffusion-based or high-frequency controllers. Several models leverage multi-modal pretraining with internet-scale vision-language corpora and robot trajectory datasets, enhancing generalization and zero-shot capabilities [297, 291, 289, 257]. This tabulated comparison serves as a reference point for researchers seeking to understand the functional diversity, domain applicability, and emerging trends in VLA design across real and simulated environments.

分解技术能够实现多时间尺度推理并提升安全性，尤其适用于需要快速反应与审慎决策并存的环境。在多阶段家庭操作基准测试中，GR00T N1模型在成功率方面较RT-1、RT-2及OpenVLA等单一模型提升了17%，同时将碰撞故障率降低了28%。

3. 自校正框架：第三种架构演进是自校正VLA模型的出现，这类模型通过显式故障检测与恢复机制对传统推理流程进行增强。SC-VLA（2024）沿用了早期端到端或分层式VLA设计中的标准快速推理路径，但新增了选择性激活的较慢校正路径——当检测到执行失败或不一致性时，该路径会重新评估决策并生成恢复动作。该框架默认行为是通过轻量级Transformer直接从融合嵌入向量预测位姿或动作。当检测到失败情况（如抓取失败或碰撞障碍物）时，模型会调用次级流程进行链式推理[281, 270]。该路径会向内部大语言模型（LLM）或外部专家系统发起查询，以诊断故障模式并生成校正策略[59]。例如，若机器人反复误判被遮挡物体，LLM可能会建议主动调整视角或重新定位夹爪。在闭环实验中，SC-VLA使任务失败率降低了35%，并在杂乱及对抗性环境中显著提高了可恢复性。

4. VLA模型的架构设计空间VLA模型展现出丰富的架构设计与功能侧重，这些特性可根据端到端与模块化流程、层级化与扁平化策略结构、以及底层控制与高层规划之间的平衡关系进行系统化分类（表1）。端到端VLA（如CLIPort[202]、RT-1[19]和OpenVLA[122]）通过单一统一网络直接将原始感官输入转化为运动指令；相比之下，以组件为核心的模型（如VLATest[237]和Chain-of-Afordance[129]）将感知、语言基础处理与动作模块解耦，从而能够对各个子模块进行针对性优化。

为应对复杂且具有长期规划需求的任务，分层架构应运而生，通过将战略决策与反应式控制分离来实现。例如，CogACT[131]和NaVILA[38]采用双层架构设计：基于大语言模型的规划器会向低层级控制器下发子目标指令，从而有效整合系统2推理与系统1执行的优势。类似地，Orion [69]将用于长期上下文聚合的QT-Former模型与生成式轨迹规划器整合到统一框架中。

低层级策略强调以扩散式控制器（如Pi-0[15]、DexGraspVLA[291]）为代表，这类控制器擅长生成平滑且多样化的运动分布，但通常会带来更高的计算成本。相比之下，高层规划器（如FAST Pi-0 Fast[171]、CoVLA[5]）专注于快速子目标生成或粗略轨迹预测，将精细控制任务委托给专用模块或传统运动规划器。像HybridVLA[142]和Helix[217]这类端到端双系统模型通过联合训练两个组件

来模糊这些差异，同时保持模块化可解释性。

表1进一步揭示了近期虚拟实验室架构（VLAs）如何平衡这些权衡关系。OpenDriveVLA[293]和CombatVLA[34]等模型在动态安全关键领域优先采用分层规划策略，而Edge VLA[20]和TinyVLA[242]这类轻量级边缘目标系统则侧重实时低层策略，但会牺牲高层级推理能力。该分类框架不仅清晰界定了VLAs的设计空间，还通过精准定位未充分探索的组合方案（如专为嵌入式部署优化的全端到端分层模型），为未来发展指明方向——这类方案有望显著提升VLAs在机器人技术、自动驾驶等领域的能力与适用性。

表1中的分类体系具有重要价值，它为比较不同VLA架构提供了清晰框架，揭示了端到端集成与分层分解等设计选择如何影响任务性能、可扩展性及适应性。通过将模型按低级策略执行与高级规划等维度进行分类，研究者能更清晰地识别现有方法的优势与局限，并发掘架构创新机遇。例如，高速水果采摘或精准喷洒等农业机器人任务受益于强调快速响应式低级控制器的架构设计，而果园导航、多行覆盖规划或长时域作物监测等应用场景则需要更强的高级规划与推理能力。该分类体系有助于为特定应用场景选择合适的VLA架构，并指导未来开发平衡响应性与认知规划的混合系统，从而加速具身人工智能的发展进程。

此外，为综合VLA模型的最新进展，表2汇总了2022至2025年间开发的代表性系统。这些模型基于早期融合、双系统处理及自校正反馈环路等架构创新，融合了多样化的设计理念与训练策略。每条表项均明确列出了模型架构组件——包括视觉编码器、语言编码器、动作解码器，以及用于验证模型能力的主要训练数据集。诸如CLIPort[202]和RT-2[299]等模型通过将语义嵌入与动作策略对齐奠定了早期基础，而近期框架如Pi-Zero、CogACT[131]及GR00 TNI[14]则引入了基于扩散机制或高频控制器的可扩展架构。多种模型采用了基于互联网规模视觉-语言语料库和机器人轨迹数据集的多模态预训练方法，从而提升了泛化能力和零样本学习能力[297, 291, 289, 257]。本表格对比为研究人员提供了参考依据，有助于他们理解视觉语言架构（VLA）在真实与模拟环境中的功能多样性、领域适用性及新兴发展趋势。

Table 1: Taxonomy of VLA models showing structured classification based on architectural paradigms and scientific priorities. We differentiate models by their support for end-to-end execution, hierarchical planning–control decomposition, or component-focused modularity, and further by their emphasis on low-level motor policies versus high-level task planners.

Model Name	Year	End-to-End	Hierarchical	Component Focused	Low-Level Policy	High-Level Planner
CLIPort [202]	2022	✓	✗	✗	✓	✗
RT-1 [19]	2022	✓	✗	✗	✓	✗
Gato [181]	2022	✓	✗	✗	✓	✗
VIMA [112]	2022	✓	✗	✗	✓	✗
Diffusion Policy [40]	2023	✓	✗	✗	✓	✗
ACT [287]	2023	✓	✗	✗	✓	✗
VoxPoser [100]	2023	✓	✗	✗	✓	✗
Seer [80]	2023	✓	✗	✗	✓	✗
Octo [218]	2024	✓	✗	✗	✓	✗
OpenVLA [122]	2024	✓	✗	✗	✓	✗
CogACT [131]	2024	✗	✓	✗	✓	✓
VLA-Test [237]	2024	✗	✗	✓	✗	✗
NaVILA [38]	2024	✗	✓	✗	✓	✓
RoboNurse-VLA [132]	2024	✓	✗	✗	✓	✗
Mobility VLA [42]	2024	✗	✓	✗	✓	✓
RevLA [48]	2024	✗	✗	✓	✗	✗
Uni-NaVid [280]	2024	✗	✓	✗	✓	✓
RDT-1B [144]	2024	✓	✗	✗	✓	✗
RoboMamba [143]	2024	✓	✗	✗	✓	✗
Chain-of-Affordance [129]	2024	✗	✗	✓	✗	✗
Edge VLA [20]	2024	✗	✗	✓	✗	✗
ShowUI-2B [139]	2024	✓	✗	✗	✓	✗
Pi-0 [15]	2024	✓	✗	✗	✓	✗
FAST (Pi-0 Fast) [171]	2025	✗	✗	✓	✓	✗
OpenVLA-OFT [121]	2025	✓	✗	✗	✓	✗
CoVLA [5]	2025	✗	✓	✗	✓	✓
OpenDriveVLA [293]	2025	✗	✓	✗	✓	✓
ORION [69]	2025	✗	✓	✗	✓	✓
UAV-VLA [191]	2025	✗	✓	✗	✓	✓
CombatVLA [34]	2025	✓	✗	✗	✓	✗
HybridVLA [142]	2025	✗	✓	✗	✓	✓
NORA [103]	2025	✓	✗	✗	✓	✗
SpatialVLA [175]	2025	✗	✗	✓	✓	✗
MoLe-VLA [283]	2025	✗	✗	✓	✓	✗
JARVIS-VLA [130]	2025	✓	✗	✗	✓	✗
UP-VLA [279]	2025	✓	✗	✗	✓	✗
Shake-VLA [120]	2025	✗	✗	✓	✓	✗
DexGraspVLA [291]	2025	✗	✓	✗	✓	✓
DexVLA [241]	2025	✗	✓	✗	✓	✓
Humanoid-VLA [53]	2025	✓	✗	✗	✓	✗
ObjectVLA [297]	2025	✓	✗	✗	✓	✗
Long-VLA [64]	2025	✓	✗	✗	✓	✗
RetoVLA [123]	2025	✗	✗	✓	✓	✗
Vlaser [256]	2025	✗	✓	✗	✓	✓
Discrete Diffusion VLA [138]	2025	✓	✗	✗	✓	✗
Being-H0 [152]	2025	✓	✗	✗	✓	✗
EgoVLA [258]	2025	✓	✗	✗	✓	✗
StereoVLA [47]	2025	✓	✗	✗	✓	✗
GeoVLA [212]	2025	✗	✓	✗	✓	✓
EfficientVLA [262]	2025	✗	✗	✓	✓	✗

表1：VLA模型分类体系，基于架构范式与科学优先级进行结构化分类。我们通过以下维度对模型进行区分：支持端到端执行能力、采用层级化规划-控制分解机制、或采用组件导向模块化设计，并进一步根据其侧重于低级运动策略还是高级任务规划器进行分类。

模型名称	年	端到端	Hie rarec 希卡尔	组 件 聚 焦	低级策略	高级规划师
CLIPort [202]	2022	✓	✗	✗	✓	✗
RT-1 [19]	2022	✓	✗	✗	✓	✗
猫[181]	2022	✓	✗	✗	✓	✗
VIMA [112]	2022	✓	✗	✗	✓	✗
传播策略[40]	2023	✓	✗	✗	✓	✗
ACT [287]	2023	✓	✗	✗	✓	✗
VoxPoser [100]	2023	✓	✗	✗	✓	✗
西尔[80]	2023	✓	✗	✗	✓	✗
Octo [218]	2024	✓	✗	✗	✓	✗
OpenVLA [122]	2024	✓	✗	✗	✓	✗
CogACT [131]	2024	✗	✓	✗	✓	✓
VLA _{Test} [237]	2024	✗	✗	✓	✗	✗
NaVILA [38]	2024	✗	✓	✗	✓	✓
RoboNurse-VLA [132]	2024	✓	✗	✗	✓	✗
移动性VLA[42]	2024	✗	✓	✗	✓	✓
RevLA [48]	2024	✗	✗	✓	✗	✗
Uni-NaVid [280]	2024	✗	✓	✗	✓	✓
RDT-1B [144]	2024	✓	✗	✗	✓	✗
RoboMamba [143]	2024	✓	✗	✗	✓	✗
资金链[129]	2024	✗	✗	✓	✗	✗
Edge VLA [20]	2024	✗	✗	✓	✗	✗
ShowUI-2B [139]	2024	✓	✗	✗	✓	✗
Pi-0 [15]	2024	✓	✗	✗	✓	✗
FAST（Pi-0 Fast）[171]	2025	✗	✗	✓	✓	✗
OpenVLA-OFT [121]	2025	✓	✗	✗	✓	✗
CoVLA [5]	2025	✗	✓	✗	✓	✓
OpenDriveVLA [293]	2025	✗	✓	✗	✓	✓
猎户座[69]	2025	✗	✓	✗	✓	✓
UAV-VLA [191]	2025	✗	✓	✗	✓	✓
CombatVLA [34]	2025	✓	✗	✗	✓	✗
HybridVLA [142]	2025	✗	✓	✗	✓	✓
NORA [103]	2025	✓	✗	✗	✓	✗
SpatialVLA [175]	2025	✗	✗	✓	✓	✗
MoLe-VLA [283]	2025	✗	✗	✓	✓	✗
Jarvis-VLA[130]	2025	✓	✗	✗	✓	✗
UP-VLA [279]	2025	✓	✗	✗	✓	✗
Shake-VLA [120]	2025	✗	✗	✓	✓	✗
DexGraspVLA [291]	2025	✗	✓	✗	✓	✓
DexVLA [241]	2025	✗	✓	✗	✓	✓
人形-VLA[53]	2025	✓	✗	✗	✓	✗
ObjectVLA [297]	2025	✓	✗	✗	✓	✗
长视场阵列望远镜[64]	2025	✓	✗	✗	✓	✗
RetoVLA [123]	2025	✗	✗	✓	✓	✗
Vlaser [256]	2025	✗	✓	✗	✓	✓
离散扩散VLA[138]	2025	✓	✗	✗	✓	✗
存在 H0[152]	2025	✓	✗	✗	✓	✗
EgoVLA [258]	2025	✓	✗	✗	✓	✗
StereoVLA [47]	2025	✓	✗	✗	✓	✗
GeoVLA [212]	2025	✗	✓	✗	✓	✓
EfficientVLA [262]	2025	✗	✗	✓	✓	✗

Table 2: **Compact summary of representative Vision–Language–Action (VLA) models.** Each row reports the primary encoders/decoders, training data, and the main distinctive capability.

Model (Ref.)	Architecture (vision / language / action)	Training data	Key strength / uniqueness
CLIPort [202]	CLIP-ResNet50 + Transporter-ResNet / CLIP-GPT / LingUNet	Self-collected [SC]	Aligns semantic CLIP features with Transporter spatial reasoning for precise SE(2) manipulation.
RT-1 [19]	EfficientNet / Universal Sentence Encoder / Transformer (discretized actions)	RT-1-Kitchen [SC]	Early large-scale transformer policy for multi-task kitchen manipulation with tokenized actions.
RT-2 [299]	ViT-22B or ViT-4B / PaLI-X or PaLM-E / symbol-tuning (action tokens)	VQA + RT-1-Kitchen	Co-finetunes internet-scale VQA with robot data, yielding emergent generalization for embodied tasks.
Gato [181]	ViT / SentencePiece / Transformer (unified token stream)	Self-collected [SC]	Generalist agent unifying robotics, language, and Atari via shared tokenization and a single transformer.
VIMA [112]	ViT + Mask R-CNN / T5 / Transformer	VIMA-Data [SC]	Prompt-driven VL grounding across multiple compositional task types (six prompt modalities).
ACT [287]	ResNet-18 / — / CVAE-Transformer	ALOHA [SC]	Temporal ensembling enables smooth bimanual imitation with fine control precision.
Octo [218]	CNN / T5-base / Diffusion Transformer	Open X-Embodiment (OXE)	Large multi-robot policy trained on 4M+ trajectories spanning many robot embodiments.
VoxPoser [100]	ViLD + MDETR / GPT-4 / MPC (LLM-guided planning)	Zero-shot	Composes LLM+VLM for constraint-aware motion planning without task-specific training.
Diffusion Policy [40]	ResNet-18 / — / U-Net or Transformer diffusion	Self-collected [SC]	Diffusion modeling captures multimodal action distributions for robust visuomotor control.
OpenVLA [122]	DINOv2 + SigLIP / Prismatic-7B / symbol-tuning	OXE + DROID	Open-source RT-2-like VLA; supports efficient LoRA adaptation and broad generalization.
π^0 (Pi-Zero) [15]	PaliGemma VLM / PaliGemma (multimodal) / 300M diffusion action model	Pi-Cross-Embodiment	Lightweight general robot controller (reported ~3B total) with strong cross-robot, open-world generalization and bi-manual skills.
π^0 -Fast [171]	PaliGemma VLM / PaliGemma / autoregressive transformer with FAST tokenization	Pi-Cross-Embodiment	High-frequency real-time control via compressed frequency-space action tokens (reported up to 15× faster inference).
OpenVLA-OFT [121]	SigLIP + DINOv2 (multi-view) / Llama-2 7B / parallel decoding + action chunking (L1 regression)	LIBERO; bimanual ALOHA	Fine-tuning recipe with parallel decoding and chunked actions; reported 97.1% LIBERO success and 26× faster inference for high-frequency bimanual control.
RDT-1B [144]	Multi-view RGB encoder / transformer language module / Diffusion Transformer (unified action space)	46 datasets (>1M episodes) + ALOHA fine-tune	1.2B diffusion foundation model for dexterous bimanual manipulation with strong language conditioning and zero-shot transfer.
Helix ¹	System 2: open-source VLM for multimodal reasoning (7–9 Hz) / integrated semantics / System 1: transformer visuomotor policy (200 Hz, full upper-body)	Figure robot E2E (pixels+language→actions)	Humanoid-focused dual-rate VLA enabling real-time high-DoF control, dexterity, and collaborative multi-robot manipulation with zero-shot generalization.
CogACT [131]	DINOv2 ViT-L/14 + SigLIP ViT-So400M/14 / Llama-2 via Prismatic-7B / DiT-Base (300M diffusion)	OXE subset; Realman & Franka tasks	Componentized VLA with diffusion action transformer; reported +59.1% real-world success vs. OpenVLA and strong adaptation to unseen robots/objects.
Chain-of-Affordance (CoA) [129]	Affordance-aware visual encoder / transformer reasoning prompts / autoregressive + diffusion policy (affordance-conditioned)	LIBERO; real+sim manipulation	Sequential affordance reasoning (object→grasp→spatial→motion) improves spatial planning and obstacle avoidance; reported stronger LIBERO performance than OpenVLA.
Edge VLA (EVLA) [20]	SigLIP + DINOv2 / Qwen2 (0.5B) / non-autoregressive joint control prediction	Bridge; OXE; 1.2M text–image pairs	Edge-optimized VLA (e.g., Jetson-class) with reported 30–50 Hz inference and OpenVLA-comparable performance under low power.
ShowUI-2B [139]	UI-guided visual token selection / interleaved V–L–A streaming / transformer GUI action predictor	256K GUI instruction-following	Compact 2B VLA for digital automation; strong screenshot grounding and GUI/web navigation with efficient token selection.
GR00T N1 [14]	NVIDIA Eagle-2 VLM / integrated high-level planning / diffusion transformer (DiT)	Human demos + robot trajectories + simulation + internet video	Generalist humanoid dual-system design combining planning and diffusion execution for dexterous multi-step control and broad embodiment generalization.
Seer [80]	Grounding-optimized visual backbone / transformer language / autoregressive action head	LIBERO	Strong visual grounding for manipulation; competitive on LIBERO but typically below newer fine-tuned variants (e.g., OpenVLA-OFT).
DiffusionVLA [240]	Transformer visual encoder / autoregressive reasoning / diffusion action head	LIBERO; factory sorting; zero-shot bin-picking	Diffusion control improves robustness and interpretability; reported weaker spatial generalization than CoA in some configurations.

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¹<https://www.figure.ai/news/helix>

表2：代表性视觉-语言-动作（VLA）模型的简明摘要。每行报告主要编码器/解码器、训练数据及核心独特能力。

型号（参考编号）	架构（愿景/语言/行动）	训练数据	关键优势/独特性
CLIPort [202]	CLIP-ResNet50 + Transporter-ResNet / clippt / LingUNet	自采[SC]	将语义CLIP特征与转运体空间推理对齐，以实现精确的SE(2)操作。
RT-1 [19]	EfficientNet / 通用句子编码器 / Transformer（离散化动作）	RT-1厨房[SC]	基于分词动作的多任务厨房操作早期大规模变压器策略
RT-2 [299]	ViT-22B 或 ViT-4B / PaLI-X 或 PaLM-E / 符号调谐（动作标记）	VQA + RT-1-厨房	通过将机器人数据与互联网规模 VQA 相结合，实现具身任务的涌现性泛化能力。
猫[181]	ViT / SentencePiece / Transformer（统一标记流）	自采[SC]	通用智能体通过共享分词机制和单一Transformer模型，实现了机器人技术、语言处理与Atari游戏机系统的统一。
VIMA [112]	ViT + 面罩 R-CNN / T5 / Transformer	VIMA -数据[SC]	基于提示的VL接地技术在多种组合任务类型（六种提示模式）中的应用。
ACT [287]	ResNet-18 / — / CVAE-Transformer	ALOHA [SC]	时间集成技术可实现精细控制精度下的双侧手部平滑模仿。
Octo [218]	CNN / T5-base / Diffusion Transformer	开放X-实施例（OXE）	基于覆盖多种机器人实现方式的400余万条轨迹训练而成的大型多机器人策略。
VoxPoser [100]	ViLD + MDETR /GPT-4/MPC（基于LLM的规划）	零样本	无需任务特定训练即可构建LLM+ VLM，用于约束感知运动规划。
传播策略[40]	ResNet-18 / — / U-Net 或 Transformer 扩散网络	自采[SC]	扩散建模能够捕捉多模态动作分布，以实现稳健的视觉运动控制。
OpenVLA [122]	DINOv2 + SigLIP / Prismatic-7B / 符号调谐	OXE + DROID	开源的RT-2类VLA；支持高效的LoRA自适应及广泛的通用性。
π^0 （Pi-Zero）[15]	PaliGemma VLM / PaliGemma（多模态）/ 300M扩散作用模型	Pi-交叉体现	轻量级通用机器人控制器（据报道总重量约3磅），具备强大的跨机器人、开放世界泛化能力及双臂操作技能。
π^0 -快速[171]	PaliGemma VLM / PaliGemma / 带有FAST分词器的自回归变换器	Pi-交叉体现	通过压缩频域动作令牌实现高频实时控制（推断速度最高可提升15倍）。
OpenVLA-OFT [121]	SigLIP + DINOv2（多视角）/ Llama-2 7B / 并行解码 + 行动分块（L1回归）	LIBERO；双手合十礼	采用并行解码与分块动作的微调方案，报告 LIBERO 成功率高达97.1%，且高频双侧手部控制任务的推理速度提升26倍。
RDT-1B [144]	多视角RGB编码器/Transformer语言模块/扩散Transformer（统一动作空间）	46个数据集（>100万次事件）+ aloha微调	1.2B扩散基础模型，适用于具备强语言条件反射和零样本迁移能力的灵巧双手操作。
螺旋 ¹	系统2：多模态推理开源 VLM（7-9Hz）/ 集成语义处理/系统1：Transformer视觉运动策略（200Hz，覆盖上半身全范围）	图 机器人（像素+语言→动作）	E2E 以人形机器人为核心的双速率视觉定位算法（VLA），可实现实时高自由度控制、高灵巧性操作，并支持无需零样本学习即可进行协作式多机器人操控。
CogACT [131]	DINOv2 ViT-L/14 + SigLIP ViT-So400M/14 / Llama-2 通过 Prismatic-7B / DiT-Base（300M扩散）	OXE 子集；Realman与Franka任务	采用扩散作用变压器的组件化VLA系统；与OpenVLA相比，实际应用成功率报告为+59.1%，且对未见过的机器人/物体具有强适应性。
供给链 (CoA) [129]	支持预算感知的视觉编码器/Transformer推理提示/自回归+扩散策略（基于功能位点条件）	LIBERO；真实+模拟操纵	顺序的 a型舞蹈 推理（物体→抓取→空间→运动）可提升空间规划与障碍物规避能力；研究显示其 LIBERO 表现优于OpenVLA。
边缘甚大阵列望远镜（EVLA）[20]	SigLIP + DINOv2/Qwen2（0.5B）/非自回归联合控制预测	桥梁；OXE；120万组文本-图像对	边缘优化型VLA（如Jetson级设备）具有30–50 Hz的推断频率报告值，在低功耗条件下性能与OpenVLA相当。
ShowUI-2B [139]	UI引导视觉标记选择/交错V-L-A流式处理/Transformer GUI动作预测器	256K图形用户界面指令跟踪器	Compact 2B VLA支持数字化自动化操作；具备强大的截图功能、稳定的图形用户界面（GUI）及网页导航功能，并支持高效的令牌选择机制。
GR00T N1 [14]	NVIDIA Eagle-2 VLM / 集成高级规划 / 扩散变压器（DiT）	人类演示 + 机器人轨迹 + 模拟 + 网络视频	通用型人形机器人双系统设计，结合规划与扩散执行机制，实现灵活多步控制及广泛的物理形态泛化能力。
西尔[80]	接地优化视觉骨干网络/Transformer语言模型/自回归动作头	LIBERO	具备强大的视觉操作基础；在 LIBERO 方面具有竞争力，但通常逊色于更新的精细优化版本（例如OpenVLA-OFT）。
DiffusionVLA [240]	变压器视觉编码器/自回归推理/扩散动作头	LIBERO；工厂分拣；零样本拣货	扩散控制可提升鲁棒性和可解释性；在某些配置中，其空间泛化能力较CoA有所减弱。

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Model (Ref.)	Architecture (vision / language / action)	Training data	Key strength / uniqueness
NaVILA [38]	CLIP + CNN / LLaMA-2 / hierarchical control: topological planner + RL locomotion	Real-world legged robot nav demos	Modular hierarchy for terrain generalization; reported 88% real-world navigation success from natural language.
RoboNurse-VLA [132]	SAM2 + RGB-D / LLaMA-2 + voice-to-text / pose regression + gripper classifier	Surgical handover videos + voice prompts	Real-time surgical tool handover robust to novel tools and dynamic operating-room scenes.
Mobility VLA [42]	Long-context ViT + goal-image encoder / T5-based instruction encoder / graph planner + visual goal localization	MINT dataset (VL instruction tours)	Topological mapping from multimodal tours enables navigation generalization across large unseen spaces.
TinyVLA [242]	FastViT / compact 128-d language / diffusion decoder (50M)	Mini-ALOHA + SC tasks	No large-scale pretraining required; reported faster inference (5×) with strong precision under tight compute budgets.
QUAR-VLA [54]	CLIP + proprioceptive embedding / BERT + grounding adapter / transformer full-body decoder	QUART (locomotion+manipulation)	Quadruped-centric VLA with strong sim-to-real transfer and fine-grained instruction alignment.
ChatVLA [295]	Phase-aligned vision encoder / Prismatic MoE LLM / unified V-L-A planner	Unified chat-action (web+robot)	Joint VQA+planning; mitigates forgetting and supports conversational task execution with manipulation.
PointVLA [125]	CLIP + 3D point cloud fusion / LLaMA-2 / transformer with spatial token fusion	Few-shot spatial tasks (real+sim)	Improves long-horizon spatial reasoning by injecting 3D structure while preserving pretrained 2D knowledge.
VLA-Cache [252]	SigLIP + token memory buffer / Prismatic-7B / transformer with dynamic token reuse	ALOHA + sim/real fusion	Caches static visual tokens for efficiency; reported 40–50% faster inference with negligible performance loss.
HybridVLA [142]	CLIP + DINOv2 / LLaMA-2 / hybrid diffusion + autoregressive ensemble	RT-X + synthetic fusion	Dynamic ensemble improves robustness in multi-arm settings and supports stronger sim-to-real generalization.
MoLe-VLA [283]	Multi-stage ViT + STAR router / CogKD-enhanced transformer / sparse transformer (dynamic routing)	RLBench + real-world manipulation	Selective layer activation yields efficiency (reported 5.6× speedup) and higher success (reported +8%).
UAV-VLA [191]	ViT for aerial imagery / GPT instruction parsing / transformer path planner	Satellite + UAV imagery instructions	Zero-shot aerial task planning with scalable language grounding for large unmapped environments.
DexGraspVLA [291]	Object-centric spatial ViT / transformer grasp reasoning / diffusion grasp controller	Dexterous grasping benchmark (sim+real)	Reported >90% zero-shot success across diverse objects; robust to lighting/background variation and unseen conditions.
GraspVLA [46]	Multi-view DINOv2 + SigLIP / VLM predicts boxes+grasps / flow-matching action expert (PAG)	SynGrasp-1B; GRIT	Synthetic-pretrained grasping VLA; improves sim-to-real transfer and supports zero-/few-shot generalization to long-tail object classes and preferences.
Interleave-VLA [62]	InternVL2.5 + OWLv2 / Qwen2.5 / continuous action predictor (OpenVLA+ π^0 -style, diffusion controller)	Open Interleaved X-Embodiment (210k eps., 11 datasets)	End-to-end interleaved image-text instruction following; reported 2–3× out-of-domain gains and zero-shot execution from sketches and novel multimodal prompts.
Long-VLA [64]	End-to-end VLA for long-horizon tasks / phase-aware input masking + transformer policy (subtask phase segmentation)	Long-horizon multi-step robot manipulation demonstrations (task sequences)	Targets long-horizon execution by explicitly separating “moving” vs. “interaction” phases, improving subtask compatibility and robustness over extended task horizons.
RetoVLA [123]	VLM-based policy / reuses register tokens as spatial context for action prediction	Real-robot manipulation on a 7-DoF arm + task-specific demonstrations	Lightweight spatial-reasoning upgrade by repurposing register tokens; reported sizable success-rate gains on complex manipulation with minimal architectural overhead.
Vlaser [256]	VLM-to-VLA pipeline / synergistic embodied reasoning + policy learning (reasoning-aware VLA fine-tuning)	Vlaser-6M embodied reasoning dataset + VLA fine-tuning data (robot demonstrations)	Bridges embodied reasoning and control: strong performance across embodied grounding/QA/planning while improving transfer to policy learning under domain shift.
Discrete Diffusion VLA [138]	Single-transformer VLA / discretized action chunks + discrete diffusion refinement (CE training; remasking)	LIBERO + SimplerEnv (Fractal/Bridge) style VLA benchmarks	Unifies diffusion-style refinement with discrete token interfaces: adaptive decoding order, error correction via remasking, and reduced autoregressive bottlenecks with strong benchmark success rates.
Being-H0 [152]	Dexterous VLA pretrained from human videos / explicit hand-motion modeling + VL grounding for action	Large-scale human manipulation videos (egocentric/third-person) + transfer to robot control	Scales dexterous manipulation by leveraging diverse human video data; improves generalization to novel tasks/scenes compared with small teleop-only robot datasets.
EgoVLA [258]	VLM pretraining on egocentric human manipulation / unified human-robot action space + robot fine-tuning	Large-scale egocentric human videos + small set of robot demonstrations	Uses abundant egocentric videos for scalable pretraining, then aligns embodiments via unified action space to enable practical robot transfer with limited robot data.
StereoVLA [47]	Stereo-enhanced VLA / geometric-semantic fusion from stereo pairs (+ auxiliary depth cues) for action decoding	Stereo robot manipulation demonstrations (stereo RGB with language-conditioned actions)	Explicitly exploits stereo geometry to improve spatial precision (depth-sensitive grasping/manipulation) and robustness to viewpoint/camera variations.
EfficientVLA [262]	Training-free VLA acceleration/compression / structured redundancy removal across VLA pipeline (token/interface optimizations)	Applies to existing VLAs (evaluation on standard manipulation benchmarks; e.g., SIMPLER-style settings)	Practical deployability: speeds up and reduces compute/memory of large VLA policies with minimal accuracy drop, enabling closer-to-real-time inference on constrained hardware.

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型号（参考编号）	架构（愿景/语言/行动）	训练数据	关键优势/独特性
NaVILA [38]	CLIP + CNN / LLaMA-2 / 分层连接 trol: 拓扑规划器 + RL 运动控制	真实世界腿部机器人导航演示	地形泛化的模块化层级结构；报告准确率为88% 基于自然语言的真实世界导航成功率
RoboNurse-VLA [132]	SAM2 + RGB-D / LLaMA-2 + 语音转文本 / 姿态回归+夹爪分类器	手术交接视频+语音 为提示台词	实时手术器械交接系统对新型器械具有鲁棒性 动态手术室场景
移动性VLA[42]	长上下文视觉目标+目标图像编码器/基于T 5的指令编码器/图规划器+视觉目标定位	MINT数据集（VL指令游览）	基于多模态路线的拓扑映射能够实现对大型未见空间的导航泛化。
TinyVLA [242]	FastViT/紧凑型128位语言/扩散解码器（50 M）	Mini-Aloha + SC 任务	无需大规模预训练；据报道，在有限计算资源下可实现更快的推理速度（提升5倍）且精度优异。
QUAR-VLA [54]	CLIP + 本体感觉嵌入 / BERT + 地点适配 器 / 变换器全身解码器	四分之一（原位） （添加/操作）	以四足机器人为核心的VLA架构，具备强大的仿真到真实环境迁移能力及精细的指令对齐机制。
ChatVLA [295]	相位对准视觉编码器/棱镜式MoE LLM/统一视觉- 运动-动作规划器	统一聊天操作（网页端+机器人）	联合 VQA 与规划；可减轻遗忘效应，并通过操作支持会话任务执行。
PointVLA [125]	CLIP + 3D点云融合 / LLaMA-2 / 带空间标 记融合的Transformer模型	少样本空间任务（真实数据+模拟数据）	通过注入三维结构同时保留预训练二维知识，从而提升长时程空间推理能力。
VLA缓存[252]	SigLIP + 令牌内存缓冲区 / Prismatic-7B / 带 动态令牌重用功能的变压器	Aloha + 双模/实融合	缓存静态视觉标记以提升效率；据报道推理速度可提升40%-50%，且性能损失可忽略不计。
HybridVLA [142]	CLIP + DINOv2/LLaMA-2/混合扩散+自回 归集成模型	RT-X + 合成融合	动态集成方法在多臂设置中增强了鲁棒性，并支持更强的模拟到真实场景的泛化能力。
MoLe-VLA [283]	多阶段ViT + STAR路由/CogK增强型变 压器/稀疏变压器（动态路由）	RLBench + 实际操作	选择性层激活可提高效率（报告速度提升5.6倍）并增加成功率（报告成功率提升8%）。
UAV-VLA [191]	航空影像ViT模型 / GPT指令解析 / Transformer路 径规划器	卫星+无人机影像操作指南	面向大规模未测绘环境的零样本航空任务规划与可扩展语言定位方法
DexGraspVLA [291]	基于对象中心的空间视觉表征/Transform- er抓取推理/扩散式抓取控制器	灵巧抓取基准测试（模拟+真实）	据报道，在多种物体上实现了超过90%的零样本成功率；对光照/背景变化及未见条件具有鲁棒性。
GraspVLA [46]	多视角DINOv2+SigLIP/ VLM 预测框+抓 取/流匹配动作专家（PAG）	SynGrasp-1B; GRIT	合成预训练抓取视觉语言表（VLA）；提升模拟到真实场景的迁移能力，并支持对长尾对象类别及偏好进行零样本/少样本泛化。
Interleave-VLA [62]	InternVL2.5 + OWLv2/Qwen2.5/连续动作预 测器（OpenVLA+ π^0 风格，扩散控制器）	开放式交错X-实施例（21万条 ϵ 值，11 个数据集）	采用端到端交错式图像-文本指令序列；实验报告显示可获得2-3倍的域外增益，并能通过草图及新颖的多模态提示实现零样本执行。
长视场阵列望远镜[64]	端到端VLA用于长时程任务 / 相位感知输入掩码 + 变换器策略（子任务：相位 分割）	长视距多步机器人MA- 操纵演示（任务序列）	通过明确区分目标来实现长期执行 “移动”阶段与“交互”阶段的对比，提升了子任务在长期任务周期中的兼容性与稳健性。
RetoVLA [123]	基于 VLM 的策略/重用注册令牌作为动作 预测的空间上下文	基于7自由度机械臂的实操操作及 任务特定演示	通过重新利用寄存器令牌实现轻量级空间推理升级；在复杂操作中报告了显著的成功率提升，且架构开销极小。
Vlaser [256]	VLM 至VLA流程/协同式内化推理+策略学 习（基于推理感知的VLA微调）	Vlaser-6M 集成了推理数据集 + VLA微调数据（机器人演示）	桥梁模型体现了推理与控制能力；在具身化接地/QA/规划任务中表现出强劲性能，同时在领域迁移条件下提升了策略学习的迁移能力。
离散扩散VLA[138]	单变压器VLA/离散动作块+离散扩散细化 （CE训练；重掩膜）	LIBERO + SimplerEnv（分形/桥接）风 格的VLA基准测试	该方法将扩散式精炼技术与离散标记接口相结合：包括自适应解码顺序、通过重掩码实现的错误校正，以及减少自回归瓶颈效应，并在基准测试中展现出优异的成功率。
存在 H0[152]	基于人类视频预训练的Dexterous VLA模型/显式手 部运动建模+动作视觉学习（VL grounding）	大规模人类操作视频（第一人称/第 三人称视角）+向机器人控制系统的 迁移	通过利用多样化的人类视频数据实现灵活尺度操控； 与仅包含小型远距操作机器人数据集相比，可提升对新任务/场景的泛化能力。
EgoVLA [258]	VLM 在以自我为中心的人类操纵数据上的预训练 统一/整合的人机动作空间 机器人微调	大规模的 自我中心的 人类 视频 + 少量机器人恶魔—— 层状结构	提供大量以自我为中心的视频，适用于可扩展的预训练。 随后通过统一动作空间对实施例进行对齐，以实现 利用有限的机器人数据进行实际的机器人转移。
StereoVLA [47]	基于立体视觉增强的VLA/几何-语义融合技 术；通过立体视觉对（+辅助深度线索）实现 动作解码	立体机器人操作演示（采用立体 RGB视觉系统及语言条件化动作）	明确利用立体几何学特性以提升空间精度（深度敏感 抓取/操作）及对视角/相机变化的鲁棒性。
EfficientVLA [262]	无需训练的VLA加速/压缩功能 跨甚大阵列结构冗余消除 流水线（令牌/接口优化）	适用于现有VLAs（评估- 标准操作的延期 基准值；例如，更简洁的设置方式	实际可部署性：加速并降低压缩比 以最低精度复现大型VLA策略的记忆特征 数据下采样技术可在受限硬件环境下实现更接近实时的推理。

3.2. Training Efficiency Advancements in Vision–Language–Action Models

VLA models have seen rapid progress in training and optimization techniques to reconcile multimodal inputs, reduce compute requirements, and enable real-time control. Key areas of advancement include:

- **Data-Efficient Learning.**

- *Co-fine-tuning* on massive vision–language corpora (e.g. LAION-5B) and robotic trajectory collections (e.g. Open X-Embodiment) aligns semantic understanding with motor skills. OpenVLA (7 B parameters) achieves a 16.5 % higher success rate than a 55 B-parameter RT-2 variant, demonstrating that co-fine-tuning yields strong generalization with fewer parameters [194, 227, 122] compared to scaling model size alone.
- *Synthetic Data Generation* via UniSim produces photorealistic scenes including occlusions and dynamic lighting to augment rare edge-case scenarios, improving model robustness in cluttered environments by over 20 % [264, 218].
- *Self-Supervised Pretraining* adopts contrastive objectives (à la CLIP) to learn joint visual–text embeddings before action fine-tuning, reducing reliance on task-specific labels. Qwen2-VL leverages self-supervised alignment to accelerate downstream grasp-and-place convergence by 12 % [177, 98].

- **Parameter-Efficient Adaptation.** Low-Rank Adaptation (LoRA) inserts lightweight adapter matrices into frozen transformer layers, cutting trainable weights by up to 70 % while retaining performance [93]. The Pi-0 Fast variant uses merely 10 M adapter parameters atop a static backbone to deliver continuous 200 Hz control with negligible accuracy loss [171].

- **Inference Acceleration.**

- *Compressed Action Tokens* (FAST) and *parallel decoding* in dual-system frameworks (e.g. GR00T N1) achieve up to 2.5× faster policy inference, reducing per-step latency to below 5 ms compared to standard autoregressive decoding in single-policy VLAs evaluated on real-time manipulation and humanoid control benchmarks. This acceleration comes at a modest cost in trajectory smoothness, manifested as slightly higher action discretization error and reduced fine-grained motion continuity under high-frequency control [14, 209].

Together, these methods have transformed VLAs into practical agents capable of handling language-conditioned, vision-guided tasks in dynamic, real-world settings.

3.3. Parameter-Efficient Methods and Acceleration Techniques in VLA Models

Beyond data-efficient learning strategies, a complementary line of research in VLA models focuses on reducing model size, memory footprint, and inference latency to enable deployment on real robotic platforms with limited computational and power resources. Unlike training-centric efficiency methods discussed earlier, the techniques in this subsection primarily target *parameter efficiency at adaptation time and runtime acceleration during policy inference*.

1. **Parameter-Efficient Adaptation via Low-Rank Modules:** Rather than full fine-tuning, many VLAs adopt parameter-efficient adaptation mechanisms such as Low-Rank Adaptation (LoRA), which was introduced earlier in the context of training efficiency. In this section, we emphasize its role in *reducing the effective parameter footprint during deployment*. For example, OpenVLA employs lightweight LoRA adapters (approximately 20 M parameters) on top of a frozen 7 B-parameter backbone, allowing task adaptation with minimal memory overhead and avoiding duplication of full model weights across tasks [93, 122]. This design enables multiple task-specialized policies to coexist on resource-constrained systems while preserving the general-purpose visual–semantic representations learned during pretraining.
2. **Quantization for Edge Deployment:** Model quantization reduces numerical precision to improve inference throughput and memory efficiency. Experiments with OpenVLA show that INT8 quantization on embedded platforms such as NVIDIA Jetson Orin preserves approximately 97 % of full-precision task success on pick-and-place benchmarks, with only minor degradation in fine-grained dexterous manipulation [194, 122]. Post-training quantization and per-channel calibration further mitigate accuracy loss under high-dynamic-range sensory inputs [164]. These optimizations enable sustained control frequencies of up to 30 Hz within strict power budgets typical of mobile robots.
3. **Model Pruning and Architectural Slimming:** Structured pruning removes redundant architectural components such as attention heads or feed-forward sublayers to reduce memory and compute requirements. Although less explored in VLAs than in standalone vision or language models, early studies on diffusion-based visuomotor policies indicate that pruning up to 20 % of convolutional vision encoders results in negligible degradation in grasp stability [40]. Similar pruning strategies applied to transformer-based VLAs (e.g. RDT-1B) can reduce memory footprint by approximately 25 % with less than 2 % drop in task success, enabling sub-4 GB deployments [144, 131].
4. **Compressed Action Tokenization:** To address inference bottlenecks arising from long-horizon control sequences, compressed action representations have been proposed. FAST reformulates continuous action trajectories into compact frequency-domain tokens, substantially reducing decoding length. The Pi-0 Fast variant achieves

3.2. 视觉-语言-动作模型在训练效率方面的改进

VLA模型在训练与优化技术方面取得了快速进展，这些技术旨在整合多模态输入、降低计算需求并实现实时控制。关键进展领域包括：

• 数据高效学习

- 在大规模视觉-语言语料库（如 LAION -5B）和机器人轨迹数据集（如OpenX-Embodiment）上进行联合微调，可使语义理解与运动技能保持一致。Open-VLA（70 亿参数）的准确率比传统方法高出 16.5%。55 B 参数RT-2变体，表明与仅缩放模型规模相比，协同微调能以更少参数实现更强的泛化能力 [194, 227, 122]。
- 通过UniSim进行合成数据生成可生成包含遮挡和动态光照的逼真场景，以增强罕见边缘场景的处理能力，使模型在复杂环境中的鲁棒性提升超过20% [264, 218]。
- 自监督预训练采用对比目标（类似 CLIP 的方法），在动作微调前学习联合的视觉-文本嵌入，从而降低对任务特定标签的依赖。Qwen2-VL 利用自监督对齐技术，使下游抓取与放置任务的收敛速度提升了 12% [177, 98]。

• **参数高效自适应。**低秩自适应（LoRA）通过在冻结的Transformer层中插入轻量级适配器矩阵，可将训练权重最多削减70%同时保持性能 [93]。Pi-0 Fast变体仅在静态主干网络上使用1000万适配器参数，即可实现持续200Hz的控制频率且精度损失可忽略不计 [171]。

• 推理加速

- **压缩动作令牌（FAST）**与双系统框架中的并行解码技术（如GR00T N1模型）可实现高达2.5 倍的策略推理速度提升，在实时操控与人形机器人控制基准测试中，相较于单策略VLAs中的标准自回归解码方法，每步延迟可降低至5毫秒以下。这种加速效果虽显著，但会带来轨迹平滑度的适度代价——表现为动作离散化误差略有增加，且高频控制下的精细运动连续性有所下降 [14, 209]。

这些方法共同将视觉语言代理（VLAs）转化为实用智能体，使其能够在动态现实场景中处理语言条件化、视觉引导的任务。

3.3. 基大天线阵模型中的参数高效方法与加速技术

除了数据高效的训练策略外，VLA模型领域还有一条互补的研究方向，专注于缩小模型规模、降低内存占用和推理延迟，从而实现在计算资源和能源有限的真实机器人平台上的部署。与前文讨论的以训练效率为中心的方法不同，本小节所述技术主要侧重于适应阶段的参数效率和策略推理过程中的运行加速。

1. **通过低秩模块实现参数高效自适应：**与传统的全量微调不同，许多视觉语言适配器（VLAs）采用了参数高效的自适应机制，例如先前在训练效率领域提出的低秩自适应（LoRA）。本节重点阐述该机制在降低部署阶段实际参数占用量方面的作用。例如，OpenVLA在冻结的70亿参数主干模型基础上，采用轻量级LoRA适配器（约2000万参数），从而以极低的内存开销实现任务自适应，并避免在不同任务间重复存储完整模型权重 [93, 122]。这种设计使得多个任务专用策略能够在资源受限的系统上共存，同时保留预训练过程中习得的通用视觉-语义表征。

2. **边缘部署中的量化技术：**模型量化通过降低数值精度来提升推理吞吐量和内存效率。OpenVLA实验表明，在 NVIDIA Jetson Orin等嵌入式平台上采用INT8量化后，抓取与放置基准测试中全精度任务的成功率仍保持约97%，精细操作能力仅出现轻微下降 [194, 122]。后训练量化与通道级校准技术进一步有效缓解了高动态范围传感输入下的精度损失 [164]。这些优化方案可实现最高达

30 在移动机器人典型的严格功耗预算范围内运行。

模型剪枝与架构精简：Struc

结构化剪枝通过移除注意力头或前馈子层等冗余架构组件，来降低内存和计算需求。尽管在视觉语言模型（VLA）中的应用不如独立视觉或语言模型广泛，但早期基于扩散机制的视觉运动策略研究表明：对卷积视觉编码器进行高达20%的剪枝，对抓取稳定性的影响可忽略不计 [40]。将类似剪枝策略应用于基于Transformer的VLA（例如 RDT -1B）后，内存占用可降低约25%，任务成功率下降幅度不足2%，从而实现低于4GB的部署规模 [144, 131]。

4. **压缩动作分词技术：**为解决长时域控制序列引发的推理瓶颈问题，学界提出了压缩动作表示方法。FAST算法将连续动作轨迹重构为紧凑的频域分词，显著缩短了解码所需长度。其改进版本Pi-0 Fast进一步提升了性能表现。

up to $15\times$ faster inference by compressing 1000 ms action windows into 16 discrete tokens, enabling control rates of up to 200 Hz on desktop GPUs [171]. This method trades minimal trajectory granularity for large speedups, making it well suited for high-frequency, reactive manipulation tasks.

5. **Parallel Decoding and Action Chunking:** Standard autoregressive VLAs decode actions sequentially, leading to cumulative latency. Parallel decoding strategies, as employed in dual-system architectures such as GROOT N1, generate groups of spatio-temporal action tokens concurrently, achieving approximately $2.5\times$ reduction in end-to-end inference latency on 7-DoF robotic arms operating at 100 Hz [14, 209]. Action chunking further abstracts multi-step routines (e.g. *pick-and-place*) into single high-level tokens, reducing inference steps by up to 40 % in long-horizon manipulation tasks such as kitchen workflows [112].
6. **Hardware-Aware Compilation and Runtime Optimization:** Finally, hardware-aware optimizations leverage compiler-level graph rewrites, kernel fusion, and accelerator-specific primitives to maximize throughput. Frameworks such as TensorRT-LLM exploit tensor cores, fused attention kernels, and pipelined memory transfers to accelerate both transformer inference and diffusion sampling. In OpenVLA-OFT, such optimizations reduce inference latency by approximately 30 % and lower energy consumption per inference by 25 % on RTX-class GPUs compared to standard PyTorch execution [121]. These system-level optimizations are critical for deploying real-time VLAs on mobile robots, aerial platforms, and humanoid systems with strict power constraints.

Discussion: Parameter-efficient adaptation and inference acceleration techniques collectively democratize VLA deployment:

- LoRA and quantization empower smaller labs and research/development programs to fine-tune and operate billion-parameter VLAs on consumer-grade hardware, unlocking cutting-edge semantic understanding for robots [93, 122].
- Pruning and FAST tokenization compress model and action representations, enabling sub-4 GB, sub-5 ms control loops without sacrificing precision in dexterous tasks [144, 171].
- Parallel decoding and action chunking overcome sequential bottlenecks of autoregressive policies, supporting 100–200 Hz decision rates needed for agile manipulation and legged locomotion [14, 209].
- Hybrid RL-SL training stabilizes exploration in complex environments, while hardware-aware compilation ensures real-time performance on edge accelerators [159, 121].

Together, these advances make it practical to embed VLA models across industrial manipulators, assistive drones, and

consumer robots, bridging the gap from research prototypes to real-world autonomy.

3.4. Applications of Vision-Language-Action Models

VLA models are rapidly emerging as foundational building blocks for embodied intelligence, integrating perception, natural language understanding, and motor control within a unified architecture. By encoding visual and linguistic modalities into shared semantic spaces and generating contextually grounded actions, VLA models enable seamless interaction between agents and their environments [131, 293]. By encoding visual and linguistic modalities into shared semantic spaces and generating contextually grounded actions, VLA models enable seamless interaction between agents and their environment. This multimodal capacity has positioned VLAs as transformative agents across a wide spectrum of real-world applications. In humanoid robotics, systems like Helix and RoboNurse-VLA combine vision, language, and dexterous manipulation to assist with domestic tasks and surgical operations, demonstrating real-time reasoning and safety-aware control [132, 242]. In autonomous vehicles, models such as OpenDriveVLA and ORION process dynamic visual streams and natural language instructions to make transparent, adaptive driving decisions in complex urban environments [69, 293]. Industrial deployments leverage VLA architectures for high-precision assembly, inspection, and collaborative manufacturing [131]. In agriculture, VLA-powered robotic systems could enable vision-guided fruit harvesting, plant monitoring, and anomaly detection, reducing labor dependency and increasing sustainability. Furthermore, recent advances in interactive augmented reality systems utilize VLA models for real-time, language-conditioned spatial navigation, guiding users in indoor and outdoor settings based on voice or visual cues [191, 74]. Across these domains, VLAs offer a unified framework for robust, adaptable, and semantically aligned task execution, marking a pivotal shift toward embodied generalist agents.

Table 3 shows the recent VLA models by summarizing their methodologies, application domains, and key innovations.

The following subsections chronologically explore the application areas in depth as shown in Figure 11.

3.4.1. Humanoid Robotics

Humanoid robots, designed to mimic the form and functionality of the human body, represent one of the most demanding yet impactful domains for the deployment of VLA models. These platforms must seamlessly perceive complex environments, understand spoken or written natural language, and perform intricate physical tasks with human-level dexterity [184, 23]. The core strength of VLA models lies in their ability to unify perception, cognition, and control into a single, end-to-end trainable framework allowing humanoid robots to interpret visual inputs (e.g., RGB-D imagery of cluttered scenes), comprehend linguistic instructions (e.g., “place the spoon in the drawer”), and generate precise motor trajectories [151, 296].

通过将1000毫秒的动作窗口压缩为16个离散标记，推理速度可提升高达15倍，从而在桌面GPU上实现高达200 Hz的控制频率[171]。该方法以极低的轨迹粒度为代价换取显著的加速效果，因此特别适用于高频、实时响应的操作任务。

5. **并行解码与动作分块：**标准自回归VLAs会按顺序解码动作，导致累积延迟。如GR00T N1等双系统架构中采用的并行解码策略，可同时生成时空动作标记组，使运行频率为100Hz的7自由度机械臂端到端推理延迟降低约2.5倍[14, 209]。动作分块技术将多步骤操作流程（如取放操作）抽象为单一高层标记，在厨房工作流等长时域操作任务中可使推理步骤减少高达40%[112]。
6. **硬件感知编译与运行时优化：**最后，硬件感知优化通过编译器级别的图重构、内核融合以及加速器专用原语来最大化吞吐量。例如 TensorRT-LLM 等框架利用张量核心、融合注意力核和流水线化内存传输技术，显著提升了Transformer推理和扩散采样的性能。在OpenVLA-OFT 中，与标准 PyTorch 运行方式相比，此类优化使 RTX 类 GPU 的推理延迟降低约 30%，单次推理能耗降低 25%[121]。这些系统级优化对于在移动机器人、航空平台及对功耗有严格限制的人形系统上部署实时 VLA 至关重要。

讨论：参数高效的自适应技术和推理加速技术共同推动了VLA的普及：

- LoRA与量化技术使小型实验室及研发项目能够在消费级硬件上对数十亿参数的VLAs进行精细调校与运行，从而为机器人解锁前沿的语义理解能力[93, 122]。
- 剪枝与FAST分词技术可压缩模型及动作表征，实现控制循环内存占用低于4GB、处理时间低于5毫秒，同时在精细操作任务中不牺牲精度[144, 171]。
- 并行解码与动作分块技术克服了自回归策略的顺序性瓶颈，能够支持敏捷操作和腿部运动所需的100–200 Hz决策频率[14, 209]。
- 混合RL-SL训练技术可稳定复杂环境中的探索行为，而硬件感知编译技术则确保边缘加速器上的实时性能[159, 121]。

这些进展共同使得将VLA模型嵌入工业机械臂、辅助无人机及消费级机器人成为现实，从而弥合了从研究原型到

实际自主应用之间的差距。

3.4. 视觉-语言-动作模型的应用

视觉语言模型（VLA）正迅速崛起为具身智能的基础构建模块，通过统一架构整合感知、自然语言理解与运动控制功能。该模型通过将视觉与语言模态编码至共享语义空间，并生成基于上下文的动作方案，实现了智能体与环境之间的无缝交互[131, 293]。这种多模态能力使VLA成为广泛现实应用场景中的变革性智能体。在人形机器人领域，Helix和RoboNurse-VLA等系统通过融合视觉感知、语言理解与灵巧操作技术，既可辅助家务任务又可执行外科手术，展现出实时推理能力和安全感知控制能力[132, 242]。在自动驾驶领域，OpenDriveVLA和Orion等模型通过处理动态视觉流与自然语言指令，能够在复杂城市环境中做出透明化、自适应的驾驶决策[69, 293]。工业应用中，VLA架构被广泛用于高精度装配、检测及协同制造场景[131]。在农业领域，基于VLA的机器人系统可实现视觉引导的水果采摘、植物监测及异常检测，有效降低人力依赖度并提升可持续性。此外，交互式增强现实系统近期取得的进展，通过实时语言条件空间导航技术，利用VLA模型根据语音或视觉线索为用户在室内外场景提供引导[191, 74]。这些应用场景中，VLA为稳健、灵活且语义对齐的任务执行提供了统一框架，标志着向具身化通用智能体发展的关键转型。

表3通过总结其方法学、应用领域及关键创新点，展示了近期VLA模型。

以下各小节按时间顺序深入探讨了应用领域，具体如图11所示。

3.4.1. 人形机器人技术

仿人机器人旨在模拟人体的形态与功能，是部署VLA模型最具挑战性且影响深远的应用领域之一。这类平台必须能够无缝感知复杂环境、理解口语或书面自然语言，并以人类级别的灵巧度执行复杂的物理任务[184, 23]。VLA模型的核心优势在于其能够将感知、认知与控制整合为一个端到端的可训练框架，使仿人机器人能够解析视觉输入（例如杂乱场景的RGB-D图像）、理解语言指令（例如“将勺子放入抽屉”），并生成精确的运动轨迹[151, 296]。

Table 3: Comparison of representative Vision-Language-Action (VLA) methodologies, application areas, and key innovations.

Reference (Year)	VLA methodology	Application area	Strength / key innovation
CogACT [131] (2024)	Modular VLA with a specialized action module based on diffusion transformers.	Industrial robotics; language-guided manipulation.	Robust action modeling with fast adaptation and strong generalization, improving task success in diverse industrial settings.
VLATest [237] (2024)	Automated large-scale testing framework for VLA manipulation evaluation.	Robotic manipulation benchmarking (robustness/reliability).	Systematic multi-scene and multi-task evaluation that exposes robustness gaps and supports targeted VLA improvements.
NaVILA [38] (2024)	Two-level VLA: high-level vision-language generates mid-level navigation commands; RL locomotion executes.	Legged navigation from natural language in cluttered real-world scenes.	Modular high/low-level split with strong generalization and high real-world success across challenging terrains.
RoboNurse-VLA [132] (2024)	Vision module (SAM2) + language module (Llama2) with real-time voice-to-action pipeline.	Surgical assistance (instrument grasp and handover).	Accurate real-time assistance, robustness to unseen tools, and stable performance in dynamic operating room conditions.
Mobility VLA [42] (2024)	Hierarchical VLA with long-context VLM for goal localization and topological graph navigation.	Multimodal instruction navigation with demonstration tours (MINT).	Leverages demonstrations for scalable navigation in large spaces; robust to novel language+image queries.
CoVLA [5] (2025)	CLIP-based vision, Llama-2 language, trajectory prediction for action.	Autonomous driving (dataset + VLA training).	Large-scale richly annotated dataset enabling interpretable scene understanding and robust path planning.
OpenDriveVLA [293] (2025)	Hierarchical alignment of 2D/3D visual tokens and language embeddings; autoregressive agent-environment-ego modeling.	End-to-end autonomous driving.	Unified semantic space and interaction modeling that improves planning and QA performance in complex traffic scenes.
ORION [69] (2025)	QT-Former for history context, LLM reasoning, generative planner for trajectory prediction.	Holistic end-to-end autonomous driving.	Aligns reasoning and action spaces with unified optimization for VQA and planning, yielding stronger closed-loop performance.
QUAR-VLA [54] (2025)	QUART-based fusion of vision and language for action generation.	Quadruped navigation, manipulation, whole-body tasks.	Tight VL-A coupling and instruction alignment with strong sim-to-real generalization for legged platforms.
TinyVLA [242] (2025)	Compact multimodal backbone with diffusion-policy decoder.	Fast, data-efficient manipulation control.	Reduced inference cost with strong generalization; improves efficiency without requiring heavy pre-training.
UAV-VLA [191] (2025)	Modular pipeline: GPT for goal extraction, VLM for object search, GPT for action generation.	UAV mission planning from language + satellite imagery.	Efficient flight/action planning with minimal prior training; supports intuitive human-UAV interaction and benchmarking.
Bi-VLA [74] (2025)	Multimodal transformer linking vision, language, and bimanual action modules.	Bimanual household manipulation.	High adaptability and robust real-world bimanual execution through explicit action-module integration.
ChatVLA [295] (2025)	Phased alignment training with Mixture-of-Experts for VL-A integration.	Unified multimodal understanding and robot control.	Reduces forgetting/interference, improving both VQA and manipulation with efficient specialization.
RoboMamba [143] (2025)	Mamba/SSM-based VLA with co-trained vision encoder and SE(3) action modeling.	Efficient robotic reasoning and manipulation.	Linear-complexity inference and minimal fine-tuning enable fast pose-aware reasoning and efficient control.
OTTER [97] (2025)	Text-aware feature extraction using frozen pre-trained VLMs.	Manipulation with zero-shot generalization.	Preserves VL semantic alignment without VLM fine-tuning via task-relevant feature selection for strong zero-shot transfer.
PointVLA [125] (2025)	Injects 3D point-cloud features into frozen VLA via modular skip-blocks.	Spatial reasoning; few-shot and long-horizon manipulation.	Adds 3D geometry while preserving 2D knowledge, improving spatial grounding without full retraining.
VLA-Cache [252] (2025)	Adaptive token caching with selective reuse of static visual tokens.	Real-time efficient manipulation inference.	Layer-wise token reuse yields major speedups with minimal accuracy loss, practical for on-robot deployment.
CombatVLA [34] (2025)	Video-action AoT training with truncated AoT for fast inference.	Real-time combat decision-making in 3D games.	Large inference acceleration with improved tactical reasoning and high success in real-time interactive environments.
HybridVLA [142] (2025)	Unified LLM with collaborative diffusion and autoregressive action policies.	Single-/dual-arm manipulation across sim and real tasks.	Adaptive action ensembling improves robustness and generalization on complex manipulations.
NORA [103] (2025)	3B-parameter VLA using Qwen-2.5-VL-3B backbone with FAST+ tokenizer.	Generalist embodied robotics (sim + real).	Low overhead with strong visual reasoning and fast action decoding; competitive with larger VLAs.
SpatialVLA [175] (2025)	Ego3D position encoding and adaptive action grids for spatially-aware VLA.	Cross-robot, multi-task, zero-shot manipulation.	Explicit 3D spatial integration and adaptive discretization improve transfer and generalization; open-sourced.
MoLe-VLA [283] (2025)	Mixture-of-Layers with dynamic layer-skipping via router and distillation.	Efficient manipulation on RL-Bench + real robots.	Selective layer activation provides strong speedups while maintaining cognition and improving success.
JARVIS-VLA [130] (2025)	Post-trained large VLMs with VL guidance and an action head for keyboard/mouse control.	Open-world visual games (e.g., Minecraft), 1k+ tasks.	Self-supervised post-training yields strong generalization and world knowledge; open-sourced task suite and models.
UP-VLA [279] (2025)	Unified VLA with joint multimodal understanding and future prediction objectives.	Embodied manipulation with precise spatial reasoning.	Joint semantics+dynamics learning improves performance on long-horizon tasks requiring fine spatial control.
Shake-VLA [120] (2025)	Modular stack with vision, speech-to-text, RAG, anomaly detection, and bimanual arms.	Bimanual cocktail mixing in clutter/noise.	Robust end-to-end deployment with reliable ingredient handling, recipe adaptation, and real-world task completion.
MoRE [285] (2025)	Sparse MoE with LoRA modules and RL-based Q-function training.	Quadruped multi-task locomotion, navigation, manipulation.	Scalable fine-tuning on mixed-quality data with strong multi-task and OOD generalization in sim and real.
DexGraspVLA [291] (2025)	Hierarchical planner (pre-trained VL) + diffusion low-level controller.	General dexterous grasping in diverse conditions.	Domain-invariant representations support strong zero-shot generalization across objects, lighting, and backgrounds.
DexVLA [241] (2025)	Plug-in diffusion action expert with embodiment curriculum learning.	General robot control: single-arm, bimanual, dexterous hand.	Cross-embodiment action modeling and fast adaptation enable strong long-horizon performance without task-specific tuning.

表3：代表性视觉-语言-动作（VLA）方法学、应用领域及关键创新点的比较。

参考文献（年份）	VLA方法学	应用区域	优势/关键创新
CogACT [131] (2024)	基于扩散变压器的专用作用模块的模块化VLA系统	工业机器人；语言引导式操作	具有快速适应性和强泛化能力的鲁棒动作建模，可提升多样化工业场景中的任务成功率。
VLA-Test [237] (2024)	适用于 VLA 的自动化大规模测试框架 操作评估	机器人操作基准测试 ing（稳健性/可靠性）	系统性多场景多任务评估方法，可揭示ro- 填补差距并支持有针对性的VLA改进措施。
NaVILA [38] (2024)	两级视觉语言模型（VLA）：高级视觉语言生成 中层导航指令；强化学习运动系统执行指令。	在杂乱的现实场景中，基于自然语言的腿部导航。	模块化高低层级分割系统，具有强大的泛化能力，并在复杂地形环境中展现出卓越的实际应用效果。
RoboNurse-VLA [132] (2024)	视觉模块（SAM2）+语言模块（Llama2）集成 实时语音转动作流程。	手术协助（器械抓取与交接）。	精准的实时辅助功能、对未见工具的鲁棒性以及动态手术室环境中的稳定性能表现。
移动性VLA[42] (2024)	支持目标长上下文 VLM 的层次化VLA 定位与拓扑图导航。	多模态指令导航 通过示范性参观活动（MINT）。	利用演示技术实现大型空间中的可扩展导航；ro- 转换为新颖的语言和图像查询。
CoVLA [5] (2025)	基于CLIP的视觉理解、Llama-2语言模型、动作 轨迹预测	自动驾驶（数据集 + VLA 训练）。	一个大规模、富含注释的数据集，支持可解释的场景理解与稳健的路径规划。
OpenDriveVLA [293] (2025)	二维/三维视觉标记的层次化对齐 以及语言嵌入；自回归代理-环境-自我建模。	端到端自动驾驶。	统一语义空间与交互建模技术，可提升计划- 复杂交通场景中的制动与质量保证（QA）性能表现。
猎户座[69](2025)	QT-Former用于历史上下文分析、LLM推理及轨迹预测 生成规划。	整体式端到端自动驾驶	通过将推理与行动空间与统一优化相结合，实现 VQA 与规划的协同优化，从而提升闭环控制性能。
QUAR-VLA [54] (2025)	基于象限的视觉与语言融合 动作生成。	四足动物导航与操控 训练，全身性任务。	VL-A耦合紧密，指令对齐精确，且具有强大的模拟到数字转换能力。 带腿平台的真实泛化问题
TinyVLA [242] (2025)	具有扩散策略解码器的紧凑型多模态骨干网络	快速、数据高效的操纵控制	推断成本降低且具有强泛化能力；无需大量预训练即可提升效率。
UAV-VLA [191] (2025)	模块化流程：采用GPT进行目标提取，VLM 用于对象搜索，GPT用于动作生成。	从LAN进行的UAV任务规划 仪表+卫星影像	高效飞行/行动规划，仅需最少前期训练；支持 直观的人机交互与无人机能基准测试
Bi-VLA [74] (2025)	多模态转换器连接视觉、语言与双侧动作模块	双手家庭操作	通过明确的动作模块整合，具备高度适应性及稳健的现实世界 双手操作能力。
ChatVLA [295] (2025)	基于专家混合模型的分阶段对齐训练方法 用于VL-A整合	统一的多模态理解 以及机器人控制。	可减少遗忘/干扰，同时提升 VQA 与操作性能。 实现高效的专业化协作。
RoboMamba [143] (2025)	基于Mamba/SSM的视觉辅助激光雷达系统（VLA）与协同训练视觉模块 编码器与SE(3)动作建模。	高效的机器人推理与 操作	线性复杂度推断与最小微调技术可实现快速 姿态感知推理与高效控制
Otter[97]（2025）	利用冻结预训练的VLM进行文本感知特征提取。	采用零样本泛化进行操作。	通过任务相关特征选择实现强零样本迁移能力，无需 VLM 微调即可保持VL语义对齐。
PointVLA [125] (2025)	将三维点云特征注入冻结的VLA中 通过模块化跳块实现。	空间推理：少样本学习与 远期操纵	在保留二维知识的同时增加三维几何结构，从而提升空间感知能力 无需完全再培训即可实施的接地操作
VLA缓存[252] (2025)	具有选择性重用的自适应令牌缓存 静态视觉标记	实时高效操作 推理	逐层令牌复用技术可在最小精度损失下实现显著加速效果 快速定位能力丧失，适用于机器人现场部署。
战斗VLA [34] (2025)	采用截断AoT的视频动作训练 快速推理	即时处理的战斗决定- 在3D游戏中进行制作。	通过改进战术推理实现的大规模推理加速 在实时交互环境中具有高度成功率。
混合型VLA [142] (2025)	具有协同扩散与au-的统一大语言模型 回退式行动策略。	单臂/双臂操作 在模拟任务与实际任务中均表现一致。	自适应动作集成技术提升鲁棒性与泛化能力 关于复杂操作
NORA [103] (2025)	采用Qwen-2.5-VL-3B模型架构及FAST+分词器 的3B参数VLA模型。	通用型机器人体现了仿生学与现实主义的结合。	具有低开销、强大的视觉推理能力及快速动作解码能力；与大型视觉语言助理（VLA）具有竞争力。
空间基大阵列望远镜 (2025)	Ego3D位置编码与自适应动作网格 适用于具有空间感知能力的基大天线阵（VLA）。	跨机器人、多任务、零样本 操作	显式三维空间积分与自适应离散化技术的改进 迁移与泛化；开源。
MoLe-VLA [283] (2025)	具有动态层跳过的多层混合结构 通过路由器及蒸馏过程实现。	对RL的高效操作 实验台+真实机器人	选择性层激活可在保持原有性能的同时实现显著加速效果 认知功能与成功改善
Jarvis-VLA[130](2025)	经过训练后的大规模视觉语言模型（VLM），配备视觉 语言引导功能及用于键盘/鼠标控制的动作头。	开放世界视觉游戏（如《我的世界》），1000+个任务。	自监督后训练可实现强大的泛化能力和全局知识；提供开源任务 套件及模型。
UP-VLA[279]（2025）	基于联合多模态理解的统一VLA 以及未来的预测目标。	采用预处理技术进行的实体化操作..... 空间推理能力	联合语义与动态学习可提升长序列任务的性能 需要精细空间控制的水平面任务。
Shake-VLA [120] (2025)	具备视觉、语音转文本及RAG功能的模块化堆栈 异常检测与双侧上肢功能评估	双手鸡尾酒调制在clut-中 通气/噪声比	具备可靠的原料处理功能，实现稳健的端到端部署。 配方调整与实际任务完成。
MoRE[285]（2025）	基于LoRA模块与强化学习的稀疏模型增强器（MoE）及Q- 功能训练	四足多任务运动能力 导航、操控	基于强多任务能力的混合质量数据可扩展微调 以及在仿真环境与真实环境中的OOD泛化能力。
DexGraspVLA [291] (2025)	分层规划器（预训练视觉语言模型）+ 扩散机制 低级控制器	双侧手部抓握能力普遍良好 反向条件	域不变表示支持强零样本泛化能力 对象、光照与背景之间的跨元素化处理
DexVLA [241] (2025)	具有课程学习体现的插件扩散行为专家	通用机器人控制：单臂、双手、灵巧手。	跨实体动作建模与快速适应机制可在无需任务特定调优的情况下 实现强大的长期性能表现。

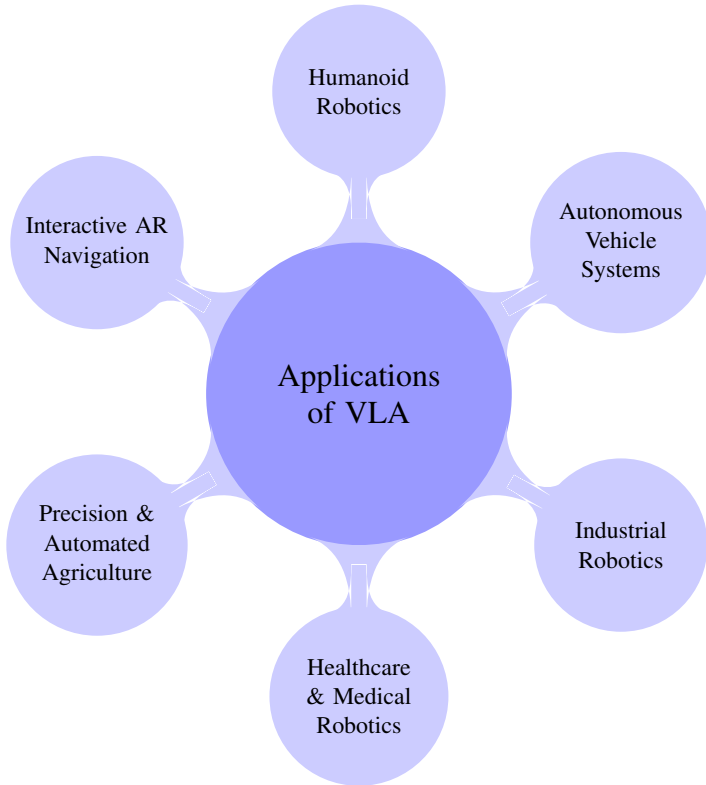


Figure 11: Mind-map of application domains for Vision–Language–Action models, with Humanoid Robotics positioned at the top and remaining domains arranged clockwise to match the order of discussion in this section.

Recent advances have significantly accelerated the deployment of VLAs in humanoid robotics. For example, Helix², a humanoid robot developed by Figure AI, leverages a fully integrated VLA model to perform full-body manipulation at high frequency, controlling arms, hands, torso, and even fine-grained finger motion in real time. The architecture follows a dual-system design: a multimodal transformer processes inputs such as language commands and vision streams, while a real-time motor policy outputs dense action vectors at 200 Hz. This allows Helix to generalize across previously unseen objects and tasks, adapting fluidly to changing environments without the need for task-specific retraining.

The key advantage of VLAs in humanoid systems is their ability to scale across diverse tasks using shared representations [8]. Unlike traditional robotic systems that rely on task-specific programming or modular pipelines, VLA-powered humanoids operate under a unified token-based framework. Vision inputs are encoded via pretrained vision-language models like DINOv2 or SigLIP, while instructions are processed using LLMs such as LLaMA or GPT-style encoders. These representations are fused into prefix tokens that capture the full context of the scene and task. Action tokens are then generated autoregressively, similar to language decoding, but represent motor commands for the robot’s joints and end-effectors.

²<https://www.figure.ai/news/helix>

This capability enables humanoid robots to operate effectively in human-centric spaces, such as households, hospitals, and retail environments. In domestic settings, VLA-powered robots can clean surfaces, prepare simple meals, or organize objects simply by interpreting voice commands [151, 296]. In healthcare, systems like RoboNurse-VLA [132] have demonstrated the ability to perform precise instrument handovers to surgeons using real-time voice and visual cues. In retail, humanoid platforms equipped with VLAs can assist with customer queries, restock shelves, and navigate store layouts without explicit pre-programming [8].

What distinguishes modern humanoid VLAs is their ability to run on embedded, low-power hardware, making real-world deployment viable. For instance, systems such as TinyVLA [242] and MoManipVLA [246] demonstrate efficient inference pipelines that run on Jetson-class GPUs, enabling mobile deployment without compromising performance. These models exploit techniques like diffusion-based policies, LoRA-based fine-tuning, and dynamic token caching to minimize compute cost while retaining high precision and generalization.

In logistics and manufacturing, VLA-enabled humanoids are already making a commercial impact. Robots like Figure 01 are deployed in warehouses to perform repetitive, physically intensive tasks such as picking, sorting, and shelving alongside human workers. Their ability to handle novel object categories and dynamically changing scenes is powered by continual learning and robust multimodal grounding [252, 131].

As VLA models continue to advance in their capacity for diverse action generation, spatial reasoning, and real-time adaptation, humanoid robots are emerging as highly capable assistants across homes, industrial settings, and public spaces. Their strength lies in their ability to unify perception, language comprehension, and motor control through a shared token-based architecture enabling seamless, context-aware behavior in unstructured human environments.

For example, as depicted in the figure 12, consider ‘Helix’, a state-of-the-art humanoid robot equipped with a next-generation VLA model. When instructed verbally, “Please take the water bottle from the fridge,” Helix activates its integrated perception system, where a foundation vision-language model (e.g., SigLIP or DINOv2) segments the visual scene to identify the refrigerator, its handle, and the bottle. The language input is processed by an LLM such as LLaMA-4, which tokenizes the instruction and fuses it with the visual context. This fused representation is passed to a hierarchical controller: the high-level policy plans the task sequence (locate handle, pull door, identify bottle, grasp), while a mid-level planner defines motor primitives, such as grasp type and joint trajectories. The low-level VLA controller, often based on diffusion policy networks, executes these actions with sub-second latency. Upon encountering variations (e.g., a tilted bottle or slippery grip), Helix’s agentic AI module performs micro-policy refinement in real time, adjusting its grip based on feedback. This example illustrates the transformative potential of VLA-enabled humanoid. From kitchens to clinics, these systems not only interpret complex instructions and execute physical tasks with dexterity but also adapt to environmental unpredictability. By em-

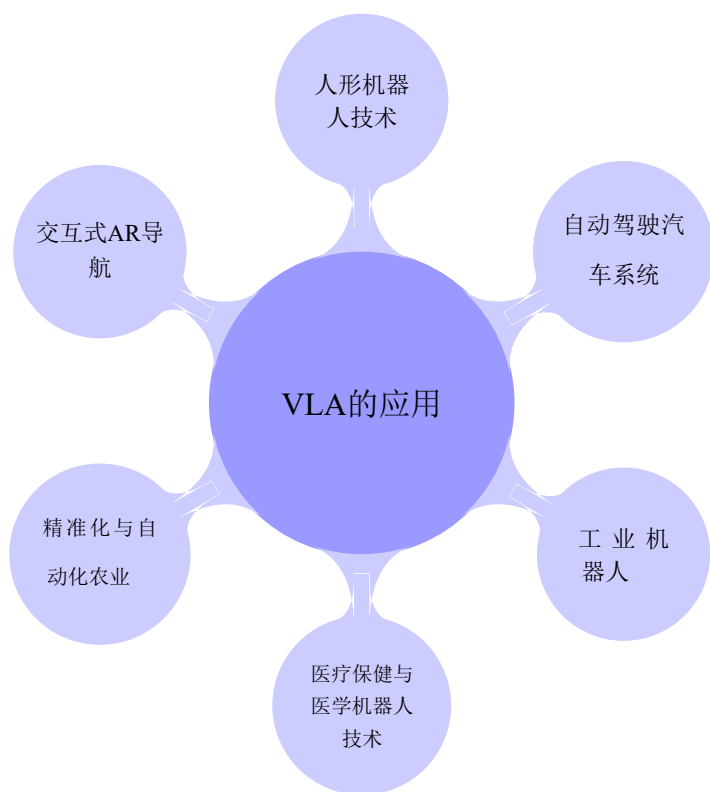


图11：视觉-语言-动作模型应用领域的思维导图，其中类人机器人技术位于顶部，其余领域按顺时针方向排列以匹配本节讨论顺序。

近年来，虚拟肢体辅助系统（VLAs）在人形机器人领域的应用取得了显著进展。以Figure AI研发的Helix²人形机器人为例，该系统采用全集成式VLAs模型实现高频次全身操控，能够实时控制手臂、手部、躯干乃至精细的手指动作。其架构采用双系统设计：多模态变换器负责处理语言指令和视觉流等输入信号，实时电机策略则以200赫兹频率输出密集的动作向量。这种设计使Helix具备跨场景泛化能力，无需针对特定任务进行重新训练，即可灵活适应环境变化，轻松应对未知物体与任务挑战。

视觉语言模型（VLAs）在人形机器人系统中的核心优势在于，它们能够通过共享表征实现跨任务场景的灵活扩展[8]。与依赖任务特定编程或模块化流程的传统机器人系统不同，基于VLAs的人形机器人采用统一的基于标记的框架运作。视觉输入通过DINOv2或SigLIP等预训练视觉语言模型进行编码，指令处理则采用LLAMA或GPT风格编码器等大语言模型。这些表征被融合成能够捕捉场景与任务完整上下文的前缀标记，随后通过自回归机制生成动作标记——其工作原理类似于语言解码过程，但实际代表的是机器人关节与末端执行器的运动指令。

这种技术能力使类人机器人能够在以人类为中心的空间环境中高效运作，例如家庭住宅、医院和零售场所。在家庭场景中，搭载视觉语言助手（VLA）的机器人仅需解读语音指令，即可完成表面清洁、简易餐食准备或物品整理等任务[151, 296]。在医疗领域，RoboNurse-VLA系统[132]已展现出通过实时语音与视觉提示实现精准器械交接给外科医生的能力。在零售行业，配备VLA的人形机器人平台无需预先编程即可协助处理顾客咨询、补货货架及导航门店布局[8]。

现代人形VLAs的独特之处在于其能够在嵌入式低功耗硬件上运行，从而实现实际场景中的部署。例如，TinyVLA[242]和MoManipVLA[246]等系统展示了高效的推理流程，这些流程可在Jetson级GPU上运行，从而实现移动部署而不影响性能。这些模型采用了基于扩散的策略、基于LoRA的微调以及动态令牌缓存等技术，在保持高精度和泛化能力的同时最大限度降低了计算成本。

在物流和制造领域，配备VLA技术的人形机器人已开始产生显著的商业影响。如图01所示的机器人被部署在仓库中，与人类工人协同执行拣选、分拣和上架等重复性、体力要求较高的任务。它们能够处理新型物体类别及动态变化的场景，这得益于持续学习能力和强大的多模态定位技术[252, 131]。

随着视觉语言模型（VLA）在多样化动作生成、空间推理及实时适应能力方面持续提升，人形机器人正逐渐成为家庭、工业场景及公共空间中高效能的智能助手。其核心优势在于通过基于共享标记的架构，将感知、语言理解与运动控制有机整合，从而在非结构化人类环境中实现无缝衔接且具备情境感知能力的行为表现。

例如，如图12所示，以配备新一代视觉语言模型（VLA）的尖端人形机器人“Helix”为例。当通过语音指令“请从冰箱取出水瓶”时，Helix会激活其集成感知系统：基础视觉语言模型（如SigLIP或DINOv2）会对视觉场景进行分割，识别冰箱、把手及水瓶；语言输入则由LLM（如LLaMA-4）处理，通过分词技术将指令与视觉上下文融合；该融合表征随后传递至分层控制器——高层策略负责规划任务序列（定位把手、拉开门板、识别水瓶、抓取），中层规划器定义运动原语（如抓取类型与关节轨迹），底层VLA控制器（通常基于扩散策略网络）以亚秒级延迟执行动作。当遇到瓶体倾斜或抓握滑动等变体情况时，Helix的智能AI模块会实时进行微策略优化，根据反馈调整抓握方式。该案例充分展现了VLA赋能人形机器人所具有的变革潜力。从厨房到诊所，这些系统不仅能解读复杂指令并以灵巧性执行物理任务，还能适应环境的不可预测性。通过em-

²<https://www.figure.ai/news/helix>



Figure 12: This figure illustrates “Helix,” a next-generation humanoid robot executing a household task using a VLA framework. Upon receiving a verbal command, Helix integrates a vision-language model (e.g., SigLIP) and a language model (e.g., LLaMA-4) to jointly perceive and interpret the environment. A hierarchical VLA controller plans and executes sub-tasks opening the fridge, grasping a bottle while an agentic AI module adapts actions in real time. This demonstrates VLA-based generalist robotics with dynamic task adaptation and safe, semantically grounded manipulation.

bedding agentic reasoning and safety alignment mechanisms, modern humanoid robots powered by VLAs are transitioning from narrow-task performers to generalist, trustworthy collaborators. As energy-efficient models like TinyVLA and MoManipVLA mature, deployment on mobile, low-power platforms becomes increasingly practical ushering in a new era of embodied, socially aligned AI.

3.4.2. Autonomous Vehicle Systems

Autonomous vehicles (AVs), including self-driving cars, trucks, and aerial drones, represent a frontier application domain for VLA models, where safety-critical decision-making demands tightly coupled perception, semantic understanding, and real-time action generation. Unlike traditional modular AV pipelines that explicitly separate perception, planning, and control, VLA frameworks explore tighter architectural coupling by jointly processing visual observations, high-level semantic cues, and internal state representations within a unified model. While such end-to-end formulations have shown promising results in simulation and controlled benchmarks for instruction-conditioned navigation and reasoning, large-scale commercial systems (e.g., Tesla Autopilot (Source Link)) continue to rely on modular or hybrid pipelines, with recent industry efforts focusing on integrating vision-language reasoning components rather than fully deploying VLA-style action generation in safety-critical driving.

VLA models empower AVs to comprehend complex environments beyond pixel-level object recognition. For instance, a self-driving car navigating an urban setting must detect traffic signs, understand pedestrian behavior, and interpret navigation commands such as “take the second right after the gas station.” These tasks involve fusing visual and linguistic signals to understand spatial relationships, predict intent, and gen-

erate context-aware driving actions. VLAs encode this information through token-based representations, where visual encoders (e.g., ViT, CLIP), language models (e.g., LLaMA-4), and trajectory decoders operate in a coherent semantic space, enabling the vehicle to reason about high-level goals and translate them into low-level motion.

A notable contribution in this direction is **CoVLA** [5], which provides a comprehensive dataset pairing over 80 hours of real-world driving videos with synchronized sensor streams (e.g., LiDAR, odometry), detailed natural language annotations, and high-resolution driving trajectories. This dataset enables training VLA models to align perceptual and linguistic features with physical actions. CoVLA employs CLIP for visual grounding, LLaMA-2 for instruction embedding, and trajectory decoders for motion prediction. This configuration allows AVs to interpret verbal cues (e.g., “yield to ambulance”) and environmental conditions (e.g., merging traffic) to make transparent and safe driving decisions.

OpenDriveVLA [293] advances the state of VLA modeling by integrating hierarchical alignment of 2D/3D multi-view vision tokens with natural language inputs. Its architecture leverages both egocentric spatial perception and external scene understanding to construct a dynamic agent-environment-ego interaction model. Through autoregressive decoding, OpenDriveVLA generates both action plans (e.g. steering angle, acceleration) and trajectory visualizations interpretable to humans. Its end-to-end framework achieves leading performance on public autonomous driving benchmarks, including planning and trajectory prediction tasks on the nuScenes and Waymo Open Motion datasets, as well as vision-language question-answering benchmarks for driving scenes, demonstrating strong robustness in urban navigation and behavioral

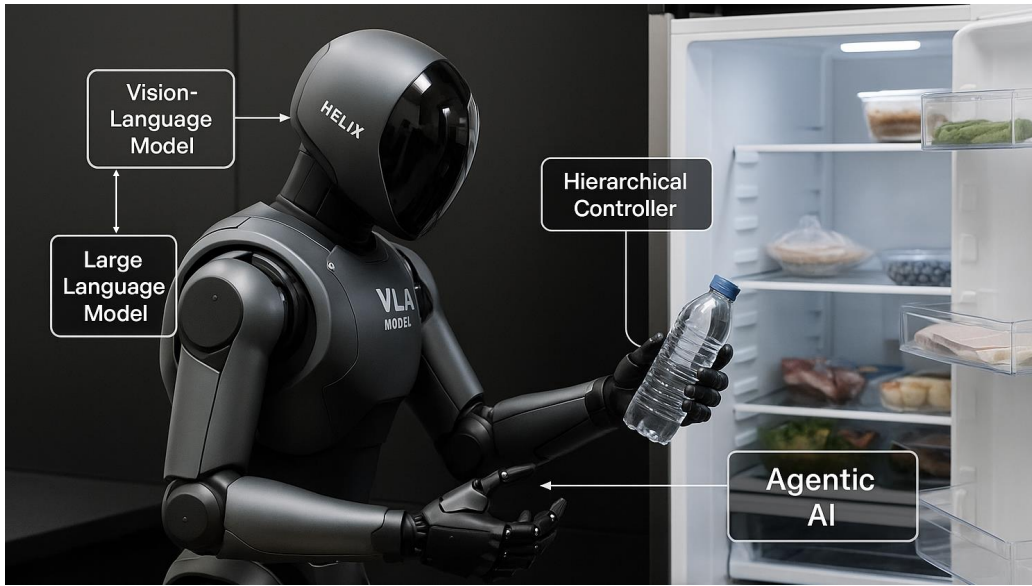


图12：本图展示了采用VLA框架执行家务任务的下一代人形机器人“Helix”。当接收到语音指令时，Helix会整合视觉语言模型（如SigLIP）与语言模型（如LLaMA-4），实现环境感知与智能解读的协同运作。基于分层式VLA控制器规划并执行开冰箱、抓取瓶子等子任务，同时智能AI模块实时调整动作方案。该案例生动展现了基于VLA框架的通用型机器人技术，具备动态任务适配能力与安全可靠的语义化操作特性。

凭借底层智能推理与安全对齐机制，采用VLAs驱动的现代人形机器人正从单一任务执行者转变为全能且值得信赖的协作伙伴。随着TinyVLA和MoManipVLA等节能模型的日益成熟，其在移动低功耗平台上的部署变得愈发可行，由此开启了具身化、社会导向型人工智能的新纪元。

3.4.2. 自动驾驶汽车系统

自动驾驶车辆（AVs），包括自动驾驶汽车、卡车和空中无人机，是VLA模型的一个前沿应用领域。在该领域中，安全关键性的决策需要感知、语义理解与实时动作生成三者紧密协同。与传统模块化自动驾驶流程（明确分离感知、规划和控制功能）不同，VLA框架通过在一个统一模型内联合处理视觉观测数据、高级语义线索和内部状态表示，实现了更紧密的架构耦合。尽管这种端到端的设计在指令驱动导航与推理的模拟及受控基准测试中已展现出良好效果，但大规模商业系统（例如特斯拉Autopilot（源链接））仍依赖模块化或混合式流程；当前行业重点正集中于整合视觉-语言推理组件，而非在安全关键驾驶场景中全面部署VLA式的动作生成技术。

视觉语言模型（VLA）使自动驾驶汽车能够超越像素级物体识别，实现对复杂环境的深度理解。例如，在城市环境中行驶的自动驾驶汽车需要识别交通标志、理解行人行为，并解析诸如“加油站后第二个右转”等导航指令。这些任务需要通过融合视觉与语言信号来理解空间关系、预测驾驶意图，并生成符合上下文的驾驶决策。VLA通过基

于标记的表征方式编码这些信息，其中视觉编码器（如ViT、CLIP）、语言模型（如LLaMA-4）与轨迹解码器协同工作于连贯的语义空间，使车辆能够对高层次目标进行推理，并将其转化为低层级的运动指令。

该领域的一项重要成果是CoVLA[5]，该数据集通过将80小时真实驾驶视频与同步传感器流（如激光雷达、里程计）、详细自然语言标注及高分辨率驾驶轨迹进行配对，构建了全面的数据集。这一数据集使VLA模型能够将感知特征与语言特征与物理动作进行关联训练。CoVLA采用CLIP进行视觉定位、LLaMA-2进行指令嵌入，并通过轨迹解码器实现运动预测。这种配置使自动驾驶系统能够解读语音指令（如“让行救护车”）和环境状况（如交通汇入点），从而做出透明且安全的驾驶决策。

OpenDriveVLA[293]通过将二维/三维多视图视觉标记的层次化对齐与自然语言输入相结合，推动了VLA建模的发展。其架构同时利用自我中心空间感知与外部场景理解，构建了一个动态的智能体-环境-自我交互模型。通过自回归解码机制，OpenDriveVLA不仅能生成动作计划（如转向角度、加速度），还能生成对人类可理解的轨迹可视化结果。其端到端框架在公共自动驾驶基准测试中表现卓越：在nuScenes和Waymo Open Motion数据集上的路径规划与轨迹预测任务，以及驾驶场景的视觉-语言问答基准测试中均取得领先成绩，充分展现了其在城市导航与行为模拟方面的强大鲁棒性。

prediction.

Another seminal model, **ORION** [69], pushes the boundaries of closed-loop autonomous driving by incorporating a QT-Former to retain long-horizon visual context, an LLM to reason over traffic narratives, and a generative trajectory planner. ORION excels at aligning the discrete reasoning space of vision-language models with the continuous control space of AV motion. This unified optimization results in accurate visual question answering (VQA) and trajectory planning, crucial for scenarios involving ambiguous human instructions or occluded obstacles (e.g., “take the exit behind the red truck”).

For example, as depicted in Figure 13, consider an autonomous delivery vehicle, “AutoNav,” operating in a dense urban environment using a next-generation VLA architecture. As AutoNav receives a cloud-based instruction “Drop off the package near the red awning beside the bakery, then return to base avoiding construction zones”, its onboard VLM (e.g., CLIP or SigLIP) parses the visual stream from multiple cameras, identifying dynamic landmarks such as bakery signs, red awnings, and traffic cones. Simultaneously, the LLM module grounded in LLaMA-4 decodes the instruction and fuses it with real-time sensory context including LiDAR, GPS, and inertial odometry. A hierarchical control stack processes these multi-modal signals via an autoregressive VLA decoder that integrates egocentric views and world-centric maps to plan adaptive paths. As the vehicle approaches the delivery location, unexpected pedestrian activity prompts an agentic submodule to trigger trajectory re-planning using a reinforcement learning-inspired policy refinement routine. At the same time, AutoNav audibly warns pedestrians and recalibrates its speed to maintain safety margins. This interplay of semantic understanding, perceptual grounding, and adaptive control exemplifies the power of VLA-based systems in achieving interpretable, human-aligned behavior in safety-critical scenarios.

This scenario illustrates how tightly integrated VLA architectures can surpass traditional perception-planning-control pipelines by enabling end-to-end semantic reasoning, rapid cross-module adaptation, and interpretable decision making. Unlike modular pipelines, where perception outputs, planner updates, and control adjustments are handled by loosely coupled components with limited semantic feedback, the VLA-based system jointly reasons over language intent, visual context, and embodied state, allowing it to dynamically replan trajectories, communicate safety-relevant intentions to humans, and adjust control policies in real time. As a result, the system exhibits greater autonomy, improved transparency through human-interpretable outputs, and enhanced decision-making agility in safety-critical environments.

In aerial robotics, VLAs enhance the capabilities of drones or UAVs in delivery and other tasks. Models such as **UAV-VLA** [191] combine satellite imagery, natural language mission descriptions, and onboard sensing to execute high-level commands (e.g., “deliver to the rooftop pad with the blue tarp”). These systems use modular VLA architectures, where a vision-language planner parses global context and a flight controller executes precise waypoints, supporting applications in logistics, disaster response, and military reconnaissance.

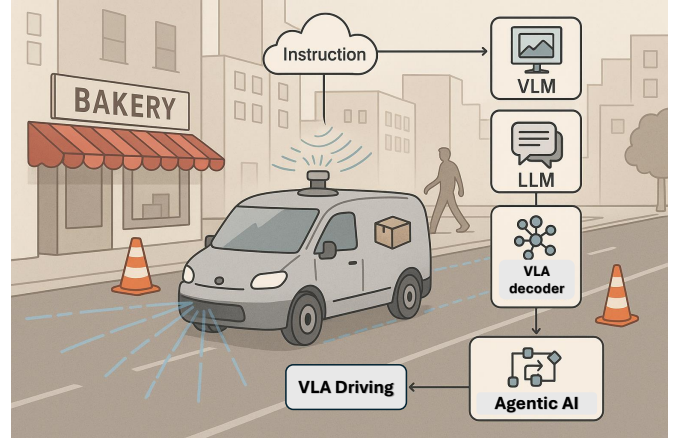


Figure 13: This illustration depicts an autonomous delivery vehicle powered by a VLA system, integrating VLMs for visual grounding, LLMs for instruction parsing, and a VLA decoder for path planning. Agentic AI enables adaptive trajectory refinement in dynamic environments, exemplifying how multi-modal integration drives safe, interpretable, and autonomous decision-making in real-world navigation tasks.

As autonomous systems increasingly operate in unstructured environments, VLAs provide a scalable, interpretable, and data-efficient alternative to traditional pipelines. By learning from large-scale multimodal datasets and modeling decision-making as token prediction, VLAs align human-level semantics with robotic motion, paving the way for safer, smarter autonomous driving and navigation technologies.

3.4.3. Industrial Robotics

Industrial robotics is undergoing a paradigm shift with the integration of VLA models, enabling a new generation of intelligent robots capable of high-level reasoning, flexible task execution, and natural communication with human operators [32, 7]. Traditional industrial robots typically operate in highly structured environments using rigid programming, often requiring extensive reconfiguration and manual intervention when adapting to new assembly lines or product variants [6, 182]. Such systems lack the semantic grounding and adaptability required for modern dynamic manufacturing settings.

VLA models, by contrast, offer a more human-interpretable and generalizable framework. Through the joint embedding of visual inputs (e.g., component layout or conveyor belt state), natural language instructions (e.g., “tighten the screw on the red module”), and robot state, VLAs can infer context and execute appropriate control commands in real-time [135, 72, 156]. Vision transformers (e.g., ViT, DINOv2), LLMs (e.g., LLaMA-4), and autoregressive or diffusion-based action decoders form the backbone of these systems, allowing the robot to parse multi-modal instructions and perform actions grounded in its environment.

One of the most significant contributions in this domain is **CogACT** [131], a modular VLA framework explicitly designed for industrial robotic manipulation. Unlike early VLAs that relied on frozen language-vision embeddings followed by direct

预测

另一项开创性模型Orion[69]通过整合QT-Former模型来保留长时程视觉上下文、利用大语言模型进行交通场景推理，并结合生成式轨迹规划器，从而突破了闭环自动驾驶技术的边界。Orion在视觉语言模型的离散推理空间与自动驾驶运动的连续控制空间之间实现了完美对接。这种统一优化机制不仅实现了精准的视觉问答（VQA）和轨迹规划，对于处理含糊指令或遮挡障碍物场景（例如“请走红色卡车后方出口”）具有关键意义。

例如，如图13所示，考虑一辆在密集城市环境中使用新一代视觉激光雷达（VLA）架构运行的自动驾驶配送车“AutoNav”。当AutoNav接收到云端指令“将包裹投放在面包店旁红色遮阳篷附近，随后返回基地避开施工区域”时，其搭载的VLM（如CLIP或SigLIP）会解析多摄像头采集的视觉流，识别面包店招牌、红色遮阳篷和交通锥等动态地标。与此同时，基于LLaMA-4架构的大型语言模型模块对指令进行解码，并将其与激光雷达、GPS和惯性里程计等实时传感器数据进行融合。分层控制堆栈通过自回归VLA解码器处理多模态信号，该解码器整合自我中心视角与世界中心地图以规划自适应路径。当车辆接近配送地点时，突发行人活动会触发智能子模块，采用强化学习启发的策略优化流程进行轨迹重规划。与此同时，AutoNav会通过语音提示提醒行人并重新校准车速以保持安全裕度。语义理解、感知基础与自适应控制之间的这种相互作用，充分体现了基于VLA（视觉语言分析）系统在安全关键场景中实现可解释且符合人类行为模式的能力。

这一场景生动展示了高度集成的VLA架构如何通过实现端到端的语义推理、快速的跨模块适应以及可解释的决策机制，超越传统感知-规划-控制流程。与模块化流程不同——后者将感知输出、规划更新和控制调整由松耦合组件处理且语义反馈有限——基于VLA的系统能够同时分析语言意图、视觉上下文和物理状态，从而实现动态轨迹重规划、向人类传达安全相关指令，并实时调整控制策略。因此，该系统在安全关键环境中展现出更高的自主性、通过可人类解读的输出提升的透明度，以及更灵活的决策能力。

在空中机器人领域，视觉语言辅助系统（VLAs）显著提升了无人机在货物投递等任务中的执行能力。诸如UAVVLA[191]等模型通过整合卫星影像、自然语言任务描述及机载传感技术，能够执行高级指令（例如“将蓝色防水布运送到屋顶停机坪”）。该系统采用模块化VLAs架构，其中视觉语言规划器解析全局环境信息，飞行控制器则精准执行航点指令，广泛应用于物流运输、灾害救援及军事侦察等领域。

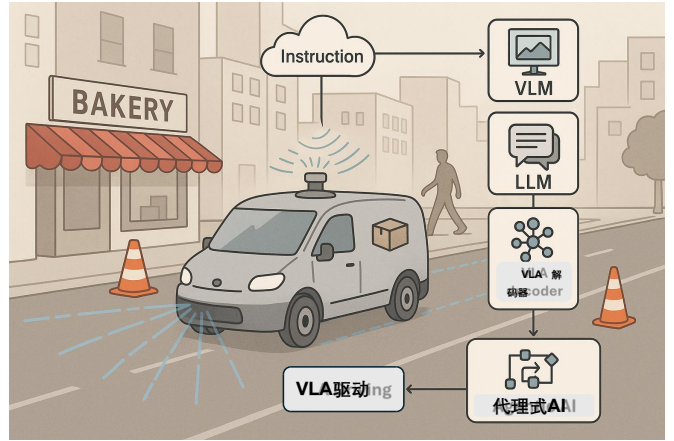


图13：该示意图展示了一种由VLA系统驱动的自主配送车辆，系统整合了用于视觉定位的视觉激光雷达（VLMs）、用于指令解析的激光激光雷达（LLMs）以及用于路径规划的VLA解码器。代理式人工智能可在动态环境中实现自适应轨迹优化，充分展现了多模态融合如何推动现实世界导航任务中安全、可解释且自主的决策过程。

随着自主系统越来越多地在非结构化环境中运行，VLAs为传统数据处理流程提供了一种可扩展、可解释且数据高效的替代方案。通过从大规模多模态数据集学习，并将决策过程建模为令牌预测，VLAs实现了人类语义与机器人运动的精准匹配，为更安全、更智能的自动驾驶及导航技术奠定了基础。

3.4.3. 工业机器人学

工业机器人领域正经历范式变革，随着虚拟实验室架构（VLA）模型的整合应用，新一代智能机器人应运而生。这些机器人具备高级推理能力、灵活的任务执行能力以及人类操作员的自然交互能力[32, 7]。传统工业机器人通常在高度结构化的环境中运行，依赖僵化的编程方式，当需要适配新装配线或产品变体时，往往需要进行大量重新配置并依赖人工干预[6, 182]。这类系统缺乏现代动态制造场景所需的语义基础与自适应能力。

相比之下，视觉语言模型（VLA）提供了一个更易于人类理解和普适化的框架。通过将视觉输入（如组件布局或传送带状态）、自然语言指令（例如“拧紧红色模块上的螺丝”）与机器人状态进行联合嵌入，VLA能够实时推断上下文信息并执行精准控制指令[135, 72, 156]。视觉Transformer（如ViT、DINOv2）、大语言模型（如LLaMA-4）以及自回归或扩散式动作解码器构成了这些系统的骨干架构，使机器人能够解析多模态指令，并根据环境条件执行相应动作。

该领域最具影响力的贡献之一是CogACT[131]，这是一个专为工业机器人操作设计的模块化视觉语言模型框架。与早期依赖固定语言-视觉嵌入后直接执行操作的视觉语言模型不同，该框架通过动态嵌入机制实现了更灵活的交互方式。

action quantization, CogACT introduces a diffusion-based action transformer that models action sequences more robustly and adaptively. The system uses a visual-language encoder (e.g., Prismatic-7B) to extract high-level scene and instruction embeddings, which are then passed to a diffusion transformer (DiT-Base) to generate fine-grained motor actions. This modular separation enables superior generalization to unseen tools, parts, and layouts while preserving interpretability and robustness under real-world constraints.

Furthermore, CogACT demonstrates rapid adaptation across different robot embodiments such as 6-DoF arms or bimanual systems through efficient fine-tuning, making it suitable for deployment across heterogeneous factory environments [131]. Empirical evaluations show that CogACT outperforms prior models like OpenVLA by over 28% in real-world task success rates [122], especially in complex, high-precision tasks such as multi-step assembly, screw fastening, and part sorting [131, 262].

As manufacturing shifts toward Industry 4.0 paradigms, VLAs promise to reduce programming overhead, support voice-commanded robot programming, and facilitate real-time human-robot collaboration on mixed-initiative tasks. While execution precision, safety guarantees, and latency optimizations remain areas of active research, the use of VLA models marks a substantial step toward autonomous, intelligent, and adaptable robots transforming the factories.

3.4.4. Healthcare and Medical Robotics

Healthcare and medical robotics represent a high-stakes domain where precision, safety, and adaptability are critical qualities that VLA models are increasingly well-suited to provide [132, 193]. Traditional medical robotic systems rely heavily on teleoperation or pre-programmed behaviors [166, 204], limiting their autonomy and responsiveness in dynamic surgical or care environments. In contrast, VLA models offer a flexible framework that integrates real-time visual perception, language comprehension, and fine-grained motor control, enabling medical robots to understand high-level instructions and autonomously perform intricate procedures or assistance tasks [131, 54, 225].

In surgical robotics, VLAs can dramatically enhance capabilities in minimally invasive surgeries [50, 230]. These systems can fuse laparoscopic video feeds [127], anatomical maps [147, 50], and voice commands into a unified tokenized representation using vision encoders (e.g., ViT, SAM-2) and language models (e.g., LLaMA, T5) [236]. For instance, as depicted in Figure 14a, in a task like “apply a suture to the left coronary artery,” the vision module identifies the anatomical target, while the language module contextualizes the instruction. The action decoder then translates the fused semantic embedding into stepwise motion commands with sub-millimeter precision. This closed-loop fusion of visual perception, language-grounded intent, and action-level control enables the robot to adaptively reposition tools, apply dynamic force feedback, and avoid critical anatomical structures, thereby reducing the need for surgeon micromanagement and minimizing the risk of human error.

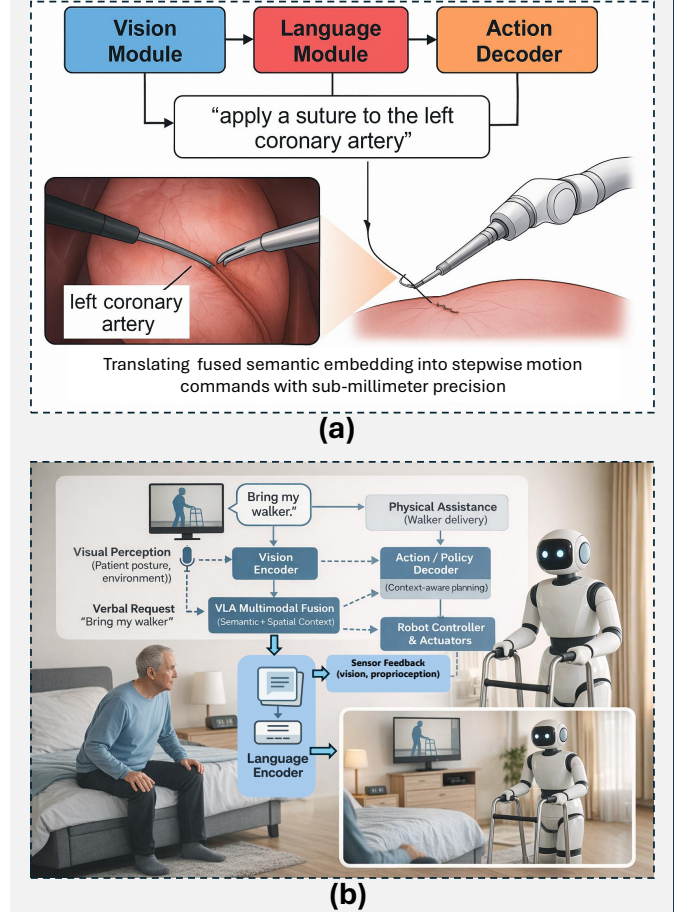


Figure 14: a) This figure illustrates a VLA surgical system executing the task “apply a suture to the left coronary artery.” The vision module identifies anatomical targets, the language model interprets the instruction, and the action decoder generates precise motor commands, enabling adaptive tool control, real-time feedback, and safe autonomous operation; b) A VLA-powered assistive robot perceives patient behavior, processes verbal requests (e.g., “bring my walker”), and autonomously executes context-aware motion plans, enabling real-time assistance in eldercare, rehabilitation, and hospital logistics without relying on predefined scripts or manual oversight.

Beyond the operating room, VLA models are powering a new generation of patient-assistive robots in elderly care, rehabilitation, and hospital logistics. These systems can autonomously perceive patient behavior, understand spoken or gestural input, and execute responsive tasks such as retrieving medication, guiding mobility aids, or notifying caregivers during emergencies. For example, as depicted in Figure 14b, a VLA-enabled robot can visually detect a patient attempting to rise from bed, interpret a verbal request such as “bring my walker,” and generate a context-appropriate motion plan to assist without predefined scripts or constant supervision.

Recent VLA frameworks such as RoboNurse-VLA [132] highlight the real-world feasibility of this approach. RoboNurse employs SAM-2 for semantic scene segmentation and LLaMA-2 for command comprehension, integrated into a real-time voice-to-action pipeline that enables robots to assist with surgical instrument handovers in operating rooms [132]. The system demonstrates robustness to diverse tools, varied lighting condi-

在动作量化领域，CogACT提出了一种基于扩散机制的动作变换器，能够更稳健且自适应地建模动作序列。该系统采用视觉语言编码器（如Prismatic-7B）提取高级场景与指令嵌入向量，随后将这些向量输入扩散变换器（DiT-Base）生成细粒度运动指令。这种模块化架构设计不仅显著提升了对未知工具、部件及场景布局的泛化能力，同时在真实场景约束条件下仍保持了模型的可解释性与鲁棒性。

此外，CogACT通过高效的微调机制，能够快速适配不同机器人形态（如六自由度机械臂或双机械臂系统），使其适用于异构化工厂环境的部署场景[131]。实证评估表明，在多步装配、螺钉紧固及零件分拣等复杂高精度任务中，CogACT的实际任务成功率较OpenVLA等传统模型提升了28%以上[122][131, 262]。

随着制造业向工业4.0模式转型，VLAs有望降低编程开销、支持语音控制机器人编程，并促进人类与机器人在混合任务中的实时协作。尽管执行精度、安全保障及延迟优化仍是当前重点研究方向，但VLA模型的应用标志着自主、智能且适应性强的机器人正在彻底改变工厂运营格局，这无疑是一大重要进展。

3.4.4. 医疗保健与医学机器人学

医疗保健与医疗机器人领域具有重大战略意义，其中精准性、安全性及适应性是关键核心要素，而视觉语言辅助（VLA）模型正日益展现出卓越的适应能力[132, 193]。传统医疗机器人系统过度依赖远程操控或预设行为模式[166, 204]，导致其在动态手术环境或护理场景中的自主决策与响应能力受限。相较而言，VLA模型构建了灵活的智能框架，通过实时视觉感知、语言理解与精细运动控制的深度融合，使医疗机器人能够准确解读高级指令，并自主执行复杂手术操作或辅助任务[131, 54, 225]。

在手术机器人领域，视觉语言助手（VLAs）能显著提升微创手术的操作能力[50, 230]。这类系统通过视觉编码器（如ViT、SAM-2）和语言模型（如LLaMA、T5）[236]，将腹腔镜视频流[127]、解剖结构图谱[147, 50]与语音指令融合为统一的标记化表示。例如图14a所示，在执行“左冠状动脉缝合”等操作时，视觉模块精准识别解剖目标，语言模块则为指令提供语境信息。动作解码器随后将融合后的语义嵌入转化为具有亚毫米级精度的分步运动指令。这种视觉感知、语言意图与动作控制的闭环融合机制，使机器人能够自主调整器械位置、实施动态力反馈控制并规避关键解剖结构，从而大幅降低外科医生的微观操作需求，最大限度降低人为失误风险。

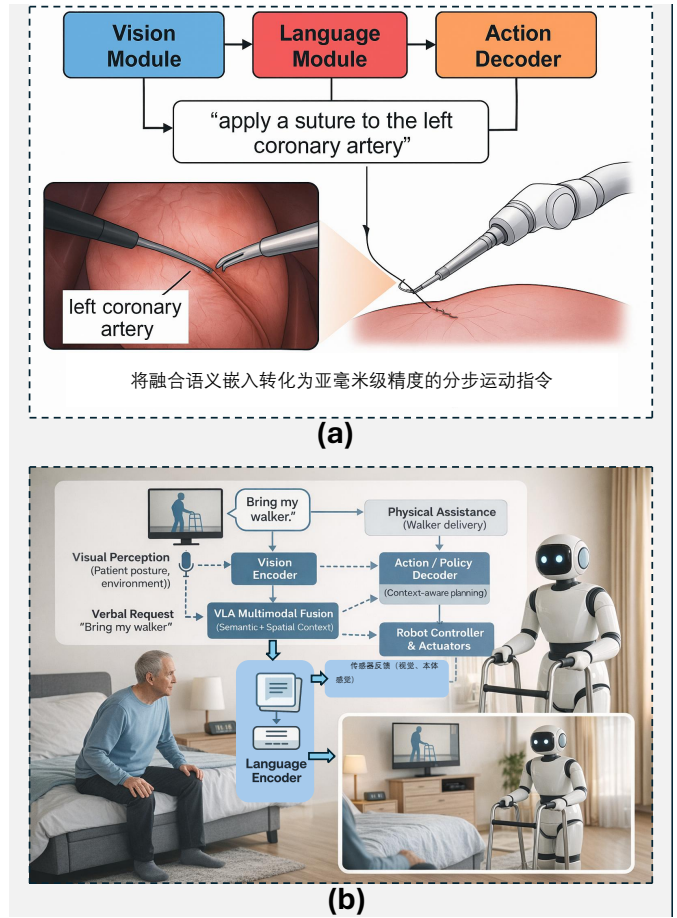


图14: a) 该图展示了VLA手术系统执行“在左冠状动脉上施加缝合线”任务的过程。视觉模块识别解剖学目标，语言模型解析指令，动作解码器生成精确运动指令，从而实现工具自适应控制、实时反馈及安全自主操作；b) 由VLA驱动的辅助机器人可感知患者行为、处理口头指令（如“给我拿助行器”），并自主执行情境感知运动规划，无需依赖预设脚本或人工监督，即可在老年护理、康复治疗及医院物流场景中提供实时辅助服务。

在手术室之外，视觉语言辅助（VLA）模型正推动着新一代患者辅助机器人在老年护理、康复治疗及医院物流领域的应用。这些系统能够自主感知患者行为、理解语音或手势指令，并执行诸如取药、引导移动辅助器具或在紧急情况下通知护理人员等响应性任务。例如图14b所示，搭载VLA技术的机器人可视觉识别患者试图下床的动作，解读诸如“请拿我的助行器”等口头指令，并根据具体情境生成相应的运动方案进行辅助，无需依赖预设脚本或持续监督。

近期开发的VLA框架（如RoboNurse-VLA[132]）充分验证了该方法在实际场景中的可行性。RoboNurse系统采用SAM-2进行语义场景分割，并运用LLaMA-2实现指令理解功能，通过集成实时语音转动作处理流程，使机器人能够协助手术室中的手术器械交接工作[132]。该系统展现出对多样化工具及复杂光照条件的强适应性。

tions, and noisy environments - common challenges in clinical settings.

Additionally, VLA architectures offer advantages in explainability and auditability, both critical in regulated medical domains [224, 146]. Scene grounding and trajectory prediction can be visualized and reviewed post-hoc [278], which could facilitate clinical trust and enabling FDA-style validation pipelines. LoRA-based fine-tuning allows adaptation to specific hospital environments or procedural workflows with minimal data and computational infrastructure [9, 229, 147].

Importantly, the multi-modal foundation of VLA models enables cross-domain transferability: the same model trained on surgical tool manipulation can be adapted to patient mobility tasks with modest retraining [56]. This modularity significantly reduces development time and cost compared to task-specific automation systems [94]. As medical robotics transitions from teleoperated assistance to semi-autonomous and collaborative systems, VLA models stand at the core of this transformation.

As also discussed earlier in other application areas, VLA’s capability in combining high-level semantic understanding with low-level control is instrumental to provide a unified solution for scalable, human-aligned, and adaptive robotic healthcare [249, 295, 279]. As healthcare systems face increasing demand and workforce shortages, VLA-driven robotics will play a crucial role in enhancing medical precision, operational efficiency, and patient-centered care.

3.4.5. Precision and Automated Agriculture

As illustrated in Figure 15, VLA models are emerging as transformative tools in precision and automated agriculture, offering intelligent, adaptive solutions for labor-intensive tasks across diverse farming landscapes [70, 191]. Unlike traditional agricultural automation systems that depend on rigid, sensor-driven pipelines requiring manual reprogramming for each task or environmental variation [220, 108], VLAs integrate multi-modal perception, natural language understanding, and real-time action generation within a unified framework [168, 84]. This unified multimodal integration enables autonomous ground robots and drones to interpret complex field scenes, follow spoken or text-based farming instructions, and generate context-aware actions such as selective fruit picking or adaptive irrigation. The ability of VLAs to dynamically adjust to occlusions, terrain irregularities, lighting variability, or varying crop types combined with training on synthetic, photorealistic datasets allows them to generalize across crop types, geographies and seasons. By leveraging action tokenization [245], transformer-based policy generation [12, 85], and techniques like LoRA fine-tuning [93], these systems are redefining the scalability and intelligence of agricultural robotics for sustainable and precision-driven farming.

In modern fruit orchards and other crop fields, VLAs can process visual inputs from RGB-D cameras, multispectral sensors, or drones to monitor plant growth, detect diseases, and identify nutrient deficiencies. Vision transformers (e.g., ConvNeXt, DINOv2) encode spatial and semantic information from visual scenes, while LLMs (e.g., T5, LLaMA) parse natural language commands such as “inspect the east plot for powdery mildew”

or “harvest ripe apples near the irrigation trench.” Through token fusion, these modalities are aligned in a shared representation space, allowing robots to execute fine-grained, context-aware actions with precision.

For instance, in fruit-picking tasks, as illustrated in Figure 15, a VLA-equipped ground robot can identify ripe produce using image-based ripeness cues, interpret user-specified criteria such as “pick only Grade A fruits,” and execute motion sequences via action tokens that control its end-effector. This approach ensures minimal crop damage, optimizes pick rates, and allows real-time adaptation to unexpected variables like occlusions or terrain shifts. In irrigation management, drones guided by VLA models can interpret field maps and verbal instructions to selectively water stressed zones, reducing water usage %.

Beyond immediate task execution, VLA models are expected to support dynamic reconfiguration and lifelong learning through closed-loop feedback mechanisms. In particular, execution outcomes, sensory observations, and task success signals collected during deployment can be logged and periodically incorporated into offline or incremental updates of the VLA policy. When combined with synthetic training data generated from photorealistic simulations of crop environments (e.g., 3D orchard renderings), such feedback-driven adaptation may enable models to progressively improve robustness to new crop varieties, pest conditions, and seasonal variations without extensive manual annotation. Parameter-efficient techniques such as LoRA adapters and diffusion-based policy refinement are anticipated to play a key role in facilitating this continual adaptation while limiting computational overhead.

Overall, the integration of VLA models into agricultural workflows is expected to offer several long-term benefits, including reduced reliance on skilled manual labor, improved yields through targeted intervention, and enhanced environmental sustainability via optimized input usage. As global food systems increasingly face climate variability and resource constraints, VLA-enabled agricultural technologies are anticipated to contribute to scalable, intelligent, and context-aware farming practices that better accommodate real-world complexity.

3.4.6. Interactive AR Navigation with Vision-Language-Action Models

Interactive Augmented Reality (AR) navigation represents a frontier where VLA models can significantly enhance human-environment interaction by providing intelligent, context-aware guidance in real-time [31, 104, 255]. In this paradigm, VLAs process continuous streams of visual data from AR-enabled devices such as smart glasses or smartphones alongside natural language queries to generate dynamic navigational cues overlaid directly onto the user’s view of the physical world. Unlike traditional GPS-based systems that rely on rigid maps and limited user input [29, 207], VLA-based AR agents interpret complex visual scenes (e.g., intersections, indoor hallways, signage) and respond to free-form instructions such as “take me to the nearest pharmacy with a wheelchair ramp” or “show the quietest route to the conference room.”

Technically, these models integrate a vision encoder (e.g., ViT, DINOv2) that extracts scene representations from RGB

电生理记录技术及嘈杂环境——临床环境中常见的挑战。

此外，VLA架构在可解释性和可审计性方面具有显著优势，这两点在受监管的医疗领域至关重要[224, 146]。场景定位与轨迹预测结果可进行可视化呈现及事后复核[278]，这有助于提升临床信任度，并支持FDA标准验证流程的构建。基于LoRA的微调技术能够以最小的数据量和计算资源，实现对特定医院环境或诊疗流程的精准适配[9, 229, 147]。

值得注意的是，VLA模型的多模态基础架构实现了跨领域迁移能力：同一模型在手术器械操作训练中获得成果，仅需少量再训练即可适配患者移动任务[56]。相较于任务专用自动化系统[94]，这种模块化设计显著缩短了开发周期并降低了成本。随着医疗机器人技术从远程操控辅助系统向半自主协作系统转型，VLA模型正成为推动这一变革的核心驱动力。

正如先前在其他应用领域所讨论的那样，VLA将高层语义理解与底层控制相结合的能力，对于提供一种可扩展、符合人类需求且具有自适应性的机器人医疗统一解决方案至关重要[249, 295, 279]。随着医疗系统面临日益增长的需求和人力资源短缺，基于VLA的机器人技术将在提升医疗精准度、运营效率及以患者为中心的护理质量方面发挥关键作用。

3.4.5. 精准农业与自动化农业

如图15所示，VLA模型正成为精准农业与自动化领域的变革性工具，为不同农业场景中的劳动密集型任务提供智能自适应解决方案[70, 191]。与依赖僵化传感器驱动流程的传统农业自动化系统不同——后者需针对每项任务或环境变化进行手动重新编程[220, 108]，VLA通过统一框架整合了多模态感知、自然语言理解与实时动作生成能力[168, 84]。这种统一的多模态集成使自主地面机器人和无人机能够解析复杂田间场景、执行语音或文本指令，并生成情境感知型操作（如选择性采果或智能灌溉）。VLA通过在合成高逼真数据集上训练，具备动态适应遮挡物、地形起伏、光照变化及作物类型差异的能力，从而实现跨作物种类、地域类型及季节条件的泛化应用。通过利用动作分词[245]、基于Transformer的策略生成[12, 85]以及LoRA微调技术[93]，这些系统正在重新定义农业机器人的可扩展性与智能水平，以实现可持续且精准导向的农业发展。

在现代果园及其他农作物种植区，视觉学习算法（VLAs）能够处理来自RGB-D相机、多光谱传感器或无人机的视觉输入数据，用于监测植物生长状况、检测病害并识别养分缺乏问题。视觉变换器（如ConvNeXt、DINOv2）可对视觉场景中的空间信息和语义信息进行编码，而大型语言模型（如T5、LLaMA）则能解析诸如“检查东侧地块是否存在白粉病”或“在灌溉沟附近采摘成熟苹果”等自然语言指令。通过令牌融

合技术，这些不同模态的数据被整合到一个共享的表征空间中，从而使机器人能够精准执行细粒度且具备上下文感知能力的操作。

例如，在水果采摘任务中（如图15所示），配备视觉激光雷达（VLA）的地面机器人能够通过图像识别成熟度特征判断果实状态，解读用户设定的筛选标准（如“仅采摘A级水果”），并通过动作指令控制末端执行器完成动作序列。该方案可最大限度减少作物损伤、提升采摘效率，并能实时应对遮挡物或地形变化等突发状况。在灌溉管理领域，基于VLA模型引导的无人机可解析田间分布图与人工指令，精准灌溉缺水区域，从而显著降低用水量占比。

除了执行即时任务外，VLA模型还应通过闭环反馈机制支持动态重构和终身学习。具体而言，部署过程中收集的任务执行结果、感官观测数据及任务成功信号均可被记录，并定期整合到VLA策略的离线或增量更新中。当结合基于作物环境照片级真实模拟（例如三维果园渲染）生成的合成训练数据时，这种基于反馈的自适应机制可使模型在无需大量人工标注的情况下，逐步提升对新作物品种、病虫害状况及季节变化的适应能力。参数高效的技术——如LoRA适配器和基于扩散的策略优化技术——预计将在持续适应过程中发挥关键作用，同时有效控制计算开销。

总体而言，将虚拟土地模型（VLA）整合到农业生产流程中，预计将带来多项长期效益：降低对熟练人工劳动力的依赖、通过精准干预措施提升作物产量、以及通过优化资源投入实现环境可持续性提升。随着全球粮食系统日益面临气候变异性与资源限制，基于VLA技术的农业创新方案有望推动规模化、智能化且具备情境感知能力的耕作实践，从而更好地适应现实世界的复杂性挑战。

3.4.6. 基于视觉-语言-动作模型的交互式AR导航系统

交互式增强现实（AR）导航技术堪称前沿领域，虚拟现实增强系统（VLA）通过实时提供智能、情境感知的引导信息，能够显著提升人机交互体验[31, 104, 255]。该技术框架下，VLA系统会实时处理来自智能眼镜、智能手机等AR设备的连续视觉数据流，并结合自然语言指令，生成动态导航提示直接叠加在用户所见的物理世界视图上。与依赖静态地图和有限用户输入的传统GPS系统不同[29, 207]，基于VLA的AR智能体能够解析复杂视觉场景（如交叉路口、室内走廊、标识系统），并响应诸如“带我去配备轮椅坡道的最近药房”或“显示前往会议室最安静的路线”等自由式指令。

从技术层面而言，这些模型整合了视觉编码器（如ViT、DINOv2），用于从RGB图像中提取场景表征。

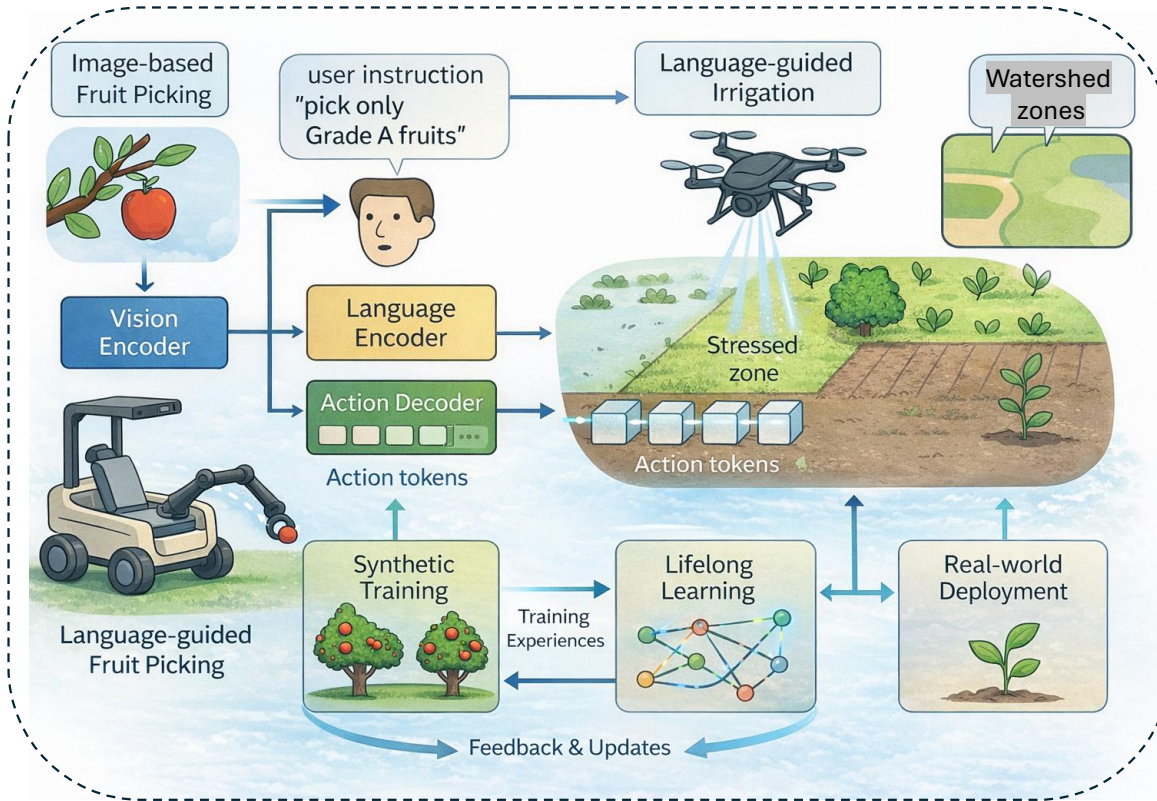


Figure 15: Conceptual illustration of VLA models in precision and automated agriculture. A ground robot combines vision encoders and language instructions (e.g., “pick only Grade A fruits”) to generate action tokens for damage-free harvesting, while aerial robots use VLA reasoning for language-guided irrigation. Synthetic training, LoRA-based adaptation, and lifelong feedback enable generalization across crops, environments, and geographies, supporting sustainable, data-driven agricultural automation.

camera frames, a language encoder (e.g., T5 or LLaMA) that processes user prompts or voice commands, and an action decoder that predicts tokenized navigation cues such as directional overlays, waypoints, or voice instructions. A transformer-based architecture fuses these modalities to reason about both the spatial layout and semantic intent, allowing the AR agent to adaptively highlight paths, landmarks, and hazards directly within the user’s field of view [211, 165]. For example, as shown in Figure 16, in a crowded airport, the VLA agent could visually identify escalators, gates, or baggage claims while understanding a query like “how do I reach Gate 22 without stairs?”, adjusting the route in response to real-time occupancy and obstacles.

VLAs are expected to support interactive instruction loops in which users can first issue high-level commands (e.g., “navigate to the pharmacy”) and subsequently refine them with additional constraints such as “avoid busy areas” or “take the scenic route.” Through context-aware feedback and iterative clarification, such interaction paradigms can improve accessibility and usability for visually impaired or cognitively challenged individuals. In logistics and indoor navigation, these systems can be integrated with IoT sensors and digital twins to guide warehouse workers, maintenance teams, or delivery robots through complex environments. Furthermore, personalized navigation can be achieved through continual fine-tuning,

where VLA models learn user preferences and local spatial layouts over time.

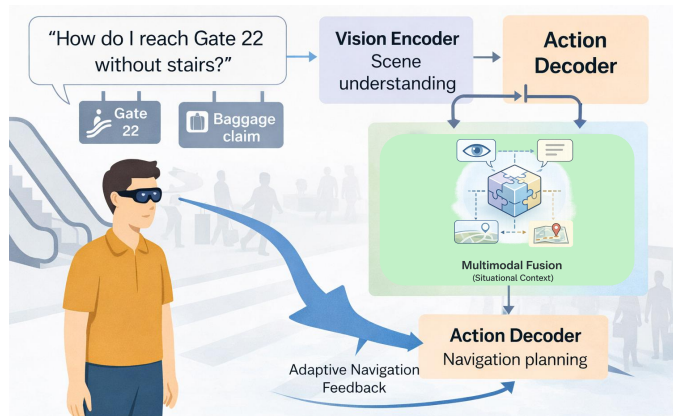


Figure 16: Showing how VLA models enable interactive AR navigation by fusing real-time visual perception, language understanding, and action planning. In dynamic environments such as airports, VLAs interpret user queries like “avoid stairs to Gate 22,” analyze visual scenes (e.g., detecting escalators), and adjust navigational paths accordingly, supporting personalized, accessible, and context-aware mobility guidance.

As AR hardware becomes more affordable and integrated into daily life, VLA-powered navigation systems will enable seamless spatial understanding, multi-modal interaction, and

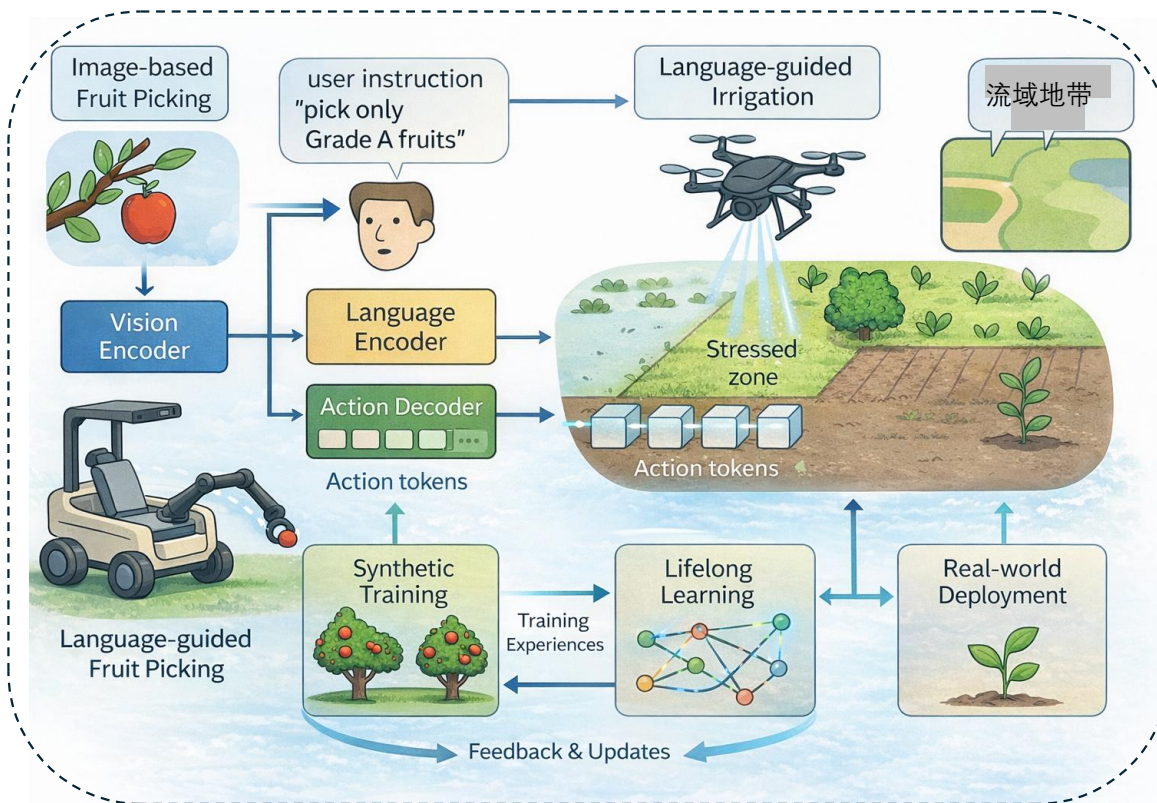


图15: VLA模型在精准农业与自动化农业中的概念示意图。地面机器人通过整合视觉编码器与语言指令（例如“仅采摘A级水果”），生成无损采收的动作指令；而空中机器人则运用VLA推理技术实现语言引导式灌溉。基于合成训练、LoRA算法的自适应学习以及终身反馈机制，该系统能够实现跨作物、跨环境及跨地域的泛化能力，为可持续发展的数据驱动型农业自动化提供支持。

相机帧、用于处理用户提示或语音指令的语言编码器（例如T5或LLaMA），以及能够预测分词化导航提示（如方向叠加信息、航点或语音指令）的动作解码器。基于Transformers的架构将这些模态融合在一起，以同时理解空间布局和语义意图，从而使AR代理能够自适应地在用户视野范围内突出显示路径、地标和危险区域[211, 165]。例如，如图16所示，在拥挤的机场中，VLA代理能够视觉识别自动扶梯、登机口或行李提取区，并理解诸如“如何不经过楼梯直接到达22号登机口？”之类的查询请求，进而根据实时占用情况和障碍物调整路线。

视觉语言助手（VLA）有望支持交互式教学循环模式：用户可先发出高层次指令（如“前往药房”），随后通过添加“避开拥堵区域”或“选择风景优美的路线”等附加约束条件进行细化调整。通过情境感知反馈与迭代式澄清机制，此类交互模式能显著提升视障人士及认知障碍群体的可访问性与使用体验。在物流与室内导航领域，这些系统可与物联网传感器及数字孪生技术深度融合，为仓库作业人员、维修团队或配送机器人提供复杂环境导航指引。此外，通过持续微调实现个性化导航功能——VLA模型

将逐步掌握用户偏好与局部空间布局特征。

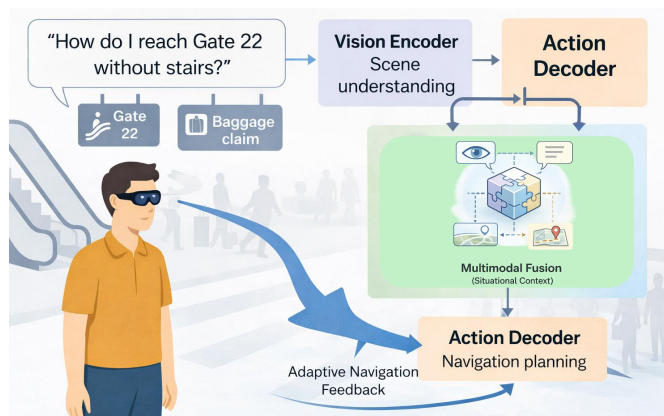


图16: 展示了VLA模型如何通过融合实时视觉感知、语言理解与行动规划实现交互式增强现实（AR）导航。在机场等动态环境中，VLA可解析用户查询（如“避开通往22号登机口的楼梯”），分析视觉场景（例如检测自动扶梯），并据此调整导航路径，从而提供个性化、无障碍且具备情境感知能力的移动引导服务。

随着增强现实（AR）硬件成本降低并融入日常生活，基于甚大阵列（VLA）的导航系统将实现无缝的空间认知、多模态交互及

autonomous guidance in public, industrial, and assistive contexts redefining how humans perceive, explore, and interact with physical spaces.

4. Challenges and Limitations of Vision-Language-Action Models

VLA models face a spectrum of interrelated challenges that impede their translation from research prototypes to robust, real-world systems. First, achieving real-time, resource-aware inference remains difficult: models like DeeR-VLA leverage dynamic early-exit architectures to cut computation 5–6× on manipulation benchmarks while preserving accuracy, yet their gains diminish in complex scenarios [269]. Similarly, UniNaVid compresses egocentric video tokens for 5 Hz navigation but still struggles under highly ambiguous instructions and longer horizons [280]. Moreover, these efficiency-driven designs often expose a trade-off between computational speed and representational coverage. When operating under aggressive compression or early-exit constraints, even advanced hybrid vision–language grounding methods exhibit limited object generalization; for example, ObjectVLA generalizes to only 64 % of novel objects, highlighting how real-time optimization can exacerbate gaps in open-world robustness [297].

Second, adapting VLA models with minimal supervision and ensuring stable policy updates under scarce, noisy data is nontrivial. ConRFT combines behavior cloning and Q-learning with human-in-the-loop fine-tuning to rapidly converge to 96.3% success over eight contact-rich tasks, yet it relies heavily on expert interventions and reward shaping [36]. Hierarchical frameworks such as Hi Robot decouple high-level reasoning from low-level execution to improve instruction fidelity, but coordinating these modules and grounding ambiguous feedback remains challenging [199]. Likewise, Tactile-language-action model model’s fusion of tactile streams with language commands achieves over 85 % success on unseen peg-in-hole tasks, but dataset breadth and real-time multi-step decoding still limit broader generalization [88].

Furthermore, ensuring safety, generalization, and end-to-end reliability in dynamic environments demands new modeling and evaluation standards. Occupancy-Language-Action models like OccLLaMA unify 3D scene understanding with action planning, yet they need to scale to richer scene dynamics and semantic consistency across modalities [239]. RaceVLA advances high-speed drone navigation through quantized, iterative control loops; however, its limited visual–physical generalization compared to larger VLAs and dedicated reasoning models raises safety concerns in unseen or rapidly changing environments [195]. Model-merging strategies in ReVLA recover lost out-of-domain visual robustness improving oriented object detection (OOD) grasp success by up to 77 % but introduce extra computation and complexity [48]. Finally, SafeVLA formulates constraints via constrained Markov decision processes to cut unsafe behavior by over 80 %, yet defining comprehensive, non-restrictive safety rules for diverse real-world tasks remains an open problem [274]. Addressing these intersecting limita-

tions is critical for VLA models to achieve reliable, autonomous operation under the full complexity of real-world robotics.

Building upon the critical limitations outlined above, it is imperative to map each challenge to targeted mitigation strategies and assess their system-level impact. Table 4 presents this map identifying core limitations, potential technical remedies drawn from recent advances, and articulating the anticipated benefits for real-world VLA deployment. For instance, addressing real-time inference constraints leverages parallel decoding and quantized transformer pipelines with hardware acceleration (e.g., TensorRT) to sustain control loop speed in drones and manipulators [129, 122, 75, 142]. Addressing multi-modal action representation via hybrid diffusion–autoregressive policies enriches a model’s capacity to produce varied, context-sensitive motor commands for complex tasks [171, 156]. To guarantee safety in open worlds, dynamic risk assessment modules and adaptive planning layers can be integrated, ensuring robust emergency stop behaviors in unpredictable settings [183, 233, 113]. Similarly, dataset bias and grounding can be minimized through curated debiased corpora and advanced contrastive fine-tuning, strengthening fairness and semantic fidelity when generalizing to novel objects and scenes [185, 17, 175]. Together, these strategies and other approaches spanning simulation-to-real transfer, tactile integration, and energy-efficient architectures frame a comprehensive roadmap for transitioning VLA research into reliable, scalable autonomy.

The remainder of this section is organized into five focused subsections, each examining a distinct cluster of VLA challenges identified in the literature. First, we analyze real-time inference constraints and the emerging methods to address them. Next, we explore multi-modal action representation alongside safety assurance in open-world settings. We then discuss dataset bias, grounding strategies, and generalization to unseen tasks, followed by an exploration of system integration complexity and computational demands. Finally, we consider robustness and the ethical implications of deploying VLAs in real-world applications.

4.1. Real-Time Inference Constraints

Despite recent progress, deploying VLA models in latency-critical settings continues to be constrained by real-time inference requirements, particularly in applications such as robotic manipulation, autonomous driving, and aerial control. VLAs typically depend on autoregressive decoding strategies, which sequentially generate action tokens based on previous predictions. While effective for many tasks, this autoregressive decoding paradigm substantially limits inference speed, typically achieving only 3–5 Hz when deployed on standard GPU-based research platforms (e.g., single high-end consumer or datacenter GPUs) for end-to-end VLA inference [67]. This rate remains substantially below the control frequencies typically required for responsive and stable robotic operation, which often range from tens of hertz for high-level planning to higher update rates for low-level feedback control, depending on the task and hardware platform. For instance, when a robotic arm manipulates delicate objects, frequent positional updates are essential to maintain accuracy and prevent damage. Models such

在公共、工业及辅助场景中实现自主引导，重新定义人类对物理空间的感知、探索及交互方式。

4. 视觉-语言-动作模型面临的挑战与局限性

VLA模型面临着一系列相互关联的挑战，这些挑战阻碍了它们从研究原型向稳健的现实世界系统转化。首先，实现实时且资源感知的推理仍然困难：诸如DecR-VLA之类的模型利用动态早期退出架构，在操作基准测试中将计算量降低5–6倍同时保持精度，但在复杂场景下其性能提升会减弱[269]。类似地，UniNaVid对以自我为中心的视频标记进行了5赫兹导航压缩，但在处理高度模糊的指令和较长时间范围时仍表现不佳[280]。此外，这些以效率为导向的设计往往在计算速度与表征覆盖度之间存在权衡。当受到激进压缩或早期退出约束时，即便是先进的混合视觉-语言映射方法也表现出有限的对象泛化能力；例如，ObjectVLA仅能对64%的新异物体进行泛化，这凸显了实时优化可能加剧开放世界鲁棒性方面的缺陷[297]。

其次，在有限监督下调整VLA模型并确保在数据稀疏且噪声较大的情况下实现稳定策略更新并非易事。ConR-FT通过结合行为克隆与Q学习，并辅以人工闭环微调，成功在八个高接触任务中快速收敛至96.3%的成功率，但其仍高度依赖专家干预和奖励塑形[36]。Hi Robot等分层框架通过将高层推理与底层执行解耦来提升指令保真度，但协调这些模块并处理模糊反馈仍具挑战性[199]。同样，触觉-语言-动作模型通过融合触觉流与语言指令，在未见过的插销-孔洞任务中取得超过85%的成功率，但数据集广度不足与实时多步解码能力仍制约其泛化能力[88]。

此外，在动态环境中确保安全性、泛化能力和端到端可靠性，需要建立新的建模与评估标准。像OccLLaMA这样的占用-语言-动作模型将三维场景理解与动作规划相结合，但其需要扩展到更丰富的场景动态特征，并实现跨模态语义一致性[239]。RaceVLA通过量化迭代控制循环推进高速无人机导航技术，但相较于大型视觉语言模型和专用推理模型，其视觉-物理泛化能力有限，这在未见场景或快速变化环境中引发了安全顾虑[195]。ReVLA中的模型融合策略通过恢复域外视觉鲁棒性提升了方向性物体检测（OOD）抓取成功率达77%，但也增加了额外计算量与系统复杂度[48]。最后，SafeVLA通过约束马尔可夫决策过程制定安全约束条件，使不安全行为减少超80%，但针对多样化现实任务制定全面且非限制性的安全规则仍是亟待解决的开放性问题[274]。解决这些相互交织的局限性对于VLA模型在真实世界机器人系统的全部复杂性下实现可

靠自主运行至关重要。

基于前文所述的关键局限性，必须将每个挑战与针对性缓解策略进行对应映射，并评估其系统层面的影响。表4呈现了该映射关系，明确了核心局限性、基于最新技术进展提出的潜在解决方案，并阐述了对实际VLA部署的预期效益。例如，通过并行解码与量化Transformer流水线结合硬件加速技术（如TensorRT），可有效解决实时推理约束问题，从而维持无人机和机械臂的控制环路速度[129, 122, 75, 142]。采用混合扩散-自回归策略处理多模态动作表征，能显著提升模型生成多样化、情境敏感型运动指令的能力，适用于复杂任务场景[171, 156]。为确保开放环境中的安全性，可集成动态风险评估模块与自适应规划层，确保在不可预测环境中实现稳健的紧急停止行为[183, 233, 113]。同样地，通过精心构建的无偏数据集和先进的对比式微调技术，可以最大限度地减少数据集偏差和接地效应，从而在泛化到新对象和场景时增强公平性和语义保真度[185, 17, 175]。这些策略与涵盖从模拟到真实场景迁移、触觉整合及节能架构等其他方法相结合，共同为将视觉自主学习（VLA）研究转化为可靠、可扩展的自主系统提供了全面的路线图。

本节余下部分将划分为五个专题子章节，每个章节分别针对文献中提出的VLA技术挑战进行系统性分析。首先，我们将解析实时推理约束条件及其新兴解决方案。其次，重点探讨开放世界场景中多模态动作表征与安全机制的协同应用。随后深入分析数据集偏差、基底策略及未见任务泛化能力，继而评估系统集成复杂度与计算资源需求。最后，我们将探讨VLA技术在实际应用中的鲁棒性表现及其伦理影响。

4.1. 实时推理约束

尽管近期取得了进展，但在对延迟要求极高的场景中部署VLA模型仍受到实时推理需求的限制，尤其是在机器人操作、自动驾驶和空中控制等应用领域。VLA通常依赖自回归解码策略，该策略会根据先前的预测结果依次生成动作令牌。虽然这种自回归解码方法适用于许多任务，但它显著限制了推理速度——在基于标准GPU的研究平台上（例如单块高端消费级或数据中心GPU）进行端到端VLA推理时，通常仅能达到3-5赫兹[67]。这一速率远低于机器人响应式稳定运行所需的控制频率：根据任务和硬件平台的不同，高级规划通常需要数十赫兹的频率，而低级反馈控制则需要更高的更新速率。例如，当机械臂操作精密物体时，频繁的位置更新对于保持精度和防止损坏至关重要。诸如

Table 4: Challenges, potential solutions, and expected impact of VLA models.

Challenge / limitation	Potential solution	Expected impact
Real-time inference constraints	Parallel decoding, quantized transformers, and hardware acceleration (e.g., TensorRT) [129, 122]; reduce autoregressive overhead [75, 142].	Enables real-time control and deployment in latency-critical domains [251, 191] (e.g., UAVs, manipulators).
Multi-modal action representation	Hybrid tokenization combining diffusion and autoregressive policies [171]; train on diverse demonstrations and multi-modal outputs [156].	Improves performance on complex, dynamic manipulation with multiple valid solution modes [74].
Safety assurance in open worlds	Dynamic risk assessment modules [183, 233]; low-latency emergency-stop and adaptive planning layers [113].	Improves reliability and safety in unpredictable settings (homes, factories, healthcare); enhances user acceptability.
Dataset bias and grounding	Curate diverse/debiased datasets [185]; stronger grounding (e.g., CLIP fine-tuning with hard negatives) [282, 17].	Improves fairness and semantic fidelity [109], and enhances generalization to novel real-world inputs [227, 288, 175].
Limited 3D perception and reasoning	Integrate depth/LiDAR; develop 3D-aware architectures; fuse point clouds with vision-language features.	Enables stronger spatial reasoning for manipulation and navigation in complex environments [129].
Cross-embodiment generalization	Train across diverse morphologies; learn embodiment-agnostic action abstractions; apply cross-domain adaptation [266].	Facilitates policy transfer across robot platforms and configurations [279, 122].
Annotation complexity and cost	Weak supervision, active learning, and synthetic data generation to reduce manual labeling [148].	Lowers development cost and accelerates scaling to new tasks/domains [233, 286].
Sim-to-real transfer gap	Domain adaptation, sim-to-real fine-tuning, and real-world calibration [210, 134].	Improves reliability and consistency when deploying beyond simulation [4, 66].
Integration of physical knowledge	Physics priors, simulation environments, and dynamics modeling in training pipelines [53].	Improves prediction and planning under real physical constraints [106].
Multi-modal integration (tactile, audio)	Fuse tactile/audio with vision and language [114]; extend multimodal transformer fusion.	Improves robustness under occlusion/ambiguity and expands the task repertoire [76, 136, 91].
Long-horizon multi-stage tasks	Hierarchical policies, memory-augmented networks, and trajectory planning modules [136].	Improves sequential planning, memory, and compositional execution [131, 286, 227, 175].
System integration complexity	Unified transformer backbones [289]; temporal alignment and sim-to-real transfer strategies [163, 277].	Enables tighter planning-control coordination and more robust transfer to physical robots [187, 208].
Energy and compute demands	Pruning, LoRA, quantization-aware training, and low-power accelerators.	Enables efficient embedded/mobile deployment [252, 283, 125, 246].
Generalization to unseen tasks	Compositional generalization, few-shot meta-learning, and task-agnostic pretraining [173, 146].	Reduces overfitting and strengthens zero-/few-shot adaptation [97, 242, 286].
Robustness to environmental variability	Domain randomization, sensor fusion, broader training datasets, and online recalibration [156].	Improves stability under changing lighting, clutter, and scene dynamics [285, 242].
Ethical and societal implications	Privacy via on-device processing/anonymization [149, 198, 252, 34]; fairness audits; regulatory and trust frameworks.	Promotes equitable and trustworthy adoption across social, medical, and labor domains [160, 180, 217, 172].

as OpenVLA [122] and Pi-0 [15] face inherent challenges with this sequential token generation approach, thereby limiting their effectiveness in dynamic environments.

Emerging solutions such as parallel decoding, exemplified by NVIDIA’s GR00T N1 model [14], aim to accelerate inference by predicting multiple tokens simultaneously. GR00T N1 achieves approximately a $2.52\times$ speedup over traditional decoding methods; however, this parallelism often introduces trade-offs in trajectory smoothness, resulting in suboptimal robot movements. Such movements are undesirable in sensitive applications like surgical robotics, where precision, adaptability and flexibility are paramount. Thus, achieving rapid inference without compromising output quality remains an open challenge.

Additionally, hardware limitations exacerbate real-time inference constraints. For example, processing high-dimensional visual embeddings, typically involving over 400 vision tokens at 512 dimensions each, requires approximately 1.2 GB/s memory bandwidth. This demand significantly exceeds the capacity of current embedded systems or edge-AI hardware such as NVIDIA Jetson platforms, thereby restricting practical deployment [82, 275]. Even with efficient quantization techniques,

which reduce the precision of floating-point operations to alleviate memory constraints, models frequently experience accuracy degradation, especially in tasks demanding sub-millimeter precision, such as bimanual robotic manipulation or medical robotics.

4.2. Multi-modal Action Representation and Safety Assurance

Multi-modal Action Representation: One significant limitation of current VLA models is accurately representing multi-modal actions, particularly in scenarios requiring continuous and nuanced control [62, 46]. Traditional discrete tokenization methods, such as those dividing actions into 256 distinct bins, inherently lack precision, creating substantial errors in fine-grained tasks like delicate robotic grasping or intricate surgical procedures [171]. For instance, during precise robotic manipulation in assembly tasks, discrete representations can result in misaligned or imprecise actions, undermining performance and reliability. On the other hand, continuous multilayer perceptron (MLP) based approaches face the risk of mode collapse [162, 232], where models converge prematurely to single action trajectories, despite multiple viable paths available. This diminishes the flexibility necessary for adaptive decision-making in

表4: VLA模型面临的挑战、潜在解决方案及预期影响

挑战/局限性	潜在解决方案	预期影响
即时处理的约束	推理 共- 并行解码、量化变换器与硬件加速（例如 TensorRT）[129, 122]；降低自回归机制带来的开销[75, 142]。	可在延迟关键领域实现实时控制与部署[251, 191]（例如：无人机、机械臂）。
多模态动作表示固定；安置	结合扩散与自回归策略的混合分词模型[171]；能够处理多样化的演示内容及多模态输出[156]。	在复杂动态操作中提升性能三重有效解模态[74]。
开放世界中的安全保证	动态风险评估模块[183, 233]；低延迟紧急响应层与自适应规划层[113]。	在不可预测环境中（家庭、工厂、医疗保健场所）提升可靠性和安全性；增强用户接受度。
数据集偏差与基准问题	整理多样化/无偏见数据集[185]；强化基础训练（例如采用硬样本负样本本进行CLIP微调）[282, 17]。	提升了公平性与语义保真度[109]，并增强了对新型现实世界输入的泛化能力[227, 288, 175]。
有限的三维感知能力推理	集成深度/激光雷达技术；开发三维感知架构；融合点云数据具有视觉-语言特征的云	支持更强的空间推理能力，以实现操作与导航功能复杂环境[129]。
跨实体化物化作用	通用的、一般的 训练跨越多种形态学特征；学习与具身性无关的动作抽象化处理；应用跨域适应技术[266]。	促进机器人平台与配置间的政策转移[279, 122]。
注释复杂性与代价	弱监督、主动学习与合成数据生成技术用于数据重构进行人工标记[148]。	降低开发成本并加速向新市场的扩展任务/领域[233, 286]。
模拟到真实传输间隙	领域适应性、仿真-真实微调及真实世界校准[210, 134]。	在实际部署（而非仅限于仿真环境）时，可提升系统的可靠性和一致性[4, 66]。
物理知识的整合边	物理学先验知识、模拟环境及动力学建模训练流程[53]。	在真实物理约束条件下提升预测与规划能力[106]。
多模态整合（触觉、听觉）	融合视觉与语言的触觉/听觉信息[114]；扩展多模态变换器融合技术。	提高了模型在遮挡/歧义条件下的鲁棒性，并扩展了任务范围[76, 136, 91]。
长期视角工作	多级 分层策略、内存增强型网络及轨迹规划模块[136]。	提升顺序规划、内存管理及组合执行性能[131, 286, 227, 175]。
系统集成复杂性	统一变压器主干结构[289]；时间对齐与同步迁移策略[163, 277]。	可实现更紧密的规划-控制协调，并向物理机器人提供更稳健的传递[187, 208]。
能源与计算需求	修剪技术、LoRA、量化感知训练及低功耗加速器。	可实现高效的嵌入式/移动设备部署[252, 283, 125, 246]。
对未见任务的泛化能力	组合泛化、少样本元学习及任务无关预训练[173, 146]。	可降低过拟合风险并增强零样本/少样本适应能力[97,242, 286]。
环境耐受性变化性	领域随机化、传感器融合、更广泛的训练数据集以及在线重新校准[156]。	在光照变化、杂乱环境及场景动态条件下提升稳定性ics [285, 242]。
伦理与社会影响行动；举措	通过设备端处理/匿名化实现隐私保护[149, 198, 252, 34]；公平性审计；监管与信托框架	促进在社会、医疗等领域的公平、可信的收养实践。以及劳动领域[160, 180, 217, 172]。

由于 OpenVLA[122]和 Pi-0[15]在这种顺序令牌生成方法中存在固有缺陷，从而限制了它们在动态环境中的有效性。

诸如并行解码等新兴解决方案（以 NVIDIA 公司的GR00T N1模型[14]为例）旨在通过同时预测多个标记来加速推理过程。GR00T N1相比传统解码方法实现了约2.52倍的加速效果，但这种并行处理常导致轨迹平滑度下降，从而产生次优的机器人运动轨迹。在手术机器人等对精度、适应性和灵活性要求极高的敏感应用场景中，此类运动轨迹显然不可取。因此，在保证输出质量的前提下实现快速推理仍是亟待解决的技术难题。

此外，硬件限制加剧了实时推理的约束。例如，处理高维视觉嵌入（通常涉及超过400个视觉标记，每个标记维度为512）需要约1.2 GB/s的内存带宽。这一需求远超当前嵌入式系统或边缘AI硬件（如 NVIDIA Jetson平台）的处理能力，从而限制了实际部署[82, 275]。即使采用高效的

量化技术（通过降低浮点运算精度来缓解内存限制），模型仍常出现精度下降的问题，尤其是在需要亚毫米级精度的任务中，例如双臂机器人操作或医疗机器人应用。

4.2. 多模态动作表示与安全保证

多模态动作表征：当前视觉语言表征（VLA）模型的一个显著局限在于难以准确表征多模态动作，尤其在需要连续且精细控制的场景中[62, 46]。传统离散分词方法（如将动作划分为256个独立区间）本质上存在精度不足的问题，在精细任务（如精密机器人抓取或复杂手术操作）中会产生显著误差[171]。例如在装配任务中的精密机器人操作中，离散表征可能导致动作对位偏差或执行不精确，从而影响系统性能与可靠性。另一方面，基于连续多层感知机（MLP）的方法存在模式坍缩风险[162, 232]——尽管存在多种可行路径，模型仍会过早收敛至单一动作轨迹。这削弱了自适应决策所需的灵活性。

highly dynamic environments. Emerging diffusion-based policies, exemplified by models like Pi-Zero and RDT-1B [144], offer richer multi-modal action representation capable of capturing diverse action possibilities. However, their substantial computational overhead, approximately three times that of conventional transformer-based decoders, renders them impractical for real-time deployment. Consequently, VLA models currently struggle with complex dynamic tasks, such as robotic navigation in densely crowded spaces or sophisticated bimanual manipulations [74, 247], where multiple strategic actions may be equally valid and contextually dependent.

Safety Assurance in Open World: Another critical challenge facing VLAs is ensuring robust safety in dynamic, unpredictable environments characteristic of real-world scenarios [39, 274]. Many current implementations depend heavily on predefined, hardcoded force and torque thresholds, significantly constraining their adaptability in encountering unforeseen or novel conditions, such as unexpected obstacles or sudden environmental changes [156]. Models used for collision prediction typically attain only about 82% accuracy in cluttered and dynamic spaces, posing serious risks in applications such as warehouse logistics or household robotics, where safety margins are minimal [288, 122]. Moreover, the essential safety mechanisms like emergency stops incorporate substantial latency often between 200 and 500 milliseconds due to comprehensive safety verifications [170, 122]. This delay, although seemingly minor, can prove hazardous in high-speed operations or critical interventions, such as automated driving or emergency robotic responses.

4.3. Dataset Bias, Grounding, and Generalization to Unseen Tasks

A significant obstacle limiting the effectiveness of VLA models is the pervasive presence of dataset bias and grounding deficiencies. Current training datasets, predominantly sourced from web-crawled repositories, frequently exhibit inherent biases [214, 119]. Studies indicate that approximately 17% of the associations within standard datasets are skewed toward stereotypical interpretations, such as disproportionately associating terms like “doctor” with male figures [222, 124]. These biases propagate through training, resulting in VLAs that produce semantically misaligned or contextually inappropriate responses when deployed in diverse environments. For instance, models such as OpenVLA have been documented to overlook approximately 23% of object references in novel settings, significantly limiting their practical utility in real-world applications where accurate interpretation of instructions is critical [122]. This grounding issue also extends to challenges in compositional generalization, where VLAs often fail when encountering rare or unconventional combinations, such as interpreting a phrase like “yellow horse” because of underrepresentation in training corpora. These shortcomings highlight an urgent need for carefully curated, balanced, and comprehensive, domain-specific datasets, coupled with advanced grounding algorithms designed to mitigate biases and enhance semantic alignment across varied contexts.

Complementing the challenges posed by dataset bias is the broader issue of generalization to unseen tasks, a critical barrier for the practical deployment of VLAs. While existing models demonstrate proficiency in familiar environments or tasks similar to their training scenarios, their performance significantly degrades, often by as much as 40%, when encountering entirely novel tasks or unfamiliar variations. For example, a VLA trained specifically on domestic tasks may struggle or fail when introduced into industrial or agricultural settings, largely due to discrepancies in object types, environmental dynamics, and operational constraints. This limitation arises primarily from overfitting to narrowly scoped training distributions and insufficient exposure to diverse task representations. Consequently, current VLAs exhibit limited generalization in zero-shot or few-shot learning scenarios, impeding their adaptability and scalability.

4.4. System Integration Complexity and Computational Demands

Integrating VLA models within dual-system architectures, which combine high-level cognitive planning (System 2) and real-time physical control (System 1), presents significant complexity in robotic applications. A primary challenge arises from temporal mismatches between these two systems. Typically, System 2 leverages LLMs such as GPT or LLaMA-4 for complex task decomposition and strategic planning. These models, due to their substantial computational requirements, often incur inference latencies on the order of ~800 ms or more when executed on standard GPU-based inference platforms (e.g., single high-end consumer or datacenter GPUs) commonly used for LLM deployment. Conversely, System 1 components responsible for low-level motor execution typically run within tightly constrained control loops implemented on real-time CPUs, microcontrollers, or dedicated robot controllers, where update intervals on the order of several milliseconds are common, depending on the platform and task. This stark discrepancy in operational cadence leads to synchronization difficulties, causing delays and potentially suboptimal execution trajectories. For example, NVIDIA’s GR00T N1 model demonstrates an effective integration of these two systems but still suffers from occasional jerkiness in motion due to asynchronous interaction, highlighting this intrinsic challenge.

Furthermore, the feature space misalignment between high-dimensional vision encoders, such as Vision Transformers (ViT), and lower-dimensional action decoders exacerbates integration complexity. When attempting to reconcile these disparate embeddings, the coherence between perceptual understanding and actionable commands can deteriorate significantly. OpenVLA [122] and RoboMamba [143], which utilize transformer-based visual processing and subsequent action decoding, illustrate these integration challenges resulting in diminished performance when ported from simulation environments to physical hardware deployments. Such discrepancies may lead to reduction in performance, primarily due to mismatches between simulated dynamics and real-world sensor noise or calibration issues [122, 49, 92].

Energy and compute demands constitute another significant barrier for VLA deployment, particularly in edge computing

在高度动态的环境中，新兴的基于扩散策略的模型（如Pi-Zero和 RDT-1B[144]）通过提供更丰富的多模态动作表征，能够捕捉多样化的动作可能性。然而，其计算开销显著高于传统Transformer解码器（约高出三倍），导致难以实现实时部署。因此，VLA模型在处理复杂动态任务时仍存在局限性，例如密集拥挤空间中的机器人导航或精密双臂操作[74, 247]——这类场景中多个战略动作可能具有同等有效性且受具体情境影响。

开放世界中的安全保障：VLA面临的另一项关键挑战是在现实场景特有的动态、不可预测环境中确保可靠的安全性[39, 274]。当前许多实现方案高度依赖预定义的硬编码力与扭矩阈值，这显著限制了其在遭遇意外障碍物或环境突然变化等未知或新型场景时的适应能力[156]。用于碰撞预测的模型在复杂动态环境中通常仅能达到约82%的准确率，这在安全裕度极小的仓库物流或家用机器人等应用中构成严重风险[288, 122]。此外，紧急停止等核心安全机制因需进行全面的安全验证，往往会产生200至500毫秒的显著延迟[170, 122]。这种延迟虽然看似微不足道亦无法证明在高速操作或关键干预措施（如自动驾驶或紧急机器人响应）中具有危害性。

4.3. 数据集偏差、基准效应以及对未见过任务的泛化能力

限制VLA模型有效性的主要障碍在于数据集普遍存在的偏差和标注缺陷。当前训练数据集主要来源于网络爬虫存储库，往往存在固有偏见[214, 119]。研究表明，标准数据集中约17%的关联关系存在刻板化解读倾向，例如将“医生”等术语与男性形象过度关联[222, 124]。这些偏见在训练过程中持续传播，导致VLA模型在多样化场景中输出语义失配或语境不恰当的响应。例如OpenVLA等模型在新场景中被记录遗漏约23%的对象引用，这严重制约了其在需要精准指令解读的现实应用中的实际效用[122]。这种知识映射问题同样体现在组合泛化能力的挑战上：当遇到罕见或非传统组合时，视觉语言助理（VLAs）往往表现欠佳——例如由于训练语料库中代表性不足，导致对“黄马”这类短语的解读存在偏差。这些缺陷凸显出两大迫切需求：一是需要精心构建、均衡且全面的领域特定数据集；二是亟需开发先进的知识映射算法，以消除系统偏见并提升跨语境语义匹配能力。

除了数据集偏差带来的挑战外，更普遍存在的问题是模型对未知任务的泛化能力不足，这成为可学习自动驾驶系统（VLAs）实际应用中的关键瓶颈。现有模型在熟悉环境或与训练场景相似的任务中表现优异，但一旦遇到全新任务或陌生场景，其性能就会显著下滑，降幅甚至可达40%。例如，专门针对家庭场景训练的VLAs在工业或农业环境中可能表现挣扎甚至失败，这主要源于物体类型差异、环境动态变化及操作限制条件的不同。这种局限性主要源于过度依赖狭窄训练数据分布，以及缺乏多样化任务表征的接触经验。因此，当前VLAs在零样本或少样本学习场景中普遍存在泛化能力不足的问题，严重制约了其适应性和扩展性。

4.4. 系统集成复杂性与计算需求

在融合高级认知规划（系统2）与实时物理控制（系统1）的双系统架构中集成VLA模型，对机器人应用而言具有显著复杂性。主要挑战源于这两个系统之间的时间错配问题。通常，系统2会利用GPT或LLaMA-4等大语言模型进行复杂任务分解与战略规划。由于这些模型计算需求极高，在标准GPU推理平台（如高端消费级GPU或数据中心GPU）上运行时，常会产生约800毫秒甚至更长的推理延迟。而负责低级运动执行的系统1组件则运行在实时CPU、微控制器或专用机器人控制器上构建的严格控制环路中，根据平台和任务需求，其更新间隔通常为数毫秒量级。这种操作节奏的显著差异会导致同步困难，引发延迟并可能产生次优执行轨迹。例如，NVIDIA的GR00T N1型号成功实现了这两个系统的有效整合，但由于异步交互的存在，其运动仍偶尔会出现顿挫现象，这凸显了该设计固有的挑战。

此外，高维视觉编码器（如视觉Transformer ViT）与低维动作解码器之间的特征空间错位会加剧集成复杂性。在尝试协调这些差异化的嵌入表示时，感知理解与可执行指令之间的连贯性可能显著下降。采用基于Transformer的视觉处理及后续动作解码技术的OpenVLA[122]和RoboMamba[143]系统，其集成挑战在从仿真环境移植至物理硬件部署时导致性能下降。此类差异主要源于模拟动力学与真实世界传感器噪声之间的不匹配，或校准问题导致的性能降低[122, 49, 92]。

能源与计算需求是甚大天线阵列（VLA）部署面临的另一重大障碍，尤其在边缘计算场景中。

contexts typical of autonomous drones, mobile robots, and wearable robotic systems. The substantial parameter counts typical of advanced VLAs (e.g., models possessing upwards of 7 billion parameters) necessitate computational resources often exceeding 28 GB of VRAM in their native form. These requirements are substantially higher than the capabilities of most current edge-oriented processors and GPUs, restricting the practical applicability of sophisticated VLAs outside specialized, high-resource environments.

4.5. Robustness and Ethical Challenges in VLA Deployment

A central barrier to the real-world deployment of VLA models lies in their limited robustness to environmental variability, which in turn raises important ethical and safety considerations. Environmental robustness refers to a system’s ability to sustain reliable perception, reasoning, and action generation under dynamically changing and partially observable conditions. In practice, real-world environments introduce significant uncertainty through factors such as fluctuating illumination, adverse weather, sensor noise, and object occlusions.

Empirical evidence highlights these limitations across multiple VLA components. For example, vision modules employed in systems such as OpenDriveVLA [293] experience accuracy degradations of approximately 20–30% in low-contrast or shadow-dominated scenes, reflecting the sensitivity of current visual encoders to challenging lighting conditions. Similarly, language understanding in VLAs such as CoVLA [5] deteriorates in acoustically noisy or semantically ambiguous settings, where instruction misinterpretation can propagate into incorrect action execution. In manipulation-centric scenarios, VLA-enabled robotic systems like RoboMamba [143] struggle in cluttered environments, frequently misestimating the pose or orientation of partially occluded objects and thereby reducing task success rates.

These robustness limitations have direct ethical implications in safety-critical deployments, as performance degradation under real-world variability can lead to unintended behaviors, reduced reliability, and loss of user trust. Addressing robustness is therefore not only a technical challenge but also a prerequisite for responsible and ethical deployment of VLA systems in human-centered environments.

5. Discussion

As illustrated in Figure 17, VLA models face a multifaceted set of challenges that span algorithmic, computational, and ethical dimensions. First, achieving real-time inference on resource-constrained hardware remains difficult due to the sequential nature of autoregressive decoders and the high dimensionality of multi-modal inputs. Second, fusing vision, language, and action into coherent policies introduces safety vulnerabilities when encountering unanticipated environmental changes. Third, dataset bias and grounding errors compromise generalization, often causing models to fail on out-of-distribution tasks. Fourth, integrating diverse components perception, reasoning, and control yields complex architectures

that are hard to optimize and maintain. Fifth, the energy and compute demands of large VLA systems hinder the deployment on embedded or mobile platforms.

Finally, limited robustness to environmental variability can result in unsafe or unreliable behavior, which in turn raises ethical and regulatory concerns related to safety assurance, accountability, privacy, and bias mitigation. Collectively, these limitations constrain the practical adoption of VLA models in real-world robotics, autonomous systems, and interactive applications. The potential solutions to these challenges are discussed below.

5.1. Potential Solutions

- Real-Time Inference Constraints.** Future research must develop VLA architectures that harmonize latency, throughput, and task-specific accuracy. One promising direction is the integration of specialized hardware accelerators such as FPGA-based vision processors and tensor cores optimized for sparse matrix operations to execute convolutional and transformer layers at sub-millisecond scales [122, 129]. Model compression techniques like Low-Rank Adaptation (LoRA) [93] and knowledge distillation can shrink parameter counts by up to 90%, reducing both memory footprint and inference time while retaining over 95% of original performance on benchmark tasks. Progressive quantization strategies that combine mixed-precision arithmetic (e.g., FP16/INT8) with block-wise calibration can further cut computation by 2–4× with minimal accuracy loss [121]. Adaptive inference architectures that dynamically adjust network depth or width based on input complexity akin to early-exit branches in DeeR-VLA [269] can reduce average compute by selectively bypassing transformer layers when visual scenes or linguistic commands are simple. Finally, efficient tokenization schemes leveraging subword patch embeddings and dynamic vocabulary allocation can compress visual and linguistic input into compact representations, minimizing token counts without sacrificing semantic richness [171]. Together, these innovations can enable sub-50 ms end-to-end inference on commodity edge GPUs, paving the way for latency-sensitive applications in autonomous drone flight, real-time teleoperation, and collaborative manufacturing.
- Multi-modal Action Representation and Safety Assurance.** Addressing multi-modal action representation and robust safety requires end-to-end frameworks that unify perception, reasoning, and control under stringent safety constraints. Hybrid policy architectures combining diffusion-based sampling for low-level motion primitives [40] with autoregressive high-level planners [242] enable compact stochastic representations of diverse action trajectories, improving adaptability in dynamic environments. Safety can be enforced via real-time risk assessment modules that ingest multi-sensor fusion streams including visual, depth, and proprioceptive data to predict

这类系统是自主无人机、移动机器人及可穿戴机器人系统的典型代表。先进的虚拟现实增强设备（VRA）通常具有庞大的参数规模（例如某些型号的参数数量超过70亿），其所需的计算资源往往超过28GB的显存（VRAM）。这些要求远超当前大多数边缘计算处理器和GPU的处理能力，从而限制了复杂VRA在非专用、高资源环境下的实际应用可行性。

4.5. 甚大天线阵列（VLA）部署中的稳健性与伦理挑战

VLA模型在实际应用中面临的主要障碍在于其对环境变化的鲁棒性有限，这进而引发了重要的伦理和安全问题。环境鲁棒性指的是系统在动态变化且部分可观察的条件下，仍能保持可靠感知、推理及行动生成的能力。实际上，现实环境会通过光照波动、恶劣天气、传感器噪声以及物体遮挡等因素引入显著的不确定性。

实证研究揭示了视觉语言助手（VLA）多个组件存在的局限性。例如，在OpenDriveVLA等系统中使用的视觉模块[293]在低对比度或阴影主导场景中会出现约20%-30%的精度下降，这反映出当前视觉编码器对复杂光照条件的敏感性。类似地，CoVLA等VLA系统在声学噪声环境或语义模糊场景中的语言理解能力会下降，指令误判可能导致错误动作执行。在以操作为中心的场景中，RoboMamba等基于VLA的机器人系统[143]在杂乱环境中表现欠佳，常因误判部分遮挡物体的姿态或方位而导致任务成功率降低。

这些鲁棒性局限性在安全关键场景中具有直接的伦理影响，因为在现实环境变化条件下性能下降可能导致非预期行为、可靠性降低以及用户信任度下降。因此，解决鲁棒性问题不仅是技术挑战，更是以人中心环境中可穿戴激光雷达（VLA）系统负责任且符合伦理部署的必要前提。

5. 讨论

如图17所示，VLA模型面临着涵盖算法、计算及伦理等多个维度的复杂挑战。首先，由于自回归解码器的顺序特性以及多模态输入的高度维度性，在资源受限的硬件平台上实现实时推理仍存在困难。其次，将视觉、语言和动作信息融合为连贯策略时，当遇到未预见的环境变化时会引入安全漏洞。第三，数据集偏差和基线误差会损害模型的泛化能力，导致其在分布外任务中表现不佳。第四，整合感知、推理和控制等多样化组件会形成复杂的架构，难以优化和维护。第五，大型VLA系统的高能耗和高计算需求

阻碍了其在嵌入式或移动平台上的部署。

最后，对环境变化的鲁棒性不足可能导致系统行为不安全或不可靠，进而引发涉及安全保障、责任归属、隐私保护及偏差缓解等方面的伦理与监管问题。这些局限性共同制约了虚拟现实增强（VLA）模型在现实世界机器人技术、自主系统及交互式应用中的实际应用。下文将探讨应对这些挑战的潜在解决方案。

5.1. 潜在解决方案

- **实时推理约束。**未来研究必须开发能够协调延迟、吞吐量与任务特定精度的VLA架构。一个有前景的方向是集成专用硬件加速器，例如基于FPGA的视觉处理器和针对稀疏矩阵运算优化的张量核心，以实现亚毫秒级卷积层与Transformer层的执行[122, 129]。模型压缩技术如低秩自适应（LoRA）[93]和知识蒸馏可将参数数量缩减高达90%，在基准任务中保持超过95%的原始性能的同时，显著降低内存占用与推理时间。结合混合精度算术（如FP16/INT8）与分块校准的渐进量化策略，可在保持最小精度损失的前提下将计算量进一步降低2-4倍[121]。类似于DecR-VLA中早期退出分支的自适应推理架构，能根据输入复杂度动态调整网络深度或宽度[269]，当视觉场景或语言指令较简单时，通过选择性跳过Transformer层可降低平均计算量。最后，基于子词块嵌入技术和动态词汇分配机制的高效分词方案，能够将视觉与语言输入压缩为紧凑表示形式，在不牺牲语义丰富度的前提下显著减少分词数量[171]。这些创新技术的协同应用，可在商用边缘GPU上实现亚50毫秒的端到端推理，为自主无人机飞行、实时远程操控及协同制造等对延迟敏感的应用场景开辟了新路径。

- **多模态动作表征与安全保障。**解决多模态动作表征与鲁棒性安全问题，需要能够在严格安全约束下统一感知、推理与控制的端到端框架。结合基于扩散的低级运动基元采样[40]与自回归式高层规划器[242]的混合策略架构，能够构建多样动作轨迹的紧凑随机表征，从而提升动态环境中的适应性。安全性可通过实时风险评估模块实现：该模块整合视觉、深度及本体感觉等多传感器融合数据流，以进行风险预测。

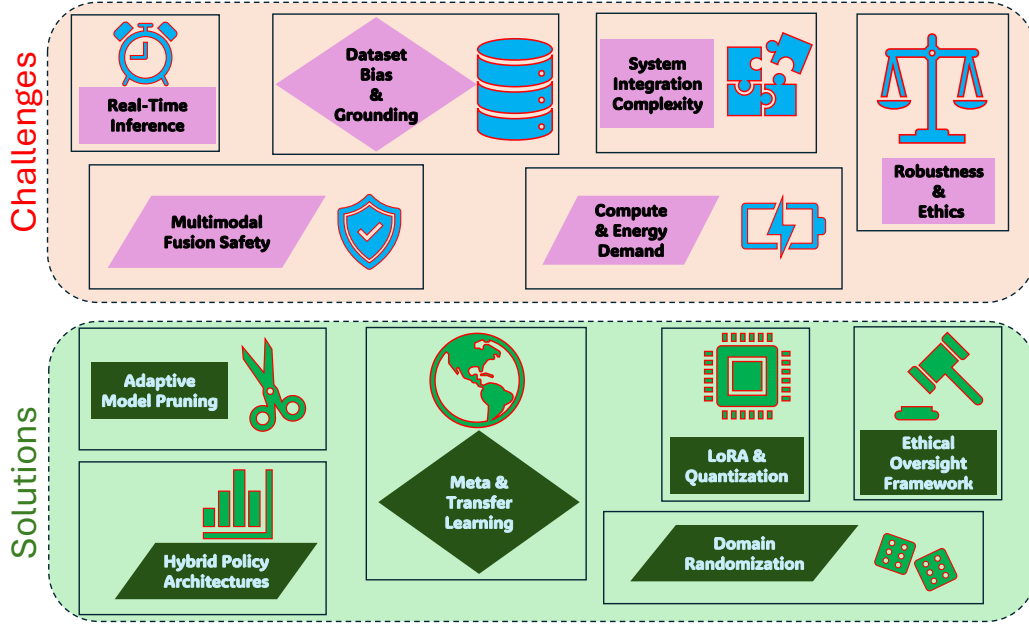


Figure 17: Figure maps six core VLA challenges; namely real-time inference, multimodal fusion safety, dataset bias, integration complexity, compute demands, and robustness/ethics; against six targeted solutions: adaptive pruning, hybrid policy architectures, meta/transfer learning, LoRA/quantization, domain randomization, and ethical oversight. This systematic alignment clarifies pathways to robust, efficient, and safe VLA deployment across broader real-world robotic domains.

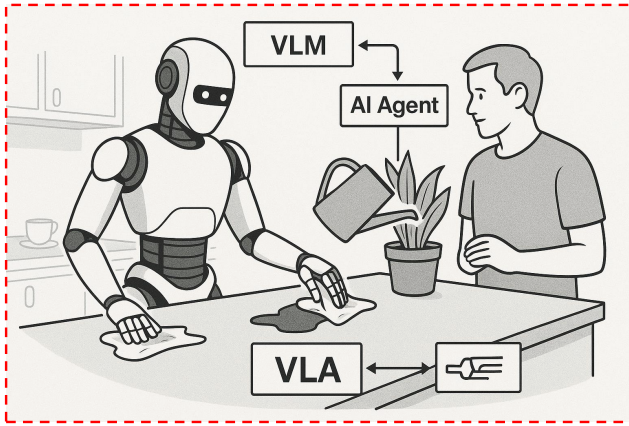


Figure 18: This conceptual illustration presents “Eva,” a future humanoid assistant powered by Vision-Language Models (VLMs), VLA frameworks, and agentic AI systems. VLMs enable semantic scene understanding and object affordance prediction, while VLAs translate language-grounded instructions into hierarchical motor plans. Agentic AI modules ensure adaptive learning, self-refinement, and interactive decision-making in open-ended environments. Together, these components represent a foundational blueprint for Artificial General Intelligence (AGI) in robotics, where perception, language understanding, planning, and safe autonomous behavior converge in real-world, socially aware tasks.

collision probability and joint stress thresholds, triggering emergency stop circuits when predefined safety envelopes are breached [183, 233]. Reinforcement learning algorithms augmented with constrained optimization (e.g., Lagrangian methods in SafeVLA [274]) can learn policies that maximize task success while strictly respecting safety constraints. Online model adaptation techniques

such as rule-based RL (GRPO) and Direct Preference Optimization (DPO) further refine action selection under new environmental conditions, ensuring consistent safety performance across scenarios [113]. Crucially, embedding formal verification layers that symbolically analyze planner outputs before execution can guarantee compliance with safety invariants, even for neural-network-based controllers. Integrating these methodologies will produce VLA systems that not only execute complex, multi-modal actions but do so with provable safety in unstructured, real-world settings.

- **Dataset Bias, Grounding, and Generalization to Unseen Tasks.** Robust generalization demands both broadened data diversity and advanced learning paradigms. Curating large-scale, debiased multi-modal datasets combining web-scale image-text corpora like LAION-5B [194] with robot-centric trajectory archives such as Open X-Embodiment [227] lays the groundwork for equitable semantic grounding. Hard-negative sampling and contrastive fine-tuning of vision-language backbones (e.g., CLIP variants) can mitigate spurious correlations and enhance semantic fidelity [17, 282]. Meta-learning frameworks enable rapid adaptation to novel tasks by learning shared priors across task families, as demonstrated in vision-language robotic navigation models [175]. Continual learning algorithms with replay buffers and regularization strategies preserve old knowledge while integrating new concepts, addressing catastrophic forgetting in VLA models [48]. Transfer learning from 3D perception domains (e.g., point cloud reasoning in 3D-VLA [288]) can endow models with stronger spatial inductive biases,

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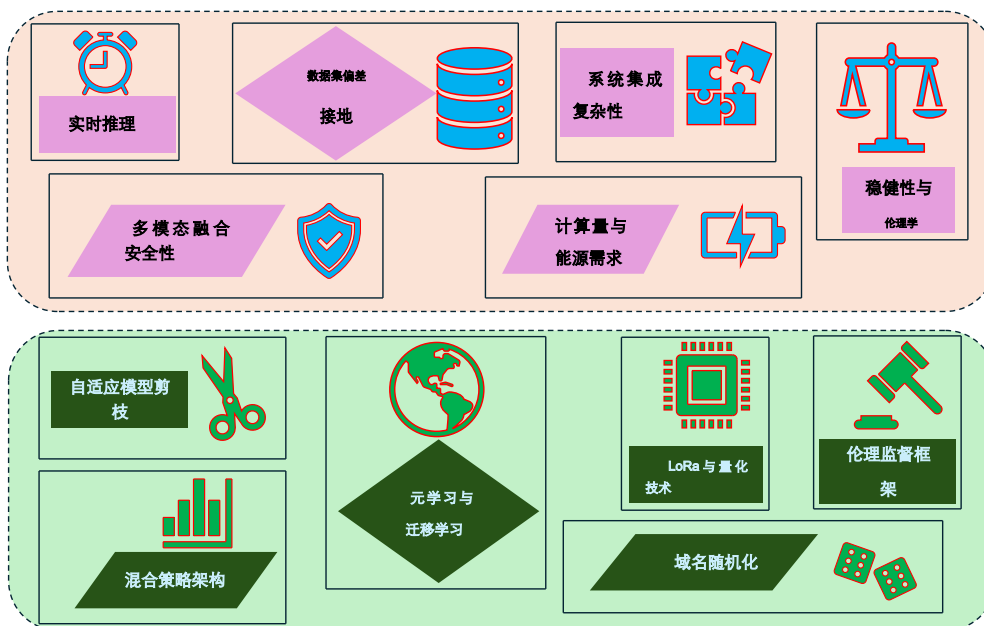


图17：该图系统性地展示了VLA（虚拟实验室架构）面临的六大核心挑战——实时推理能力、多模态融合安全性、数据集偏差问题、集成复杂度、计算资源需求以及鲁棒性与伦理考量——并针对六种针对性解决方案进行对比分析：自适应剪枝算法、混合策略架构、元学习/迁移学习技术、LoRA/量化技术、领域随机化方法及伦理监督机制。这种系统性对比清晰揭示了如何在更广泛的现实机器人应用场景中，实现稳健、高效且安全的VLA部署路径。

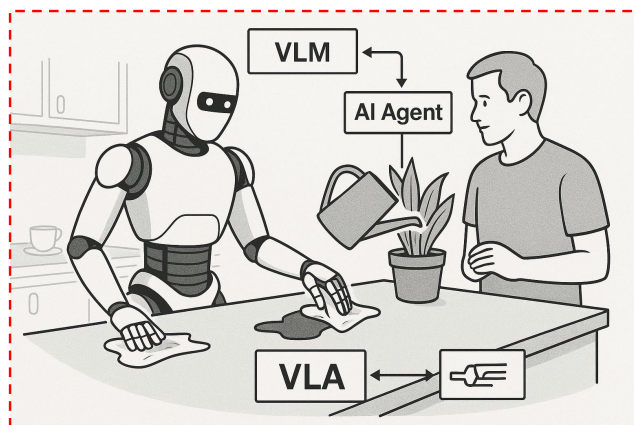


图18：本概念图展示了名为“伊娃”的未来人形助手，其核心架构基于视觉语言模型（VLMs）、视觉语言辅助框架（VLA）及智能代理AI系统。视觉语言模型能够实现场景语义理解与物体可用性预测，视觉语言辅助框架则将语言指令转化为分层运动规划方案。智能代理AI模块确保系统在开放环境中具备自适应学习能力、自我优化机制及交互式决策能力。这些组件共同构成了机器人领域人工通用智能（AGI）的基础架构蓝图，实现了感知能力、语言理解、规划决策与安全自主行为在现实场景中融合，助力社会感知型任务的完成。

术，可在新环境条件下持续优化动作选择，确保不同场景下安全性能的一致性[113]。关键在于，在执行前通过形式化验证层对规划器输出进行符号化分析，可确保即使基于神经网络的控制器也能符合安全不变量要求。整合这些方法论将构建出不仅能执行复杂多模态动作，还能在非结构化真实场景中实现可验证安全性的VLA系统。

- **数据集偏差、基准化与未见任务的泛化。**强健的泛化能力既需要扩展数据多样性，也需要先进的学习范式。通过整合大规模、无偏见的数据集——例如结合 LAION -5B[194]这类网络规模的图像-文本语料库与 Open XEmbodiment[227]这类以机器人为核心的轨迹档案库——为实现公平的语义基准化奠定了基础。硬负样本采样与视觉-语言骨干网络（如CLIP变体）的对比微调能够消除虚假相关性并提升语义保真度[17, 282]。元学习框架通过跨任务家族共享先验知识，实现了对新任务的快速适应，这在视觉-语言机器人导航模型[175]中得到了验证。采用重放缓冲区和正则化策略的持续学习算法既能保留旧知识，又能整合新概念，有效解决了VLA模型中的灾难性遗忘问题[48]。来自三维感知领域的迁移学习（例如3D-VLA中的点云推理[288]）可赋予模型更强的空间归纳偏置能力。

碰撞概率与联合应力阈值会在预设安全裕度被突破时触发紧急停止电路[183, 233]。结合约束优化技术的强化学习算法（例如SafeVLA中的拉格朗日方法[274]）能够学习出在严格遵守安全约束的前提下最大化任务成功率的策略。基于规则的强化学习（GR-PO）和直接偏好优化（DPO）等在线模型自适应技

thereby improving robustness to out-of-distribution scenarios. Finally, simulation-to-real (sim2real) fine-tuning with domain randomization and real-world calibration such as dynamic lighting, texture, and physics variations ensures that policies learned in synthetic environments transfer effectively to physical robots [4, 66]. These combined strategies will empower VLAs to generalize confidently to unseen objects, scenes, and tasks in real-world deployments.

- System Integration Complexity and Computational Demands.** To manage the intricate orchestration of multi-modal pipelines under tight compute budgets, researchers must embrace model modularization and hardware–software co–design. Low-Rank Adaptation (LoRA) adapters can be injected into pre-trained transformer layers, enabling task-specific fine-tuning without modifying core weights [93]. Knowledge distillation from large “teacher” VLAs to lightweight “student” networks, guided by mutual information–based objectives that encourage the student to match the teacher’s intermediate representations and action distributions, produces compact models with 5–10× fewer parameters while retaining 90–95% of task performance [121]. Mixed-precision quantization augmented by quantization-aware training can compress weights to 4–8 bits, cutting memory bandwidth and energy consumption by over 60% [122]. Hardware accelerators tailored for VLA workloads supporting sparse tensor operations, dynamic token routing, and fused vision–language kernels can deliver sustained 100+ TOPS throughput within a 20–30 W power envelope, meeting the demands of embedded robotic platforms [171, 242]. Toolchains like TensorRT-LLM [129] and TVM can optimize end-to-end VLA graphs for specific edge devices, fusing layers and precomputing static subgraphs. Emerging architectures such as TinyVLA demonstrate that sub-1 B parameter VLAs can achieve near-state-of-the-art performance on manipulation benchmarks with real-time inference, charting a path for widespread deployment in resource-constrained settings.
- Robustness to Environmental Variability in VLA Deployment.** Ensuring robust VLA performance in real-world settings requires targeted technical interventions to handle environmental uncertainty and long-term system drift. Domain randomization and synthetic data augmentation pipelines, such as UniSim’s closed-loop sensor simulator, generate photorealistic variations in lighting, occlusion, and sensor noise, thereby improving resilience to distributional shifts [264]. In addition, adaptive recalibration modules that dynamically adjust perception thresholds and control gains based on real-time feedback can mitigate performance degradation caused by sensor aging or changing operating conditions. Together, these approaches aim to enhance the stability and reliability of VLA systems under diverse and evolving deployment scenarios.
- Ethical, Privacy, and Societal Considerations in VLA**

Deployment. Beyond technical robustness, the deployment of VLA systems raises important ethical and societal challenges that require governance-oriented solutions. Bias auditing tools are needed to identify skewed demographic or semantic distributions in training data, followed by corrective strategies such as adversarial debiasing and counterfactual data augmentation [185, 282]. Privacy-preserving inference mechanisms, including on-device processing, homomorphic encryption for sensitive data streams, and differential privacy during training, are critical for safeguarding user data in domains such as healthcare and smart homes [160, 180]. Furthermore, transparent impact assessments, stakeholder engagement, and workforce upskilling initiatives can help manage socioeconomic effects, while regulatory frameworks and industry standards are essential to ensure accountability and responsible VLA adoption.

5.2. Future Roadmap

The future of VLA-based systems are expected to evolved at the intersection of increasingly capable multi-modal foundations, agentic reasoning, and embodied continual learning. Over the next decade, we anticipate several converging trends that will propel VLAs from capable yet brittle task specialists toward dependable, generalist robotic intelligence. However, this trajectory will be shaped by persistent constraints/limitations highlighted earlier: (i) real-time inference bottlenecks in closed-loop control, (ii) incomplete multi-modal action representations and weak safety assurance, (iii) dataset bias and grounding failures under distribution shift, (iv) integration complexity across perception-memory-reasoning-control, (v) high compute and energy demands that hinder edge deployment, and (vi) robustness, transparency, and ethical concerns in open-world settings. To address these issues holistically, Figure 19 summarizes a system-level research roadmap, while Figure 18 provides an intuitive conceptual illustration of how VLMs, VLA architectures, and agentic AI modules may co-evolve toward embodied AGI in robotics.

Multi-modal foundation models as the “cortex” for embodied perception: Today’s VLA stacks often rely on a vision-language backbone coupled to task-specific policy heads, which limits reuse of general knowledge and increases retraining cost across domains. A plausible next step is a unified multi-modal foundation model trained on web-scale image, video, text, and interaction/affordance traces to function as a shared “cortex” that encodes not only static semantics but also dynamics, contact priors, and commonsense physical knowledge [295, 289, 272]. Such a cortex can reduce failure modes caused by shallow correlations by grounding language in object-centric representations and persistent scene structure [206]. As emphasized in Figure 18, this foundation-model cortex would enable robots to segment environments into actionable entities (objects, regions, affordances) and provide stable semantic anchors to downstream planners and controllers. Yet, to prevent overconfidence and hallucinated grounding, these models must incorporate calibrated uncertainty and evidence-linked reasoning

从而显著提升了对分布外场景的鲁棒性。最后，通过结合领域随机化与真实世界校准（如动态光照、纹理及物理参数变化）的仿真到真实（sim2real）微调技术，确保在合成环境中学习到的策略能有效迁移至物理机器人[4, 66]。这些综合策略将使虚拟现实增强系统（VLAs）能够自信地泛化到真实部署场景中未见过的物体、场景及任务。

- **系统集成复杂度与计算需求。**为在有限计算预算下管理多模态流水线的复杂协调，研究人员必须采用模型模块化设计与软硬件协同设计策略。低秩自适应（LoRA）适配器可直接注入预训练Transformer层，无需修改核心权重即可实现任务特定微调[93]。通过基于互信息的目标函数引导知识蒸馏过程——从大型“教师”VLAs向轻量级“学生”网络迁移知识，促使学生网络匹配教师网络的中间表征与动作分布，最终生成参数量减少5-10倍、任务性能保持90-95%的紧凑模型[121]。结合量化感知训练的混合精度量化技术可将权重压缩至4-8位，使内存带宽与能耗降低超60%[122]。专为VLA工作负载设计的硬件加速器，通过支持稀疏张量运算、动态令牌路由以及融合视觉-语言内核，可在20-30瓦功耗范围内持续提供100+ TOPS吞吐量，完全满足嵌入式机器人平台的需求[171, 242]。TensorRT-LLM[129]和 TVM 等工具链可针对特定边缘设备优化端到端VLA图结构，实现层融合与静态子图预计算。TinyVLA等新兴架构证明，参数量小于1B的VLAs在实时推理的操控基准测试中已能接近行业顶尖水平，为资源受限场景下的大规模部署开辟了道路。
- **大规模可见光雷达（VLA）部署中的环境变异性鲁棒性研究。**为确保VLA系统在实际应用环境中的稳定性，需要采取针对性技术干预措施来应对环境不确定性及系统长期漂移问题。领域随机化技术与合成数据增强流程（如UniSim闭环传感器模拟器）能够生成光照条件、遮挡效果及传感器噪声的逼真变化，从而提升系统对分布偏移的适应能力[264]。此外，基于实时反馈动态调整感知阈值与控制增益的自适应重校准模块，可有效缓解传感器老化或运行条件变化导致的性能下降。这些技术手段共同致力于提升VLA系统在多样化及动态变化的部署场景中的稳定性与可靠性。

- **VLA部署中的伦理、隐私与社会考量。**除了技术稳健

性之外，VLA系统的部署还带来了重要的伦理和社会挑战，需要以治理为导向的解决方案。需要使用偏差审计工具来识别训练数据中存在偏斜的人口统计或语义分布，随后采取对抗性去偏和反事实数据增强等纠正策略[185, 282]。在医疗保健和智能家居等领域，保护用户数据的关键在于采用隐私保护推理机制，包括设备端处理、敏感数据流的同态加密以及训练过程中的差分隐私[160, 180]。此外，透明的影响评估、利益相关者参与以及员工技能提升计划有助于管理社会经济影响，而监管框架和行业标准对于确保问责制和负责任地采用VLA至关重要。

5.2. 未来路线图

基于视觉语言模型（VLA）的系统未来发展趋势，预计将依托日益强大的多模态基础架构、智能体推理能力和具身化持续学习技术实现融合。未来十年间，我们预计会出现多个融合趋势，推动VLA从具备特定任务能力但存在局限性的专家型机器人，向可靠通用型机器人智能体系迈进。然而这一发展路径仍将受到前文所述持续存在的制约因素影响：(i)闭环控制中的实时推理瓶颈，(ii)多模态动作表征不完整与安全保障机制薄弱，(iii)数据集偏移导致的偏差问题及基线失效现象，(iv)感知-记忆-推理-控制系统的集成复杂度，(v)阻碍边缘部署的高计算能耗与能源需求，以及(vi)开放世界场景中系统鲁棒性、透明度与伦理争议。为系统性解决这些问题，图19总结了系统级研究路线图，图18则直观展示了视觉语言模型、VLA架构与智能体AI模块如何协同进化，最终实现机器人领域的具身化通用人工智能。

*多模态基础模型作为具身感知的“大脑皮层”：*当前的视觉语言架构通常依赖于视觉语言主干网络与任务特定策略头的耦合，这限制了通用知识的复用性并增加了跨领域再训练成本。一个合理的下一步是构建统一的多模态基础模型，通过网络规模的图像、视频、文本及交互/可用性轨迹进行训练，使其作为共享“大脑皮层”不仅编码静态语义，还能处理动态变化、接触先验及常识性物理知识[295, 289, 272]。此类大脑皮层可通过将语言嵌入以物体为中心的表征和持久场景结构中，减少由浅层关联导致的失败模式[206]。如图18所示，该基础模型大脑皮层可使机器人将环境分割为可操作实体（物体、区域、可用性），并为下游规划器和控制器提供稳定的语义锚点。然而为避免过度自信与幻觉性定位，这些模型必须整合校准后的不确定性及证据关联推理机制。

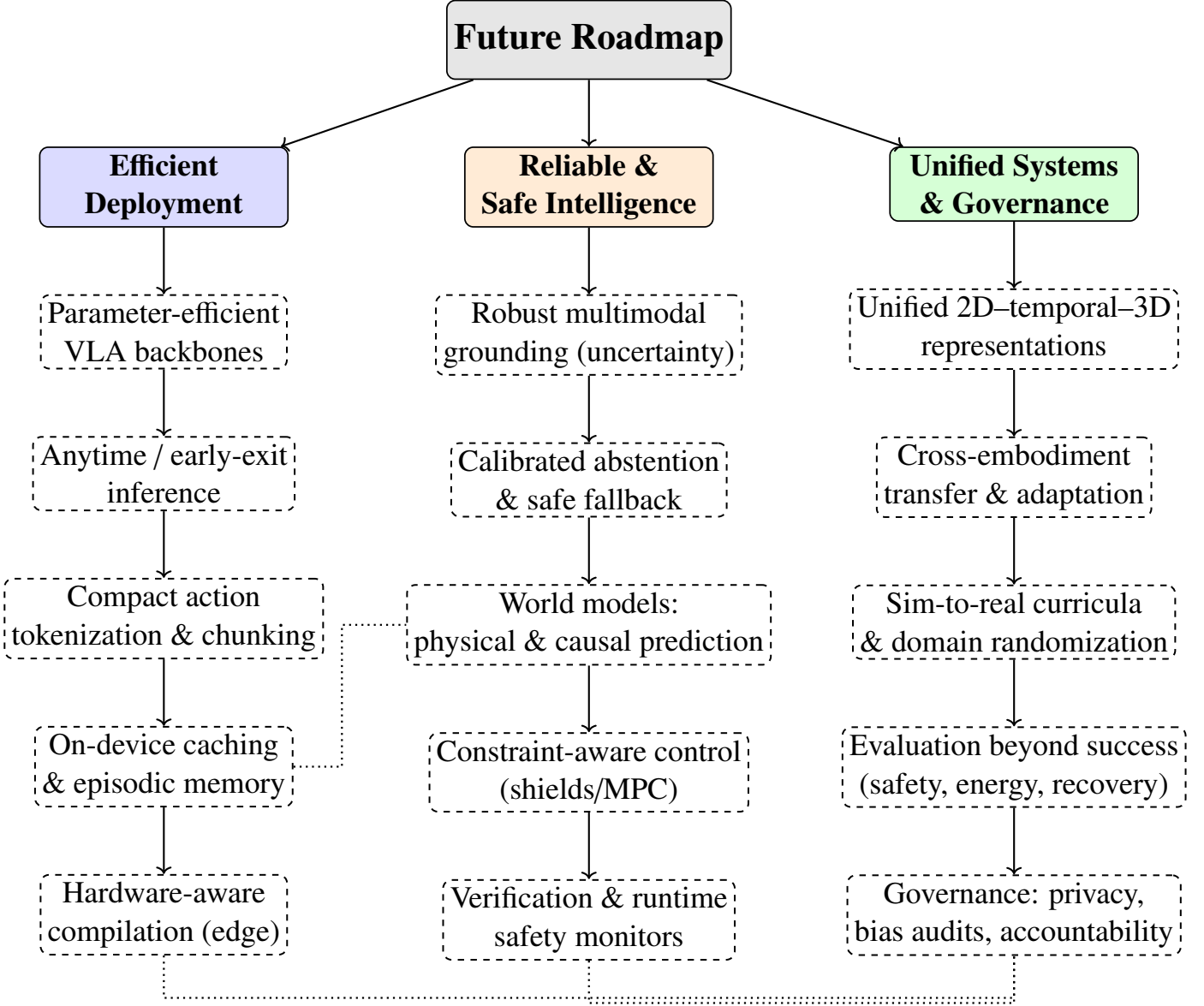


Figure 19: **Future research roadmap for Vision-Language-Action (VLA) models (Fig. 19).** Dashed boxes denote prioritized directions addressing the core limitations of VLA models identified in this work: (i) *Efficient Deployment* (latency, compute, energy), (ii) *Reliable & Safe Intelligence* (grounding, uncertainty, safety assurance), and (iii) *Unified Systems & Governance* (2D-temporal-3D integration, transfer, evaluation, and responsible deployment).

to ensure that perception-driven plans remain verifiable under occlusion, clutter, and ambiguous instructions [145].

Agentic, self-supervised, lifelong learning and continual adaptation.: A defining limitation of current VLAs is their static nature: policies trained once are deployed unchanged, despite operating in non-stationary environments. Future VLAs should adopt agentic learning loops where models propose exploration objectives, hypothesize outcomes, and self-correct through simulated and real rollouts, enabling continual skill growth over months or years [43, 90]. This direction is expected to mitigate distribution shift, dataset bias, and long-horizon brittleness by allowing models to adapt over time; however, continual policy updates introduce new risks. In particular, repeated on-line or incremental learning can overwrite previously acquired

competencies (catastrophic forgetting), induce unintended behavioral regressions, and increase susceptibility to noisy, adversarial, or unintended environmental feedback that may corrupt the learned policy [292, 263]. Therefore, lifelong learning must be coupled with replay and safety-aware updates, modular adapters, and verification-informed policy revisions [121]. Within the roadmap of Figure 19, this agentic lifelong learning paradigm falls naturally at the intersection of Reliable & Safe Intelligence and Unified Systems & Governance, where continual learning is treated as a controlled, auditable lifecycle process rather than an ad hoc fine-tuning step.

Hierarchical, neuro-symbolic planning for scalability and interpretability.: Scaling from low-level motor primitives to long-horizon objectives requires explicit hierarchy [261, 265].

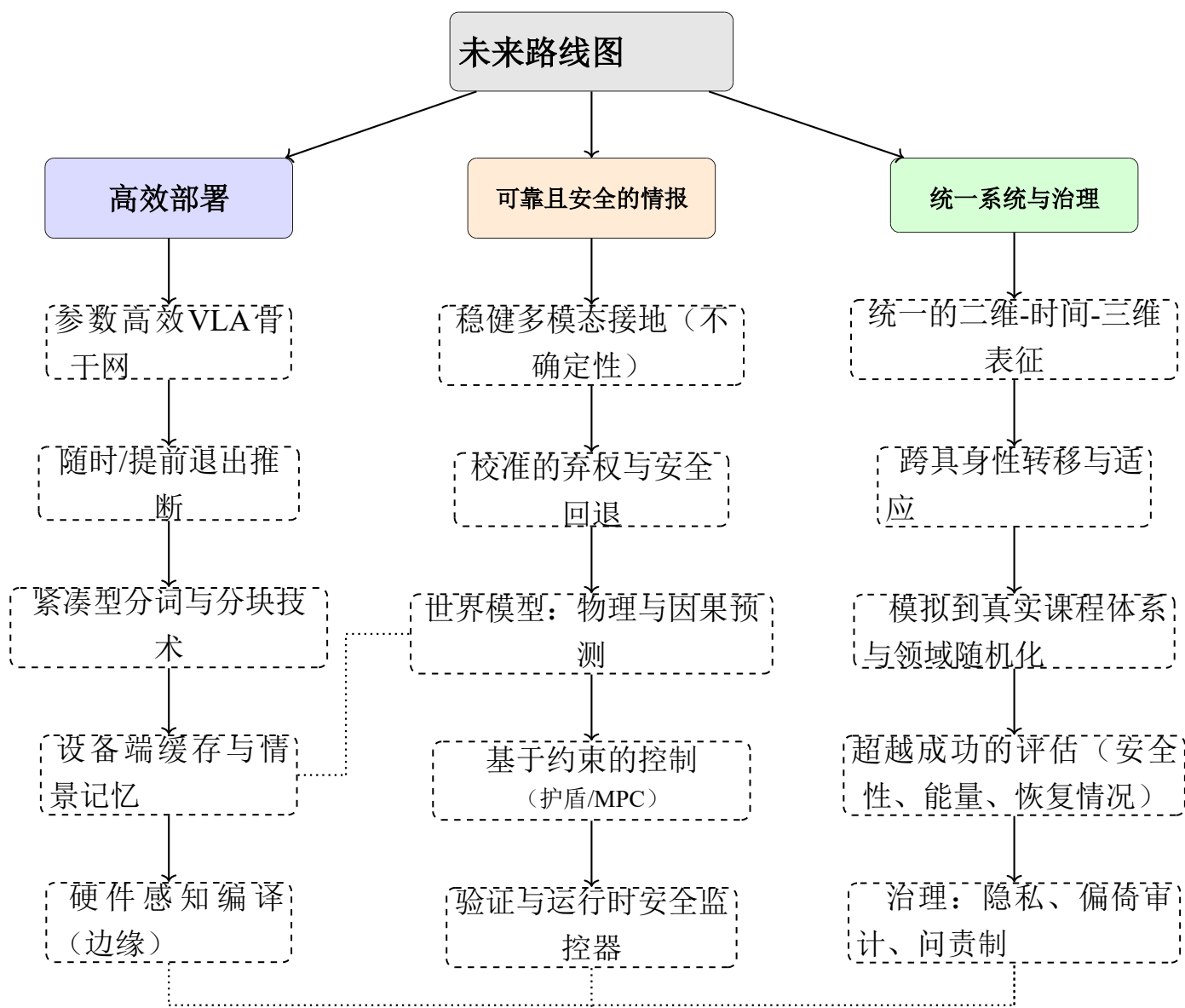


图19: 视觉-语言-动作 (VLA) 模型未来研究路线图 (图19)。虚线框标示了针对本研究中识别的VLA模型核心局限性所确定的优先方向: (i) 高效部署 (延迟、计算、能耗), (ii) 可靠与安全智能 (基底化、不确定性、安全保证), 以及 (iii) 统一系统与治理 (二维-时序-三维集成、迁移、评估及负责任部署)。

为确保基于感知驱动的计划在遮挡、杂乱环境及模糊指令条件下仍具有可验证性[145]。

代理式学习、自我监督、终身学习与持续适应: 当前虚拟智能助手 (VLA) 的核心局限在于其静态特性: 即便在非平稳环境中运行, 单次训练的策略也会原封不动地部署使用。未来的VLA应采用代理学习循环机制, 使模型能够提出探索目标、假设结果并通过模拟与真实场景部署进行自我修正, 从而实现数月乃至数年的持续技能提升[43, 90]。这种发展方向有望通过模型随时间适应性调整来缓解分布偏移、数据集偏差及长期脆弱性问题, 但持续策略更新也带来新风险。特别是重复在线学习或增量学习可能导致先前习得能力被覆盖 (灾难性遗忘)、引发非预期行

为退化, 并增加对噪声干扰、对抗性环境或意外反馈的敏感性, 这些因素可能破坏已学策略[292, 263]。因此, 终身学习必须与回放机制、安全感知更新、模块化适配器及验证驱动策略修订相结合[121]。在图19所示的发展路线图中, 这种自主型终身学习范式自然位于“可靠与安全智能”与“统一系统与治理”的交汇点——在此框架下, 持续学习被视为一个受控且可审计的生命周期过程, 而非临时性的微调步骤。

基于可扩展性与可解释性的层次化神经符号规划: 从低级运动基本单元到长期目标的扩展需要明确的层次结构[261, 265]。

Next-generation VLA systems will likely use language-grounded planners (LLM-style modules fine-tuned for affordances and constraints) that decompose goals into structured sub-tasks, followed by mid-level skill policies and low-level controllers that ensure compliant motion [94, 253]. This neuro-symbolic blend helps bridge integration complexity by imposing interfaces that are easier to debug, monitor, and certify [201]. Importantly, hierarchy also enables selective verification: high-level plans can be checked for constraint violations (unsafe steps, forbidden regions) [192, 71], while low-level trajectories can be shielded by control barrier functions, MPC, and runtime safety monitors [188, 196]. These components align with Figure 19 under the pillar of *Reliable & Safe Intelligence*, where safety is enforced through both planning-time constraints and execution-time guards rather than post hoc evaluation alone [101, 274].

Real-time adaptation via world models and physical/causal reasoning:. Robust deployment in unstructured settings demands that VLAs maintain internal predictive models of objects, contacts, and dynamics. World models that forecast near-term state transitions and failure likelihoods can support counterfactual evaluation (“if I push here, what collides?”) and rapid corrective actions when reality deviates from expectation (e.g., slipping grasp, unexpected friction) [203]. This capability is central to safe manipulation, navigation, and human-robot interaction, where small errors compound quickly [156]. Yet, world models must be efficient enough for on-board use and consistent with multi-sensor evidence. Thus, a major future direction is hardware-aware, memory-efficient predictive modeling such as temporal token compression [267, 216], event-driven state updates [250, 226], and hybrid physics-learning models that combine differentiable physical simulators with learned dynamics to enable control-relevant update rates [102, 52]. In Figure 19, these needs appear jointly in the *Efficient Deployment* pillar (real-time constraints) and the *Reliable & Safe Intelligence* pillar (physics/causality for grounded decision-making).

Efficiency and scalability: bridging generality with edge deployment:. A central barrier to VLA adoption remains the mismatch between the computational footprint of large multi-modal backbones and the latency/energy constraints of closed-loop control [268, 238]. Future VLAs should prioritize parameter-efficient designs (structured sparsity, low-rank adaptation, modular experts) that preserve generalization while reducing inference cost [65, 197]. Beyond training-time efficiency, anytime/early-exit policies can allocate computation adaptively, ensuring that safety-critical steps retain high fidelity while routine steps use cheaper pathways [41, 274]. Equally important is action-space efficiency: compact action tokenization and chunked control representations shorten autoregressive horizons, enabling higher control rates without sacrificing temporal smoothness [171]. These model-level choices must be paired with hardware-aware compilation across GPU/NPU/edge accelerators, including quantization-aware scheduling and memory-optimized attention kernels [65, 167]. Compute-aware caching and episodic memory further re-

duce redundant forward passes and improve responsiveness in long-horizon tasks [27, 133]. Collectively, these directions operationalize the *Efficient Deployment* pillar of Figure 19 and directly address the compute/energy limitations emphasized earlier.

Cross-embodiment transfer and morphology-agnostic skill representations:. The approach of training separate VLAs for each robot morphology is unlikely to scale. A key future theme is embodiment-agnostic policy learning, where skills are expressed in abstract action spaces (e.g., contact goals, affordance-point manipulation, task-space constraints) that transfer across wheeled platforms, quadrupeds, and humanoids [275, 118]. Meta-learning and few-shot calibration can allow rapid bootstrapping on new robots with minutes of data rather than weeks of training [35]. This direction also mitigates dataset bias by enforcing invariances across embodiments and environments, but it demands standardized representations, common interfaces, and reproducible evaluation protocols [118, 275]. In Figure 19, this embodiment-agnostic skill learning paradigm sits under *Unified Systems & Governance*, linking architectural unification with principled transfer and benchmarking.

Evaluation beyond task success: safety, recovery, and resource-aware metrics:. Progress in VLA requires measurement that reflects deployment realities. Task success alone does not clearly represent failure severity, temporal inconsistency, unsafe near-misses, and energy inefficiency. Future benchmarks should quantify safety violations, uncertainty calibration, recovery behavior, temporal coherence, energy consumption, and downstream utility under human constraints [274, 78]. Moreover, evaluations should report compute budgets, dataset composition, and deployment conditions to enable fair comparisons and diagnose bias-driven gains [215, 273, 118]. Such measurement is not merely scientific hygiene: it is the foundation for auditing and governance, and it determines whether a system can be responsibly deployed at scale [274, 275]. This motivation underlies the evaluation branch of Figure 19 and complements the conceptual deployment narrative in Figure 18.

Safety, ethics, and human-centered alignment as first-class design objectives:. As VLAs gain autonomy, built-in safety and value alignment become critical. Future systems should integrate real-time risk estimators that assess potential harm before executing high-risk actions, request natural-language confirmation under ambiguity, and maintain transparent logs for accountability [274, 118]. Privacy-aware sensing, bias audits, and human-in-the-loop oversight must be embedded into the lifecycle, especially for assistive robotics and safety-critical autonomy [284, 235]. Regulatory-aligned evaluation protocols and standardization efforts will be essential to translate VLA advances into trusted real-world systems [169, 294, 234]. This governance perspective is explicitly captured in Figure 19 and is implicitly reflected by the socially aware humanoid setting illustrated in Figure 18.

下一代VLA系统可能会采用基于语言的规划器（针对操作可能性和约束条件进行微调的LLM风格模块），将目标分解为结构化子任务，随后通过中层技能策略和低层控制器确保运动合规性[94, 253]。这种神经符号混合架构通过构建更易调试、监控和验证的接口，有效降低了系统集成复杂度[201]。重要的是，层次结构还支持选择性验证：高层规划可检查约束违反情况（如不安全步骤、禁区）[192, 71]，而低层轨迹可通过控制屏障函数、MPC和运行时安全监测器进行防护[188, 196]。这些组件与图19中可靠与安全智能支柱下的架构相契合，其中安全性通过规划时约束和执行时保护机制双重保障，而非仅依赖事后评估[101, 274]。

通过世界模型与物理/因果推理实现实时自适应：在非结构化环境中实现稳健部署需要VLAs保持对物体、接触点及动态关系的内部预测模型。能够预测短期状态转换与故障概率的世界模型，可支持反事实评估（“若在此处推压，会碰撞什么？”）并在现实偏离预期时（如抓握滑动、意外摩擦）快速采取纠正措施[203]。该能力对安全操作、导航及人机交互至关重要，因微小误差会迅速累积[156]。但世界模型必须具备足够的车载使用效率，并与多传感器证据保持一致。因此，未来主要发展方向包括硬件感知型、内存高效的预测建模技术，例如时序标记压缩[267, 216]、事件驱动状态更新[250, 226]，以及结合可微分物理模拟器与学习动态模型的混合物理学习模型，以实现控制相关更新速率[102, 52]。在图19中，这些需求同时出现在高效部署支柱（实时约束）和可靠与安全智能支柱（物理/因果关系用于基于现实的决策制定）中。

效率与可扩展性：弥合通用性与边缘部署的鸿沟：VLA应用的核心障碍仍在于大型多模态骨干网络的计算需求与闭环控制的延迟/能耗限制之间的不匹配[268, 238]。未来的VLA应优先采用参数高效设计（结构化稀疏性、低秩适应、模块化专家模型），在降低推理成本的同时保持泛化能力[65, 197]。除训练时效率外，随时/提前退出策略可实现计算资源的自适应分配，确保安全关键步骤保持高保真度，常规步骤则采用更低成本的路径[41, 274]。动作空间效率同样重要：紧凑的动作分词与分块控制表示可缩短自回归时间范围，在不牺牲时间平滑性的情况下实现更高控制频率[171]。这些模型层面的选择需与针对GPU/NPU/边缘加速器的硬件感知编译技术相结合，包括量化感知调度和内存优化注意力核[65, 167]。计算感知缓存和情景记忆进一步减少了冗余前向传播，并提高了长时域任务中的响

应速度[27, 133]。这些研究方向共同实现了图19中高效部署支柱的理论化，并直接解决了前文强调的计算/能量限制问题。

跨形态迁移与形态无关技能表征：针对每种机器人形态单独训练VLAs的方法难以实现规模化应用。未来关键方向是形态无关策略学习，即通过抽象动作空间（如接触目标、可操作点操控、任务空间约束）来表达技能，这些技能可在轮式平台、四足机器人和类人机器人之间迁移[275, 118]。元学习与少样本校准技术可使新机器人仅需数分钟数据即可快速启动训练，而非数周训练周期[35]。该方向通过强制形态与环境间的不变性来缓解数据集偏差，但需要标准化表征、通用接口及可复现的评估协议[118, 275]。图19中，这种形态无关技能学习范式归属于统一系统与治理范畴，将架构统一性与原理性迁移及基准测试相结合。

超越任务成功指标的评估：安全性、恢复情况及资源消耗

相关指标：VLA的发展需要能够反映实际部署情况的测量指标。仅凭任务成功并不能清晰反映失败严重性、时间不一致性、不安全的险些失败事件以及能效低下问题。未来的基准测试应量化安全违规行为、不确定性校准、恢复行为、时间一致性、能耗以及在人类约束条件下的下游效用[274, 78]。此外，评估报告应包含计算预算、数据集构成和部署条件，以便进行公平比较并诊断由偏差驱动的改进效果[215, 273, 118]。此类测量不仅是科学规范问题：它是审计与治理的基础，并决定了系统能否在大规模场景下得到负责任的部署[274, 275]。这一动机构成了图19中评估部分的核心内容，并与图18中的概念性部署论述相辅相成。**安全、伦理及以人为本的协同目标作为首要设计目标：**随着VLAs获得自主性，内置的安全性价值协同变得至关重要。未来系统应整合实时风险评估器，在执行高风险操作前评估潜在危害，在存在歧义时请求自然语言确认，并保持透明日志以确保问责机制[274, 118]。隐私感知传感技术、偏差审计机制以及人工环路监督必须贯穿整个生命周期，尤其对于辅助机器人和安全关键型自主系统而言[284, 235]。与监管要求相匹配的评估协议及标准化工作，对于将虚拟现实辅助技术（VLA）成果转化为可信的现实系统至关重要[169, 294, 234]。这一治理视角在图19中得到明确体现，图18所展示的社会感知型人形机器人场景则隐含了这一理念。

Cross-cutting themes: continual learning, failure recovery, interaction, and control fidelity. Across all pillars in Figure 19, several cross-cutting themes are expected to shape the next decade of VLA research. First, continual and lifelong learning must be safe, auditable, and resistant to forgetting, enabling long-term adaptation without destabilizing deployed systems [290]. Second, failure detection and recovery should be treated as first-class capabilities, incorporating introspective monitoring, uncertainty-aware perception, and structured recovery behaviors when execution deviates from expected outcomes [116, 259]. Third, improving the precision and reliability of action generation remains critical: while VLA-based planners enable flexible, language-conditioned decision making, their trajectory accuracy and control stability currently lag behind conventional analytical approaches such as model predictive control, sampling-based motion planning, and feedback-linearized controllers. As a result, hybrid architectures that combine VLA-driven high-level planning with classical or learned low-level controllers are expected to play a central role in achieving both semantic flexibility and control-level precision. Finally, human alignment and interaction require mechanisms for intent clarification, shared autonomy, and explainable action rationales, supporting trust and usability across diverse real-world settings [111, 110].

In summary, Figure 19 emphasizes that closing the gap between laboratory demonstrations and robust real-world deployment will require coordinated advances in efficiency, safety, data and generalization, system integration, evaluation, and governance. Complementarily, Figure 18 illustrates how these advances may converge toward generalist embodied agents across multiple platforms including mobile robots, manipulators, assistive systems, and humanoids where multi-modal perception, hierarchical planning, continual adaptation, and human-aligned safety are integrated within a unified intelligence stack. Addressing these directions collectively is expected to transform VLAs from promising research prototypes into dependable, broadly applicable embodied systems rather than solutions limited to any single robot morphology.

6. Conclusion

In this comprehensive review, we systematically evaluated the recent developments, methodologies, and applications of Vision-Language-Action (VLA) models published over the last three years. Our analysis began with the foundational concepts of VLAs, defining their role as multi-modal systems that unify visual perception, natural language understanding, and action generation in physical or simulated environments. We traced their evolution and timeline, detailing key milestones that marked the transition from isolated perception-action modules to fully unified, instruction-following robotic agents. We highlighted how multi-modal integration has matured from loosely coupled pipelines to transformer-based architectures that enable seamless coordination between modalities.

Next, we examined tokenization and representation techniques, focusing on how VLAs encode visual and linguistic information, including action primitives and spatial semantics.

We explored learning paradigms, detailing the datasets and training strategies from supervised learning and imitation learning to reinforcement learning and multi-modal pretraining that have shaped VLA performance. In ‘adaptive control and real-time execution’ section, we discussed how modern VLAs are optimized for dynamic environments, analyzing policies that support latency-sensitive tasks. We then categorized major architectural innovations, surveying over 50 recent VLA models. This discussion included advancements in model design, memory systems, and interaction fidelity. We further studied strategies for training efficiency improvement, including parameter-efficient methods like LoRA, quantization, and model pruning, alongside acceleration techniques such as parallel decoding and hardware-aware inference. Our analysis of real-world applications highlighted both the promise and current limitations of VLA models across six domains: humanoid robotics, autonomous vehicles, industrial automation, healthcare, agriculture, and augmented reality (AR) navigation. Across these settings, VLAs demonstrated strong capabilities in high-level semantic reasoning, instruction-following, and task generalization, particularly in structured or partially controlled environments. However, their effectiveness was often constrained by real-time inference latency, limited robustness under environmental variability, and reduced precision in long-horizon or safety-critical control when compared to conventional analytical planning and control pipelines. Moreover, application-specific adaptations and extensive data curation were frequently required to achieve reliable performance, underscoring challenges in scalability and deployment. These findings suggest that while VLAs are well-suited for semantic decision making and flexible task specification, hybrid architectures that integrate VLA reasoning with classical or learned low-level controllers remain essential for practical, real-world operation.

In addressing challenges and limitations, we focused on five core areas: real-time inference, multi-modal action representation and safety, bias and generalization, system integration and compute constraints, and ethical deployment. We proposed potential solutions drawn from current literature, including model compression, cross-modal grounding, domain adaptation, and agentic learning frameworks. Finally, our discussion and future roadmap articulated how the convergence of VLMs, VLA architectures, and agentic AI systems is steering robotics toward artificial general intelligence (AGI). This review provides a unified understanding of VLA advancements, identifies unresolved challenges, and outlines a structured path forward for developing intelligent, embodied, and human-aligned agents in the future.

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贯穿性主题：持续学习、故障恢复、交互性与控制保真度：在图19的所有支柱中，若干贯穿性主题预计将主导未来十年VLA研究方向。首先，持续终身学习必须具备安全性、可审计性及抗遗忘特性，以实现长期适应性调整而不导致系统失稳[290]。其次，故障检测与恢复应被视为核心能力，需结合内省式监控、不确定性感知及结构化恢复行为，当执行结果偏离预期时及时响应[116, 259]。第三，提升动作生成的精确度与可靠性仍至关重要：尽管基于VLA的规划器能实现灵活的语言条件决策，但其轨迹精度与控制稳定性目前仍落后于模型预测控制、基于采样的运动规划及反馈线性化控制器等传统分析方法。因此，将VLA驱动的高层规划与经典或学习型低层控制器相结合的混合架构，有望在实现语义灵活性与控制精度方面发挥核心作用。最后，人类对齐与交互需要具备意图澄清、共享自主权以及可解释的动作逻辑机制，从而在多样化的实际应用场景中提升信任度与可用性[111, 110]。

综上所述，图19强调，要弥合实验室演示与实际应用之间的差距，需要在效率提升、安全性增强、数据整合与泛化能力、系统集成、评估机制及治理框架等方面实现协同突破。与此同时，图18展示了这些技术进步如何汇聚成通用型具身智能体，其应用场景涵盖移动机器人、机械臂、辅助系统及类人机器人等多个平台。这类智能体通过统一智能架构整合多模态感知、分层规划、持续适应能力及符合人体工学的安全设计。若能系统性推进这些方向，虚拟实验室助理系统将从具有潜力的研究原型，蜕变为可靠且广泛适用的具身智能系统，而非局限于单一机器人形态的解决方案。

6. 结论

在这篇全面综述中，我们系统评估了过去三年间发表的视觉-语言-动作（Vision-Language-Action, VLA）模型的最新进展、研究方法及应用实践。我们的分析首先从VLA的基础概念入手，明确其作为多模态系统的核心功能——即在物理或模拟环境中整合视觉感知、自然语言理解与动作生成能力。我们梳理了该领域的演进历程与时间线，详细阐述了从孤立的感知-动作模块向完全统一、能执行指令的机器人智能体转型过程中的关键里程碑。同时，我们重点指出：多模态整合技术已从松散耦合的处理流程发展为基于Transformer架构的解决方案，实现了不同模态之间的无缝协同。

接下来，我们重点研究了分词与表征技术，深入解析视觉语言模型（VLAs）如何编码视觉信息与语言信息，包括动作基元和空间语义。通过系统梳理学习范式，详细阐

述了从监督学习、模仿学习到强化学习及多模态预训练等数据集与训练策略对VLAs性能的影响。在“自适应控制与实时执行”章节中，我们探讨了现代VLAs如何针对动态环境进行优化，重点分析了支持延迟敏感任务的策略设计。随后对50余种最新VLAs模型进行架构创新分类研究，涵盖模型设计、内存系统及交互保真度等领域的技术突破。进一步研究了提升训练效率的策略，包括LoRA参数优化方法、量化技术、模型剪枝等参数高效方法，以及并行解码、硬件感知推理等加速技术。我们对真实应用场景的分析揭示了VLA模型在六个领域（人形机器人、自动驾驶汽车、工业自动化、医疗健康、农业及增强现实导航）中的应用潜力与现存局限。在这些场景中，VLA模型展现出强大的高级语义推理能力、指令执行能力和任务泛化能力，尤其在结构化或部分受控环境中表现突出。然而其实际效能常受限于实时推理延迟、环境变化时的鲁棒性不足，以及与传统分析规划控制流程相比，在长时域控制或安全关键控制场景中精度较低。此外，要实现可靠性往往需要针对具体应用场景进行定制化调整并进行大量数据整理，这凸显了系统可扩展性与部署实施的挑战。研究结果表明，尽管VLA模型在语义决策制定和灵活任务定义方面具有优势，但将VLA推理与经典或学习型低级控制器相结合的混合架构，仍是实际应用中不可或缺的核心要素。

在应对挑战与局限性时，我们聚焦五大核心领域：实时推理、多模态动作表征与安全机制、偏差与泛化能力、系统集成与计算约束，以及伦理部署。基于现有文献，我们提出了包括模型压缩、跨模态基底化、领域适应性及智能体学习框架在内的潜在解决方案。最后，通过讨论与未来路线图，我们阐明了视觉语言模型（VLM）、视觉语言架构（VLA）与智能体AI系统的融合如何推动机器人技术向人工通用智能（AGI）迈进。本综述系统梳理了VLA技术进展，识别了尚未解决的挑战，并为未来开发智能、具身化且符合人类需求的智能体勾勒出清晰的发展路径。

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Statement on AI Writing Assistance

ChatGPT and Perplexity were utilized to enhance grammatical accuracy and refine sentence structure; all AI-generated revisions were thoroughly reviewed and edited for relevance.

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声明

作者声明无利益冲突。

关于人工智能写作辅助的声明

本研究采用ChatGPT和Perplexity技术提升语法准确性并优化句子结构；所有人工智能生成的修订内容均经过全面审阅并根据相关性进行编辑。

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